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(54) **VEHICLE HAVING ADJUSTABLE COMPRESSION AND REBOUND DAMPING**

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See application file for complete search history.

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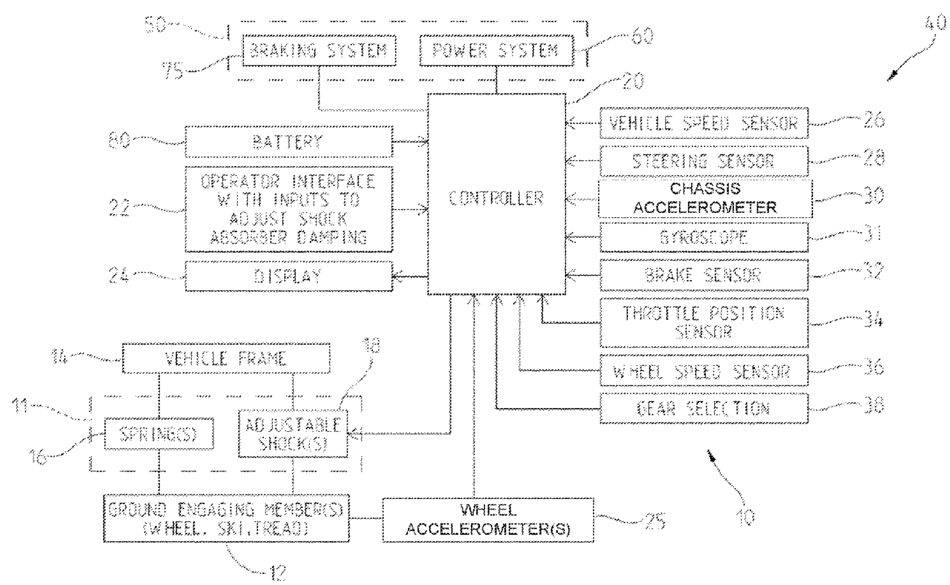
(57) **ABSTRACT**

A damping control system for a vehicle having a suspension located between a plurality of ground engaging members and a vehicle frame are disclosed. The vehicle including at least one adjustable shock absorber having an adjustable damping characteristic.

36 Claims, 31 Drawing Sheets



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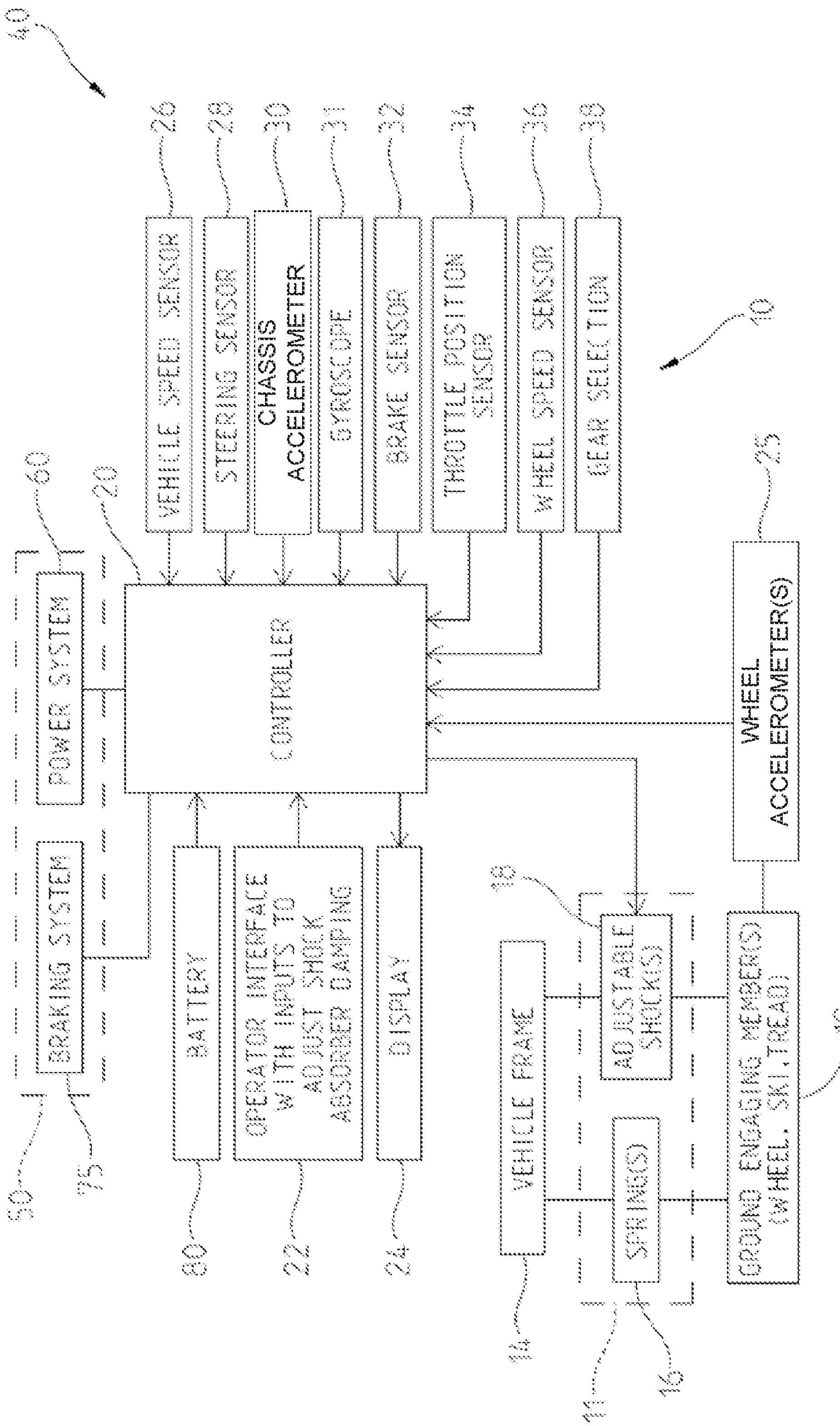
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GEAR SELECTION
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36
38
CONTROLLER
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POWER SYSTEM
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BRAKING SYSTEM
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BATTERY
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OPERATOR INTERFACE WITH INPUTS TO ADJUST SHOCK ABSORBER DAMPING
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DISPLAY
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WHEEL ACCELEROMETER(S)
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ADJUSTABLE SHOCK(S)
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GROUND ENGAGING MEMBER(S) (WHEEL, SKI, TREAD)
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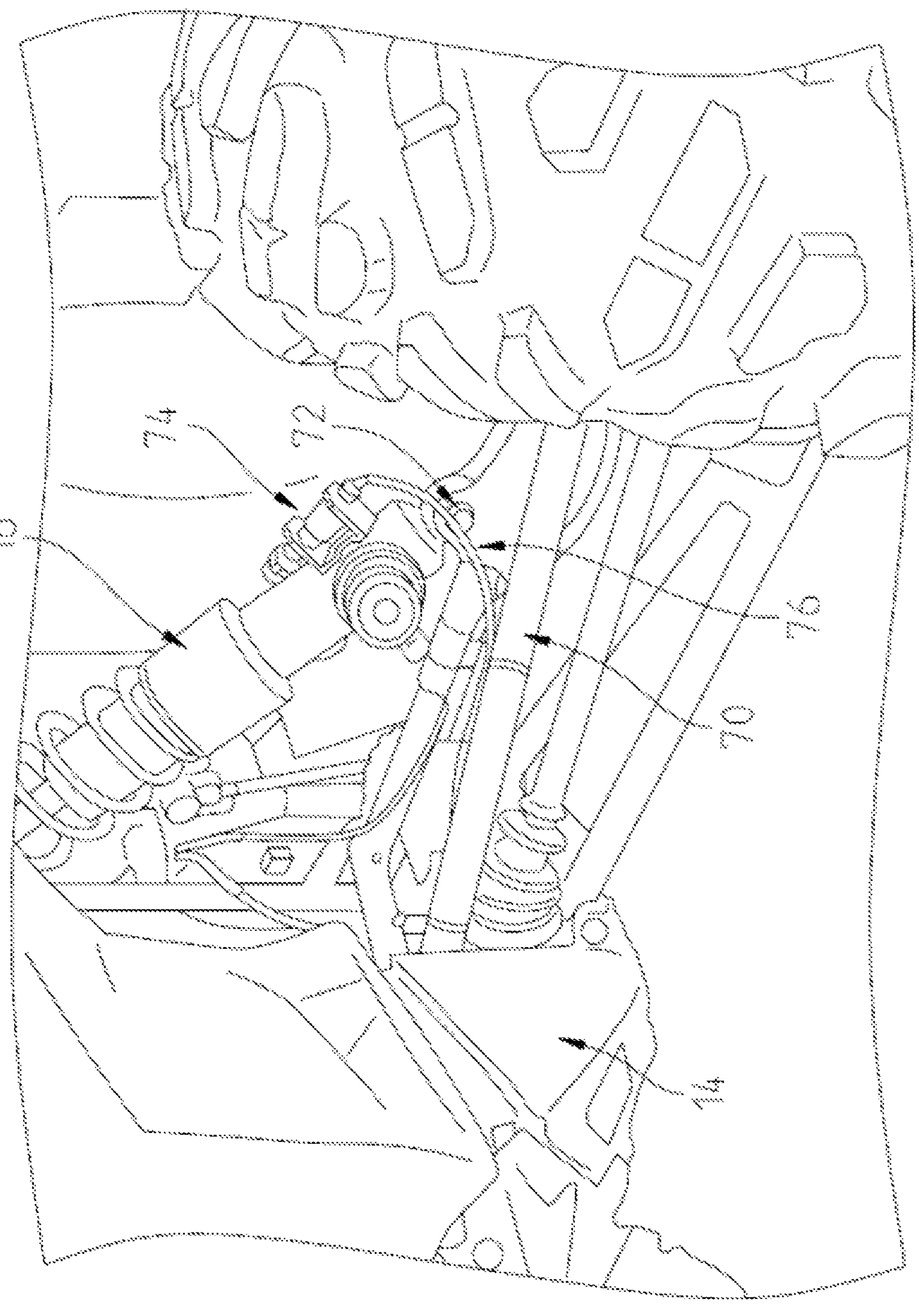
Fig. 1



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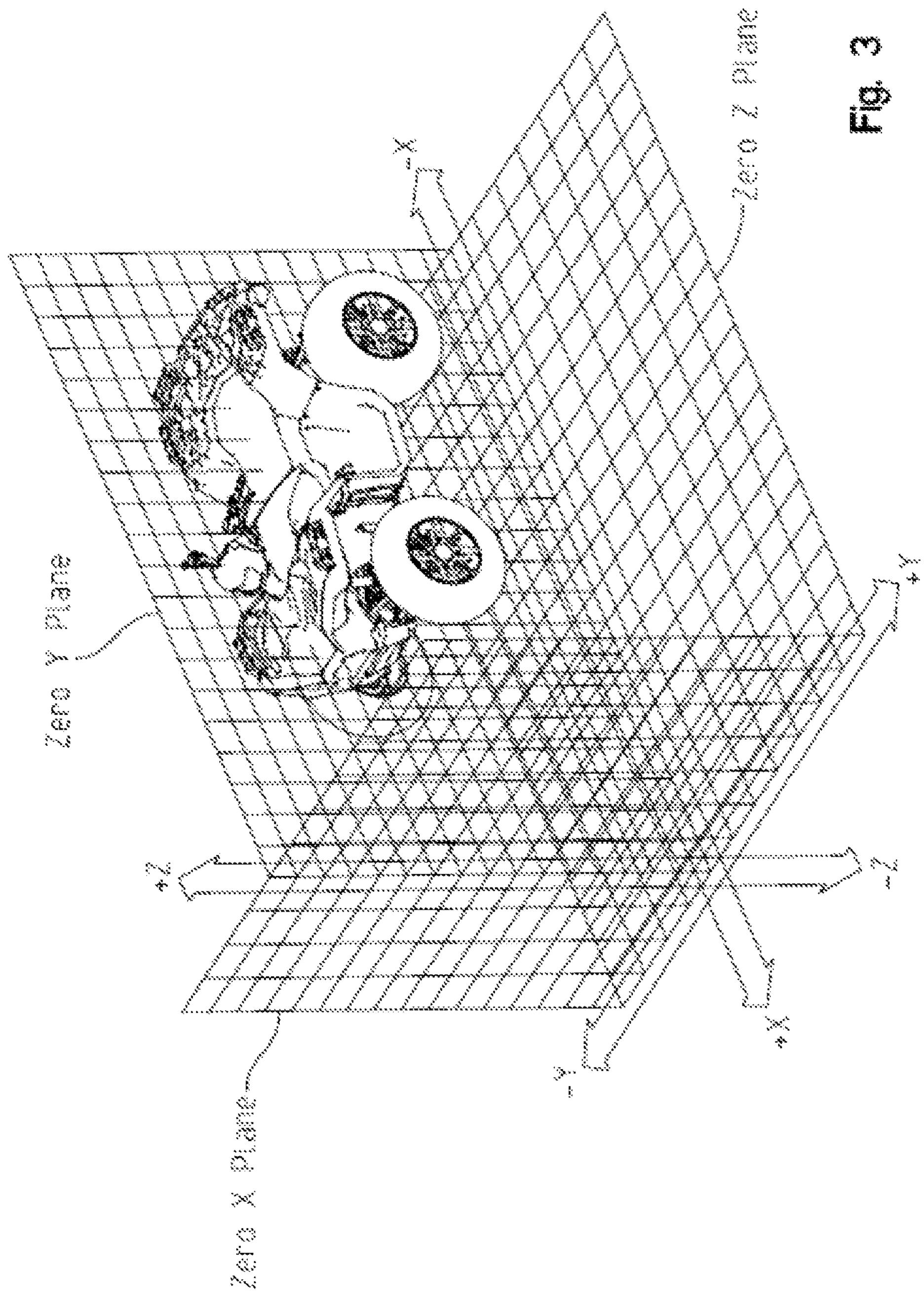

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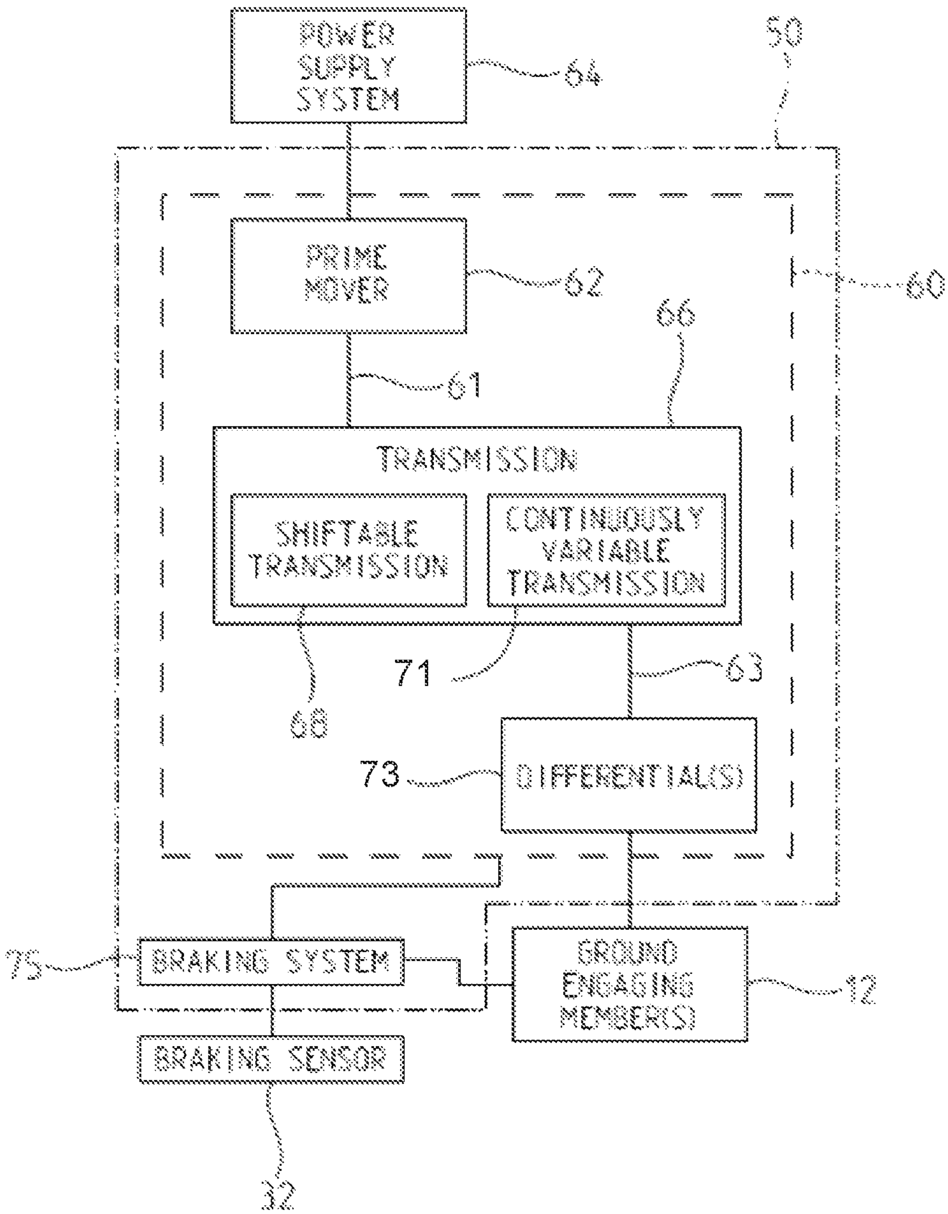

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SHIFTABLE TRANSMISSION
CONTINUOUSLY VARIABLE TRANSMISSION
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63
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BRAKING SYSTEM
GROUND ENGAGING MEMBER(S)
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BRAKING SENSOR
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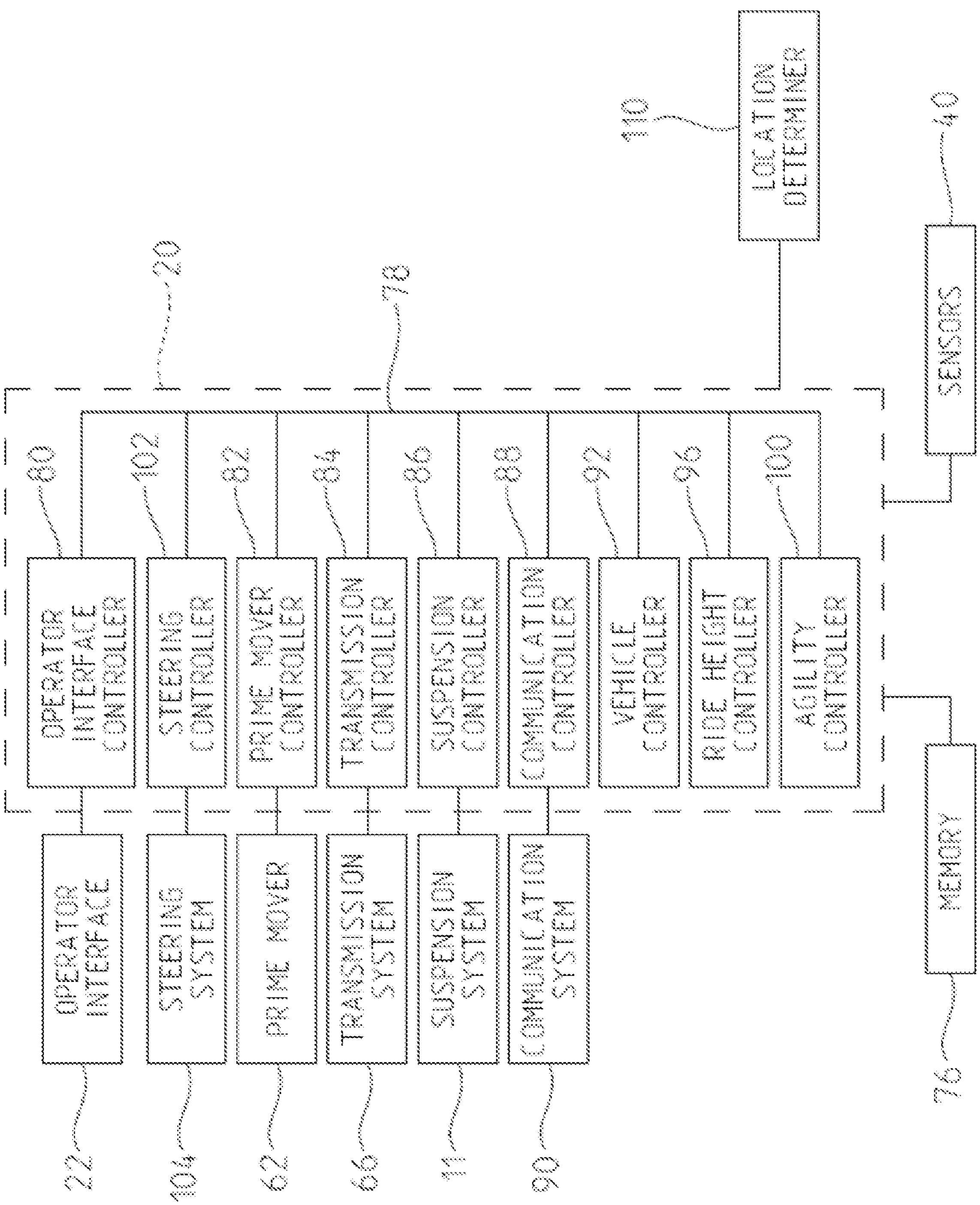
Fig. 4

Fig. 5



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SUSPENSION CONTROLLER
86
COMMUNICATION CONTROLLER
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VEHICLE CONTROLLER
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RIDE HEIGHT CONTROLLER
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AGILITY CONTROLLER
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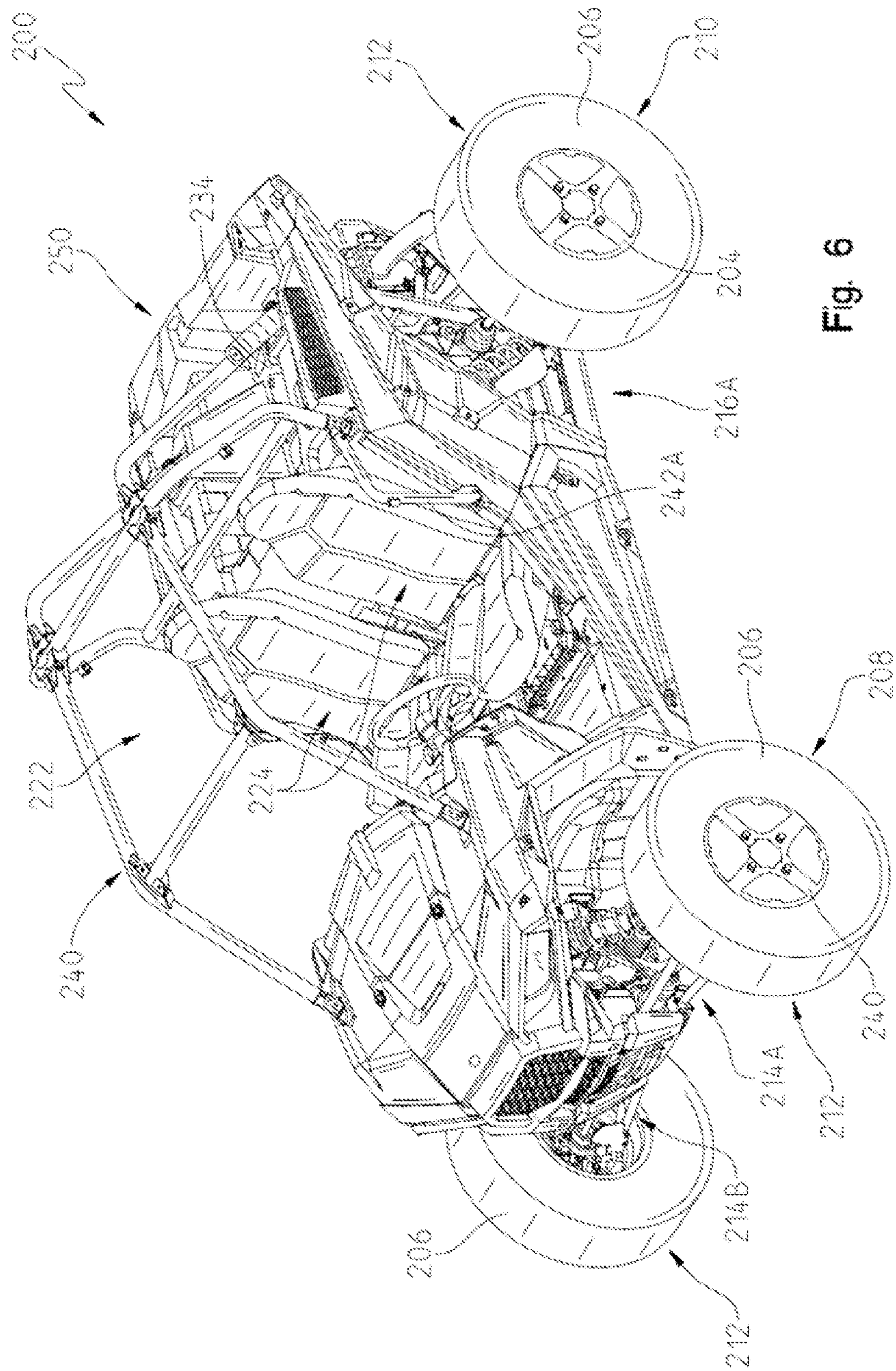
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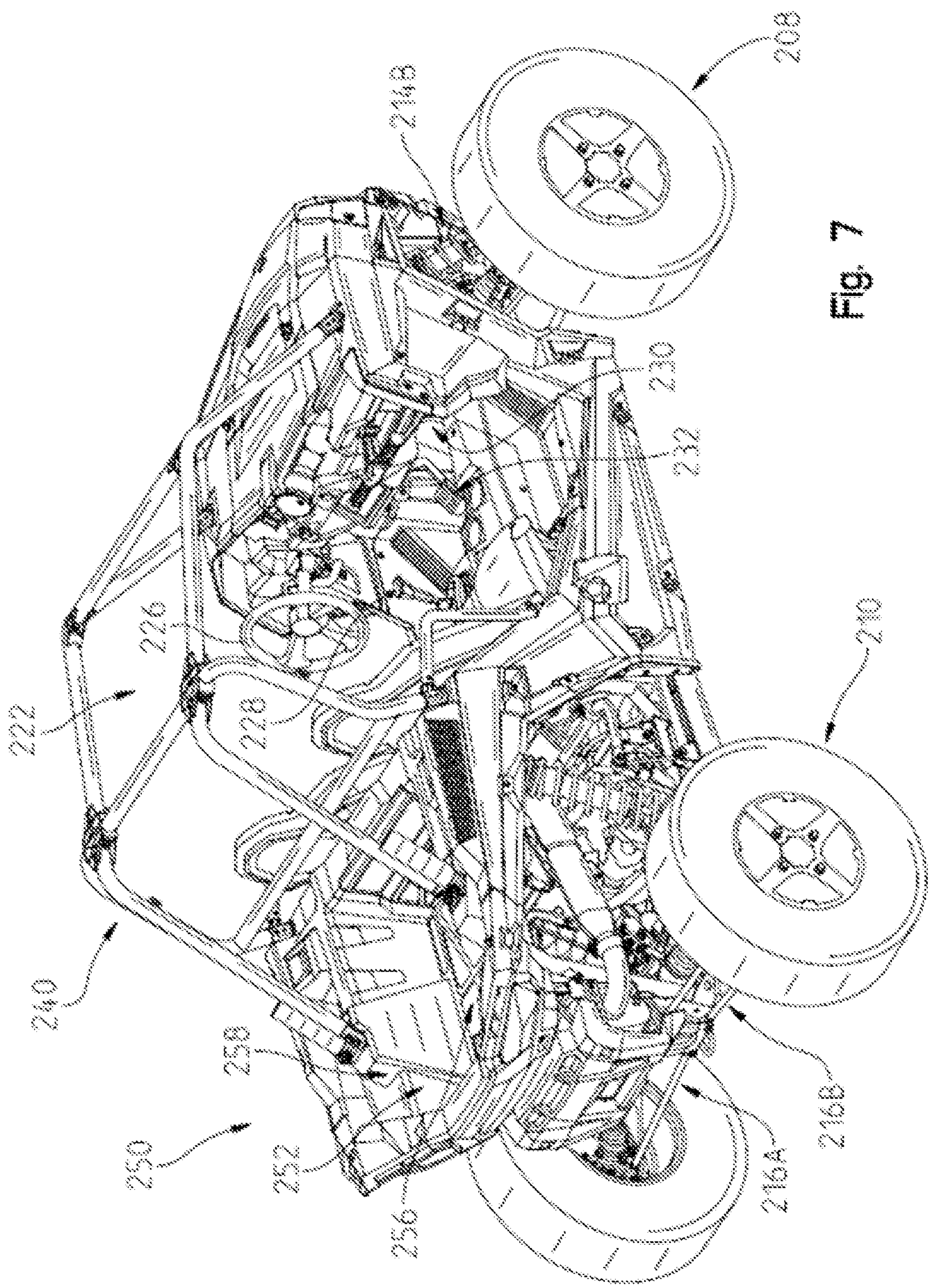
Fig. 6



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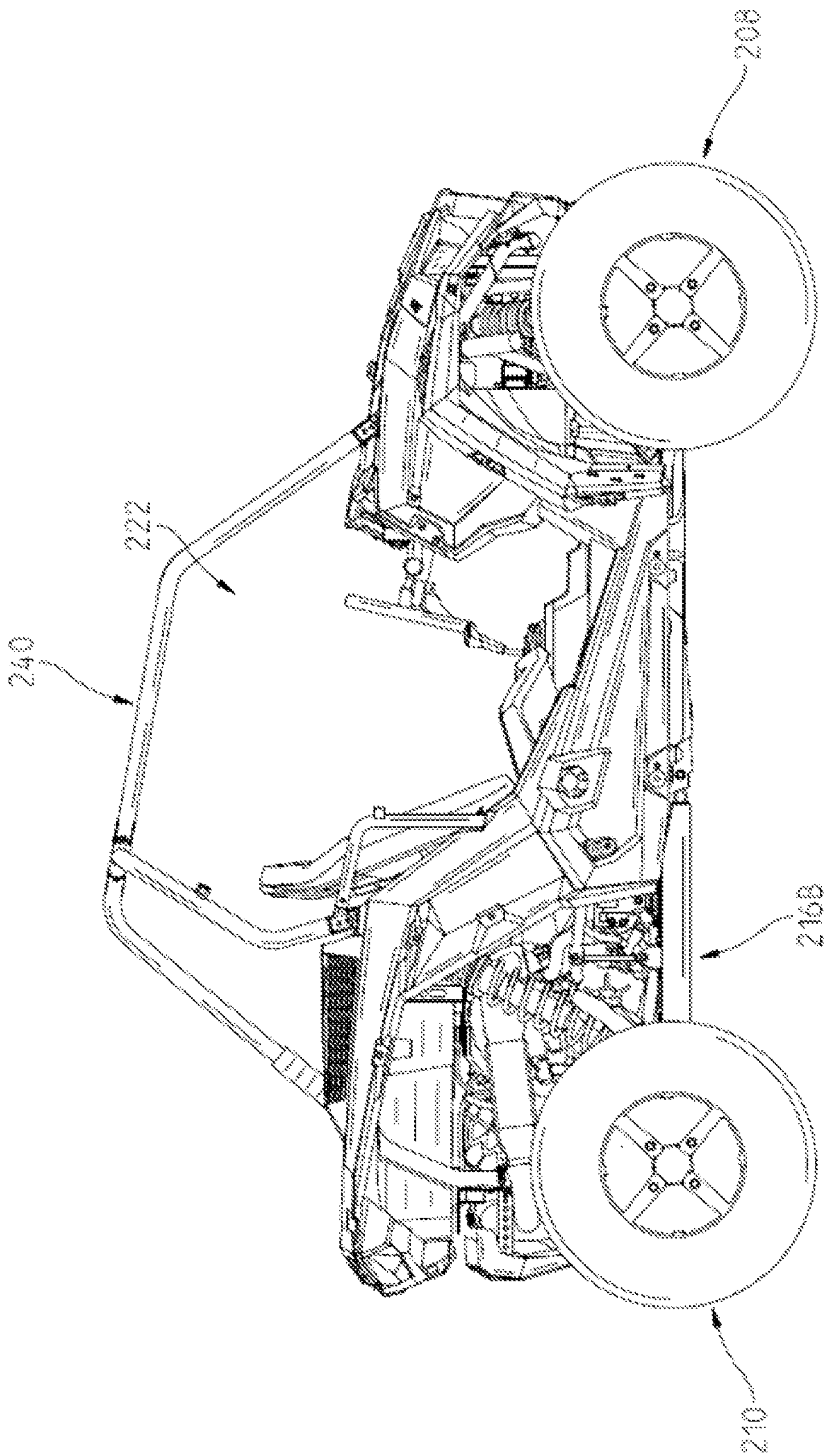

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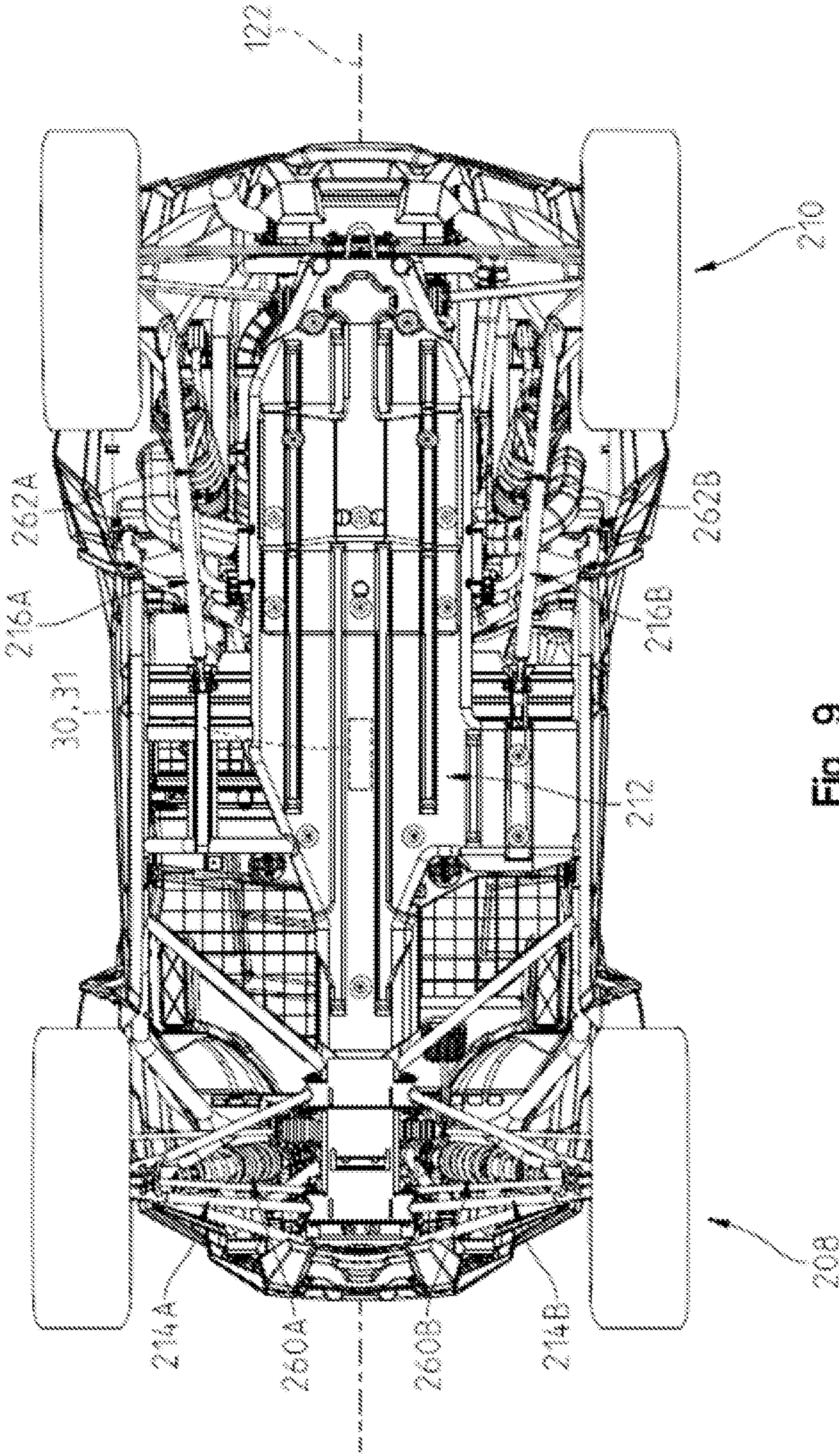

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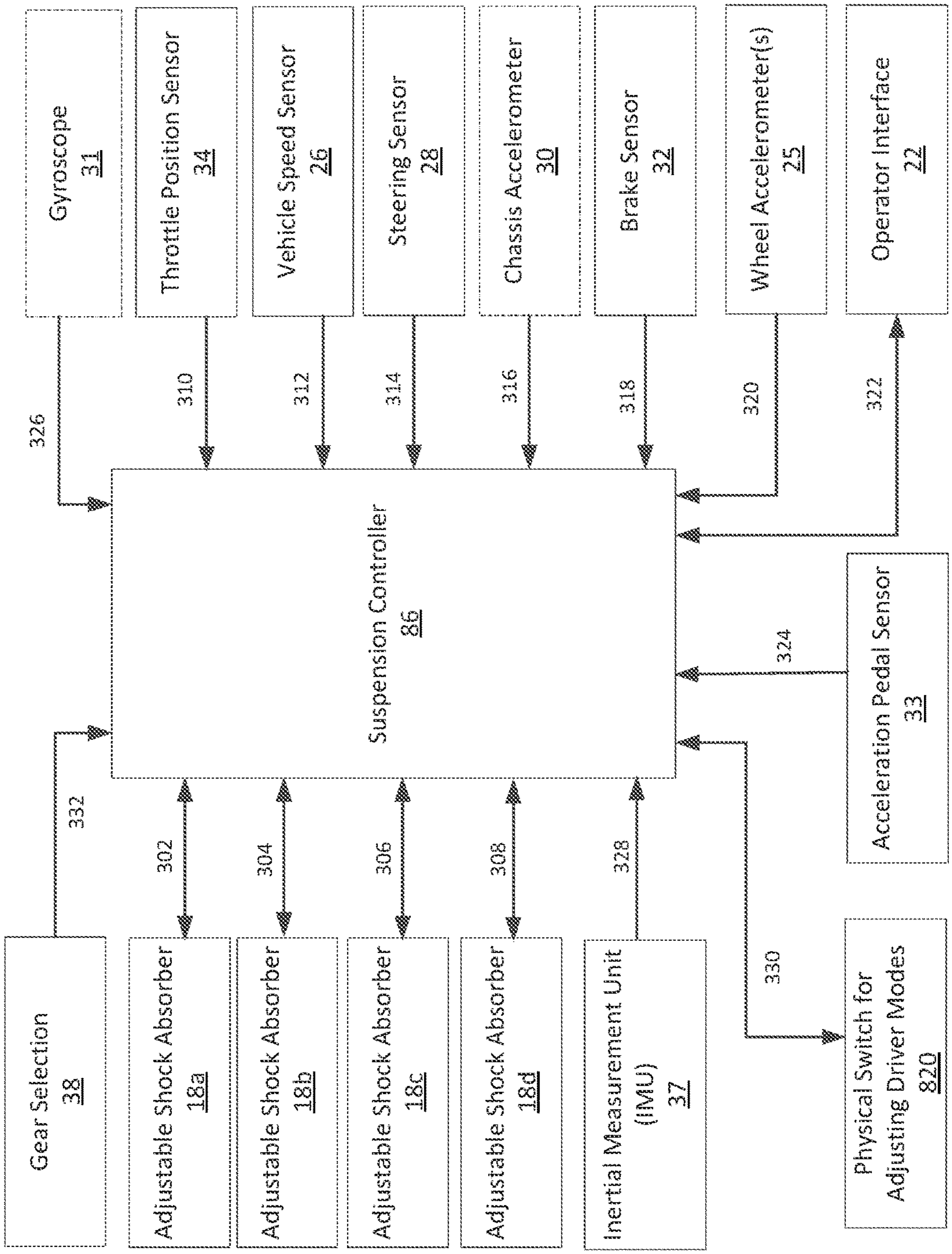
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Vehicle Speed Sensor 26
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Brake Sensor 32
Wheel Accelerometer(s) 25
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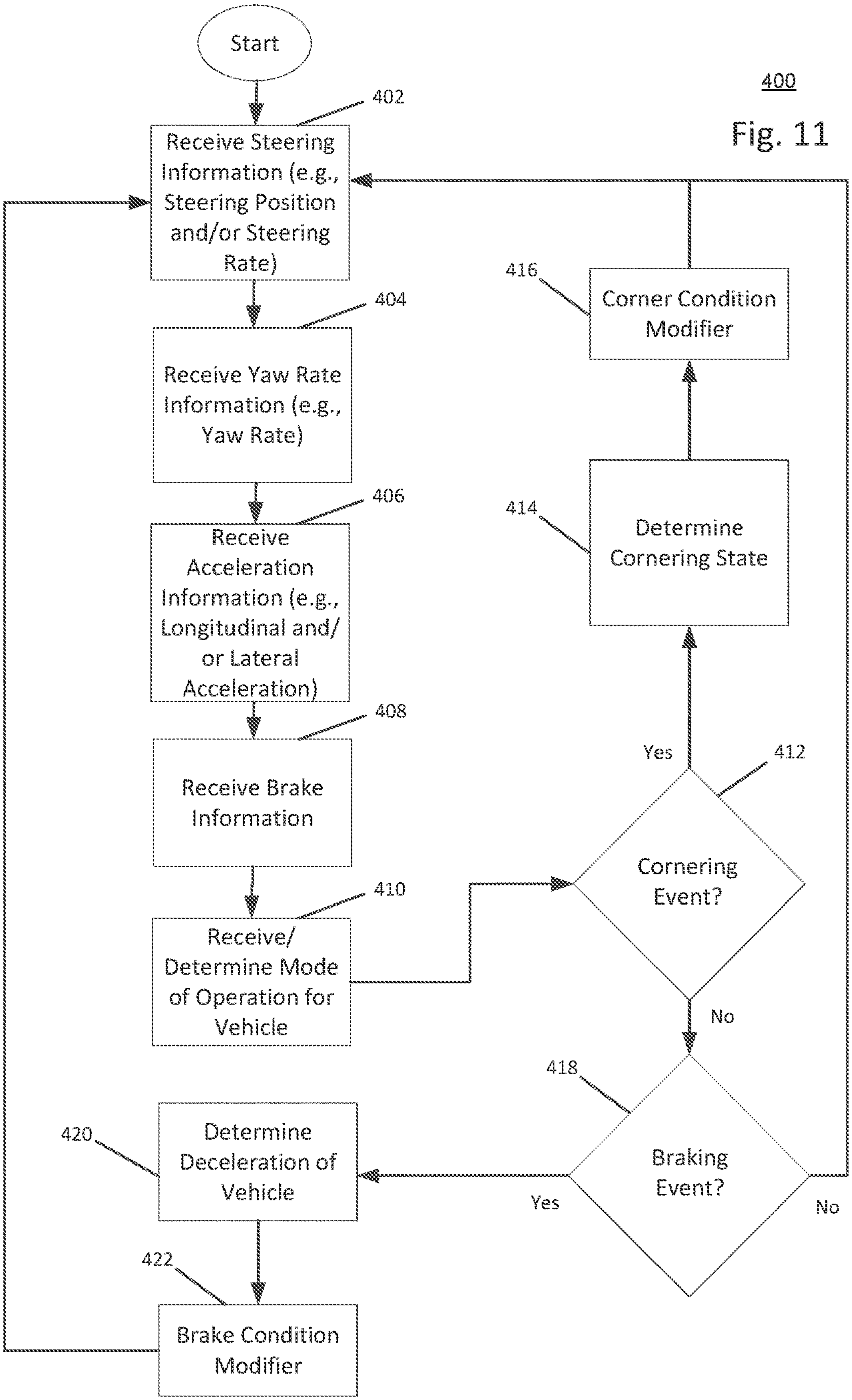
310
312
314
316
318
320
322
Suspension Controller 86
324
Acceleration Pedal Sensor 33
332
302
304
306
308
328
330
Gear Selection 38
Adjustable Shock Absorber 18a
Adjustable Shock Absorber 18b
Adjustable Shock Absorber 18c
Adjustable Shock Absorber 18d
Inertial Measurement Unit (IMU) 37
Physical Switch for Adjusting Driver Modes 820

300
FIG. 10



Start
400
Fig. 11
402
Receive Steering Information (e.g., Steering Position and/or Steering Rate)
404
Receive Yaw Rate Information (e.g., Yaw Rate)
406
Receive Acceleration Information (e.g., Longitudinal and/ or Lateral Acceleration)
408
Receive Brake Information
410
Receive/ Determine Mode of Operation for Vehicle
412
Cornering Event?
Yes
No
414
Determine Cornering State
416
Corner Condition Modifier
418
Braking Event?
Yes
No
420
Determine Deceleration of Vehicle
422
Brake Condition Modifier

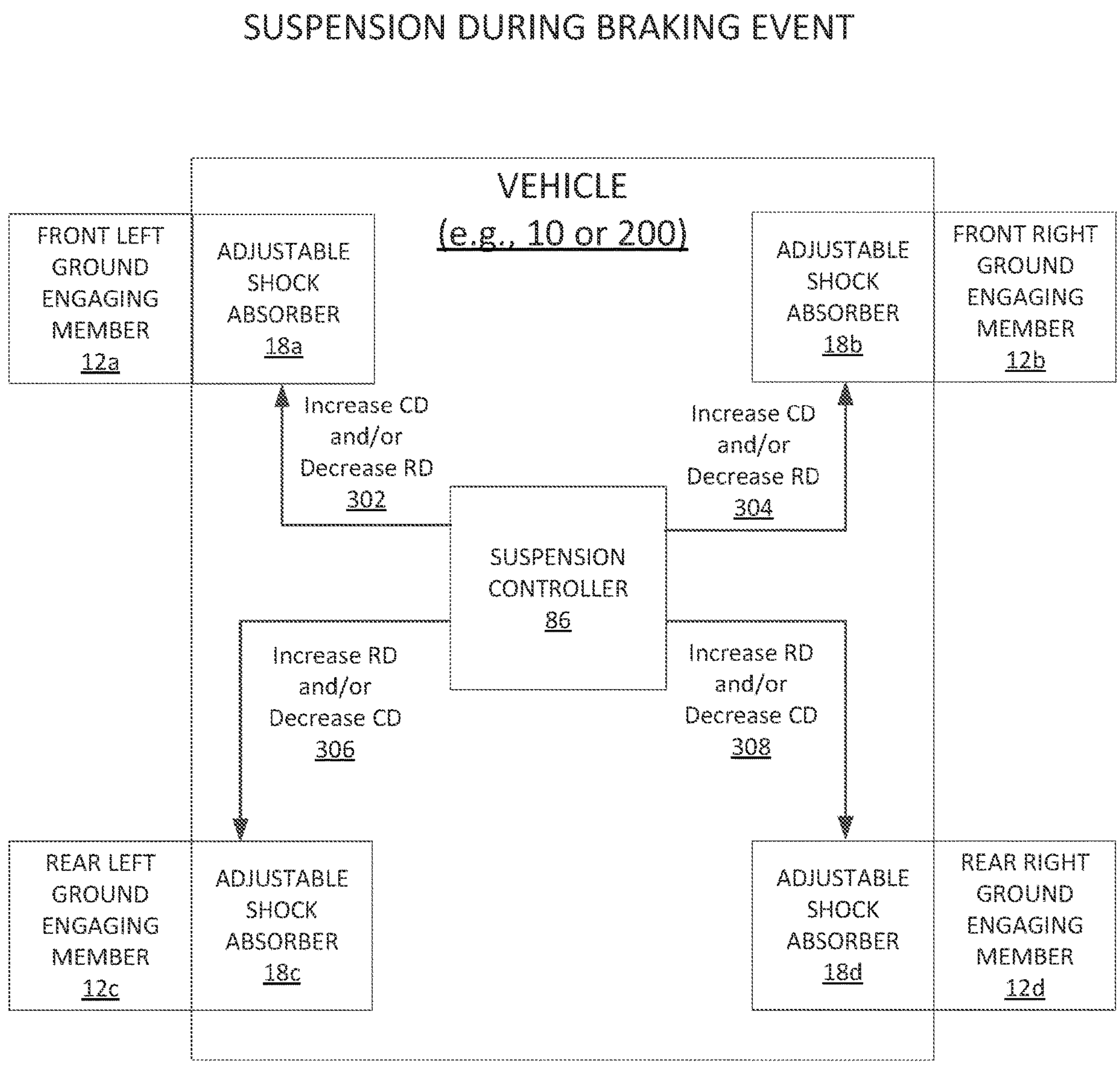

SUSPENSION DURING BRAKING EVENT
VEHICLE
(e.g., 10 or 200)
FRONT LEFT GROUND ENGAGING MEMBER 12a
ADJUSTABLE SHOCK ABSORBER 18a
ADJUSTABLE SHOCK ABSORBER 18b
FRONT RIGHT GROUND ENGAGING MEMBER 12b
Increase CD and/or Decrease RD 302
Increase CD and/or Decrease RD 304
SUSPENSION CONTROLLER 86
Increase RD and/or Decrease CD 306
Increase RD and/or Decrease CD 308
REAR LEFT GROUND ENGAGING MEMBER 12c
ADJUSTABLE SHOCK ABSORBER 18c
ADJUSTABLE SHOCK ABSORBER 18d
REAR RIGHT GROUND ENGAGING MEMBER 12d

Fig. 12

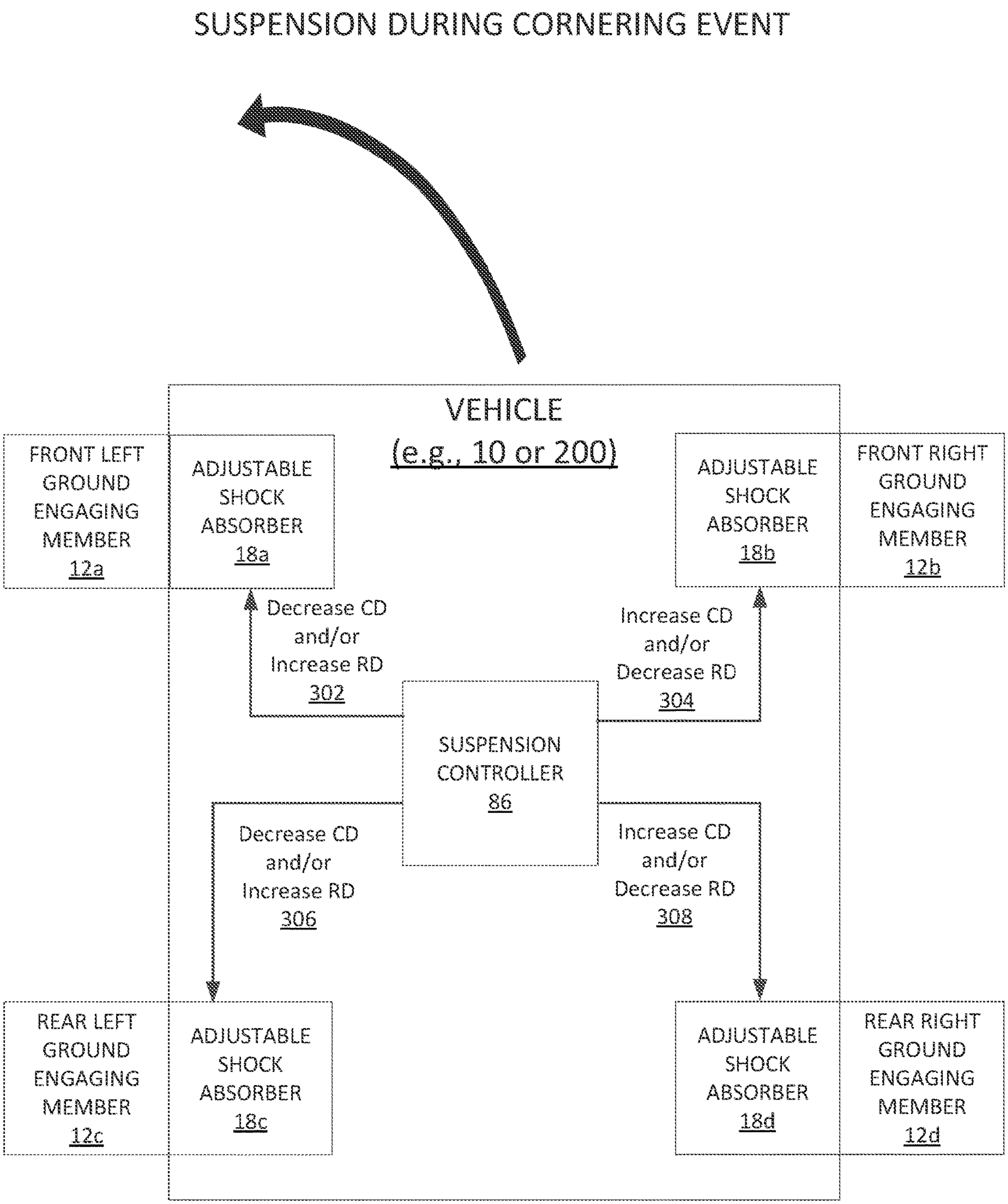

SUSPENSION DURING CORNERING EVENT
VEHICLE
(e.g., 10 or 200)
FRONT LEFT GROUND ENGAGING MEMBER 12a
ADJUSTABLE SHOCK ABSORBER 18a
ADJUSTABLE SHOCK ABSORBER 18b
FRONT RIGHT GROUND ENGAGING MEMBER 12b
Decrease CD and/or Increase RD 302
Increase CD and/or Decrease RD 304
SUSPENSION CONTROLLER 86
Decrease CD and/or Increase RD 306
Increase CD and/or Decrease RD 308
REAR LEFT GROUND ENGAGING MEMBER 12c
ADJUSTABLE SHOCK ABSORBER 18c
ADJUSTABLE SHOCK ABSORBER 18d
REAR RIGHT GROUND ENGAGING MEMBER 12d

Fig. 13

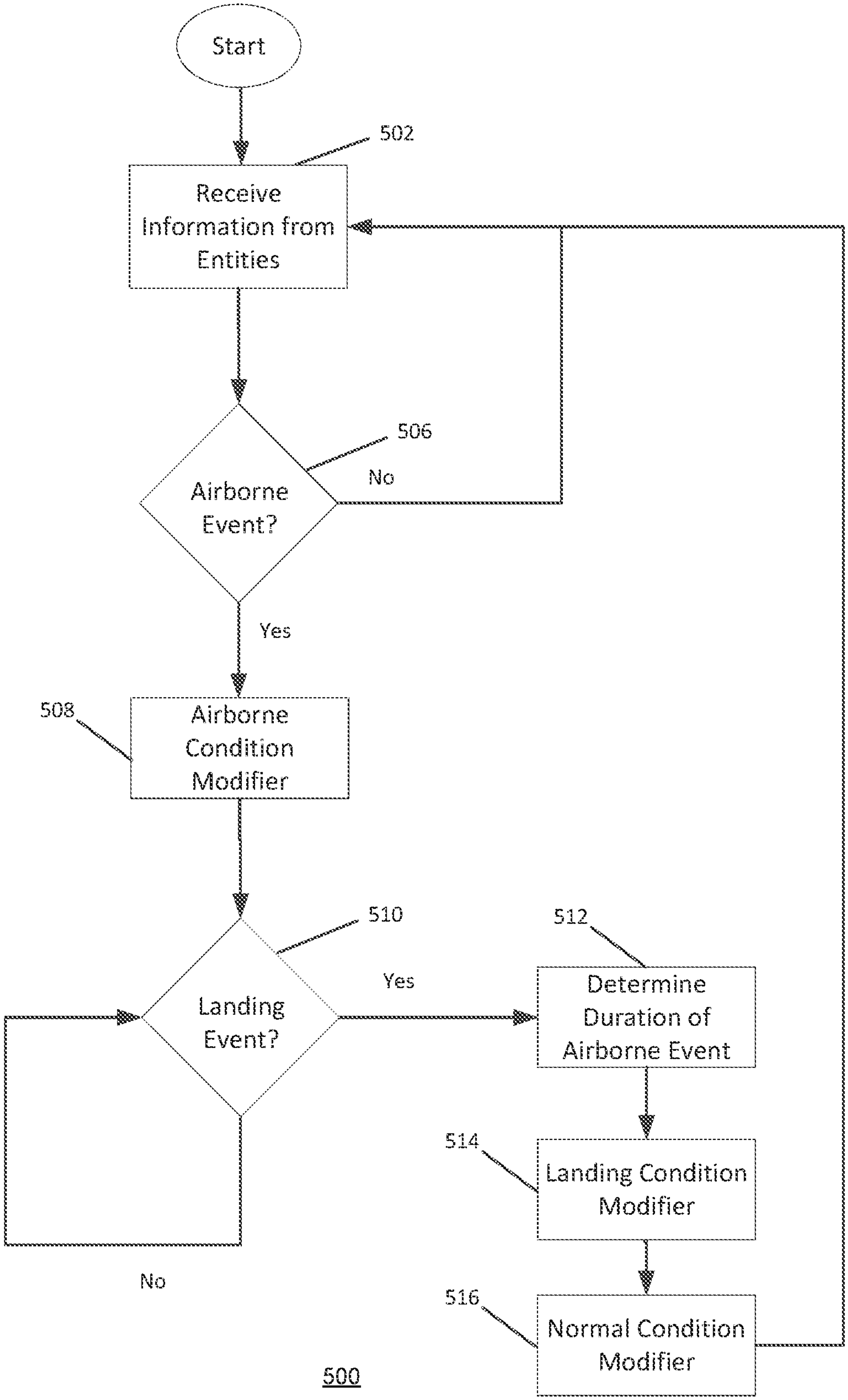

Start
502
Receive Information from Entities
506
Airborne Event?
No
Yes
508
Airborne Condition Modifier
510
Landing Event?
Yes
No
512
Determine Duration of Airborne Event
514
Landing Condition Modifier
516
Normal Condition Modifier
500

Fig. 14

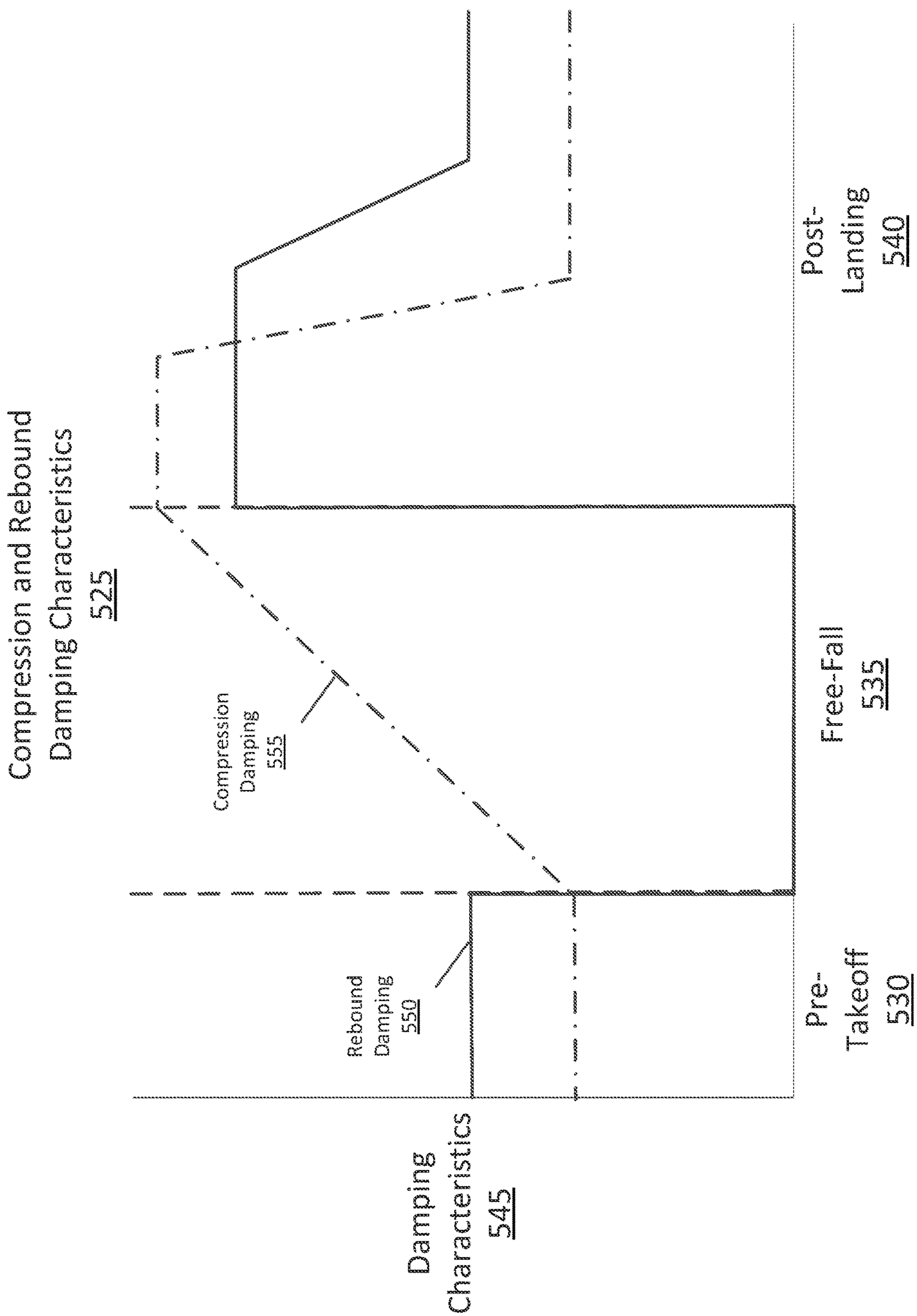

Compression and Rebound Damping Characteristics 525
Compression Damping 555
Rebound Damping 550
Damping Characteristics 545
Pre-Takeoff 530
Free-Fall 535
Post-Landing 540

Fig. 15

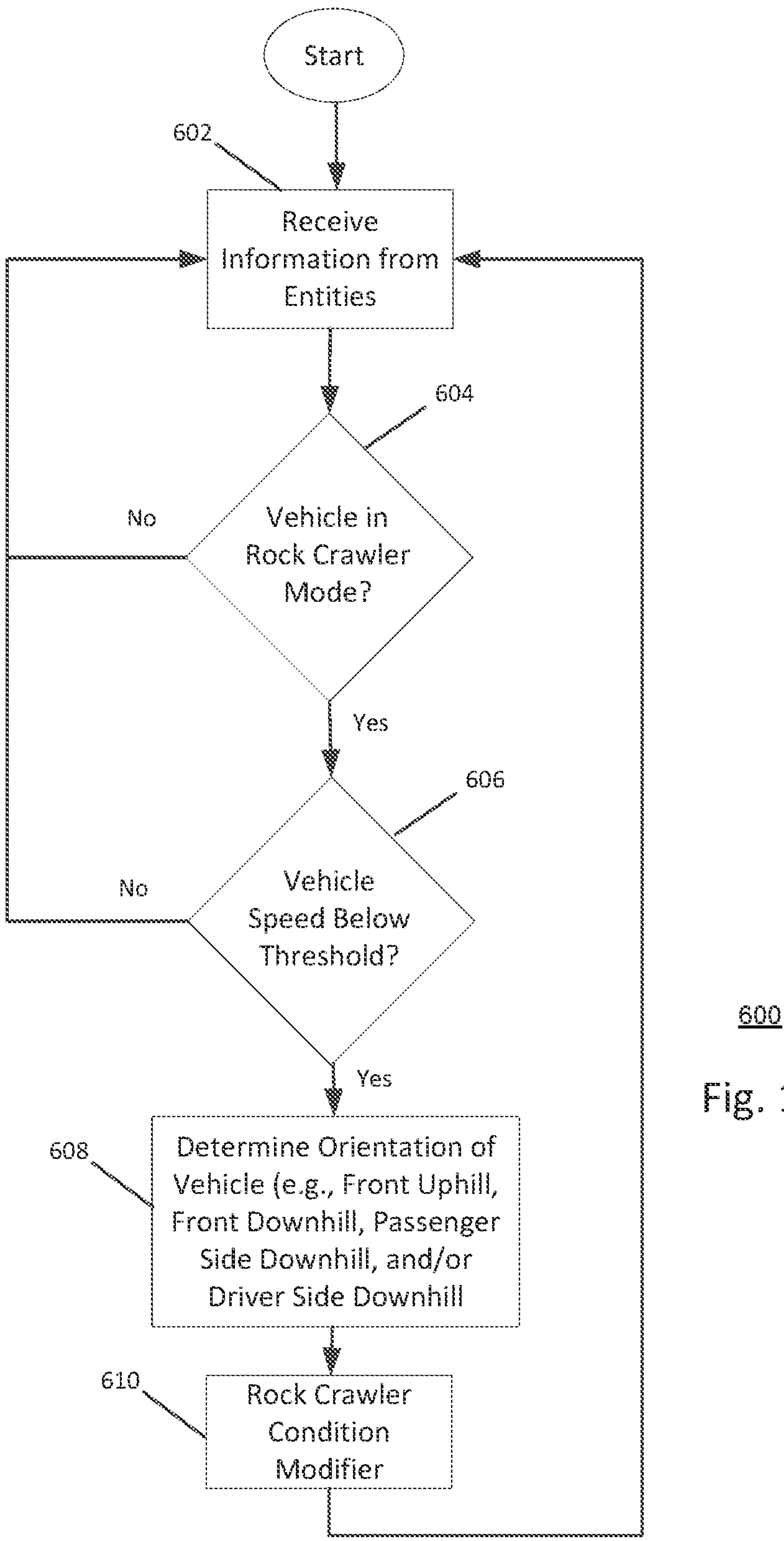

Start
602
Receive Information from Entities
604
Vehicle in Rock Crawler Mode?
No
Yes
606
Vehicle Speed Below Threshold?
No
Yes
608
Determine Orientation of Vehicle (e.g., Front Uphill, Front Downhill, Passenger Side Downhill, and/or Driver Side Downhill
610
Rock Crawler Condition Modifier
600

Fig. 16

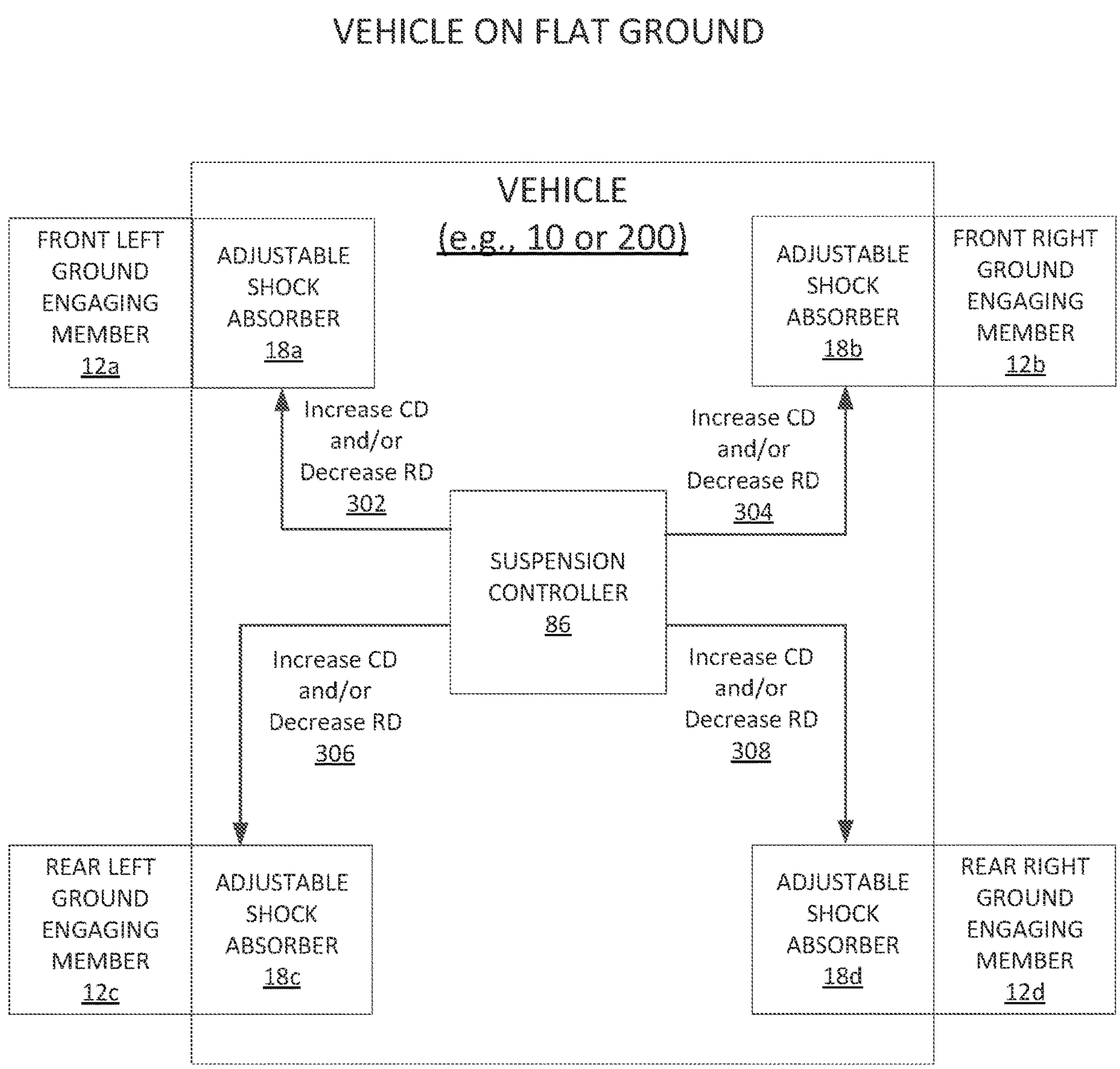

VEHICLE ON FLAT GROUND
VEHICLE
(e.g., 10 or 200)
FRONT LEFT GROUND ENGAGING MEMBER 12a
ADJUSTABLE SHOCK ABSORBER 18a
ADJUSTABLE SHOCK ABSORBER 18b
FRONT RIGHT GROUND ENGAGING MEMBER 12b
Increase CD and/or Decrease RD 302
Increase CD and/or Decrease RD 304
SUSPENSION CONTROLLER 86
Increase CD and/or Decrease RD 306
Increase CD and/or Decrease RD 308
REAR LEFT GROUND ENGAGING MEMBER 12c
ADJUSTABLE SHOCK ABSORBER 18c
ADJUSTABLE SHOCK ABSORBER 18d
REAR RIGHT GROUND ENGAGING MEMBER 12d

Fig. 17

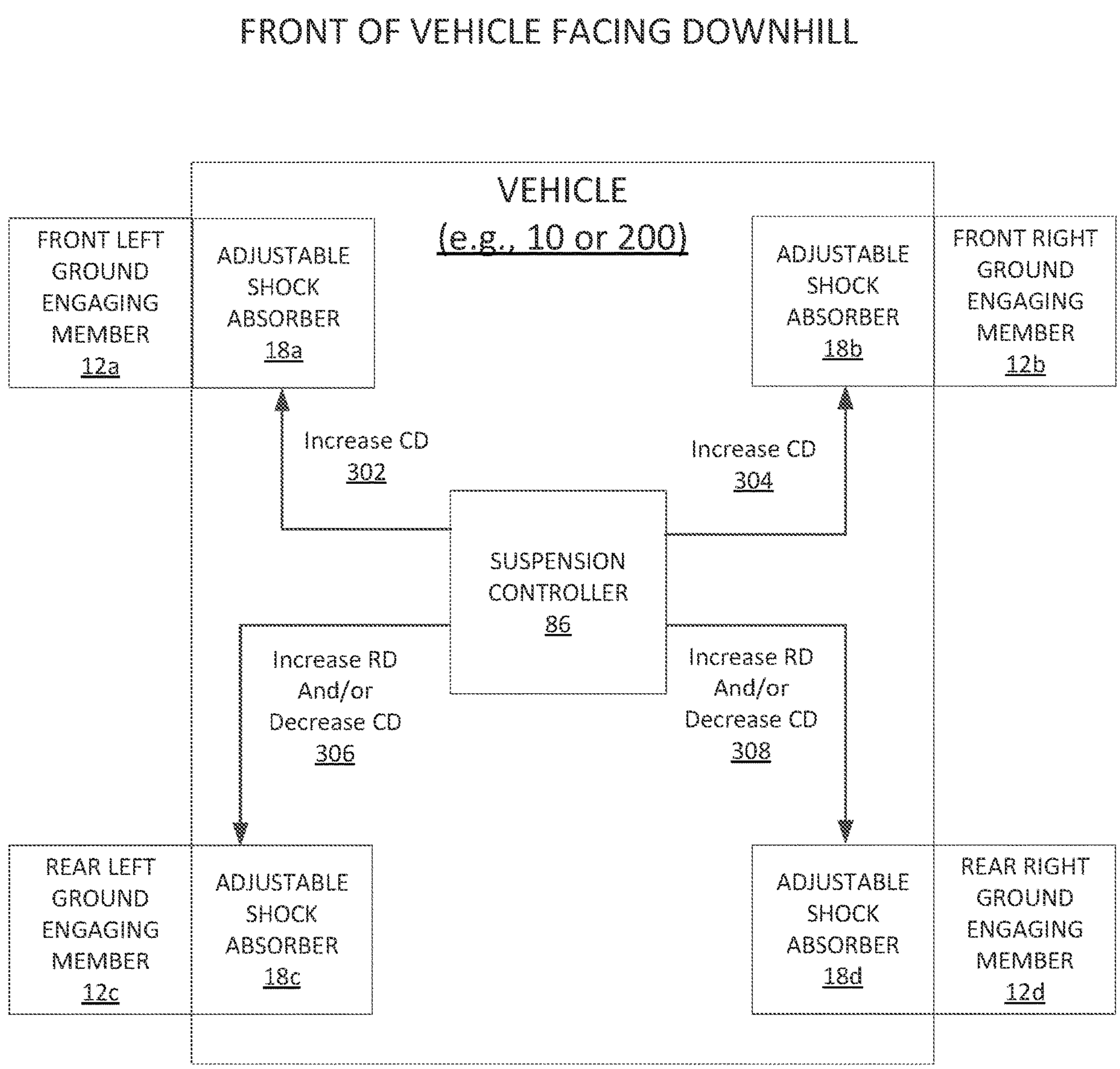

FRONT OF VEHICLE FACING DOWNHILL
VEHICLE
(e.g., 10 or 200)
FRONT LEFT GROUND ENGAGING MEMBER 12a
ADJUSTABLE SHOCK ABSORBER 18a
ADJUSTABLE SHOCK ABSORBER 18b
FRONT RIGHT GROUND ENGAGING MEMBER 12b
Increase CD 302
Increase CD 304
SUSPENSION CONTROLLER 86
Increase RD And/or Decrease CD 306
Increase RD And/or Decrease CD 308
REAR LEFT GROUND ENGAGING MEMBER 12c
ADJUSTABLE SHOCK ABSORBER 18c
ADJUSTABLE SHOCK ABSORBER 18d
REAR RIGHT GROUND ENGAGING MEMBER 12d

Fig. 18

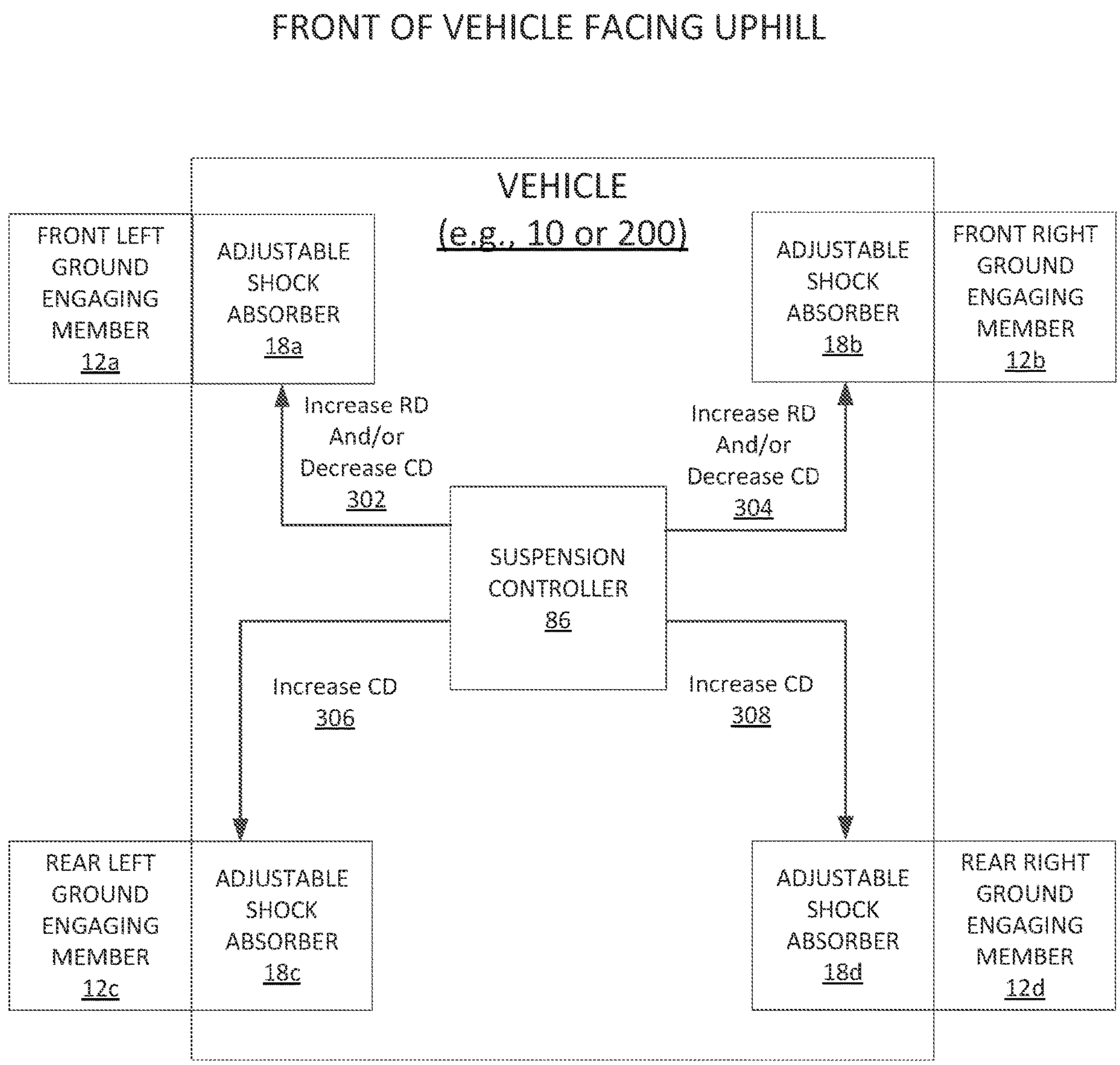

FRONT OF VEHICLE FACING UPHILL
VEHICLE
(e.g., 10 or 200)
FRONT LEFT GROUND ENGAGING MEMBER 12a
ADJUSTABLE SHOCK ABSORBER 18a
ADJUSTABLE SHOCK ABSORBER 18b
FRONT RIGHT GROUND ENGAGING MEMBER 12b
Increase RD And/or Decrease CD 302
Increase RD And/or Decrease CD 304
SUSPENSION CONTROLLER 86
Increase CD 306
Increase CD 308
REAR LEFT GROUND ENGAGING MEMBER 12c
ADJUSTABLE SHOCK ABSORBER 18c
ADJUSTABLE SHOCK ABSORBER 18d
REAR RIGHT GROUND ENGAGING MEMBER 12d

Fig. 19

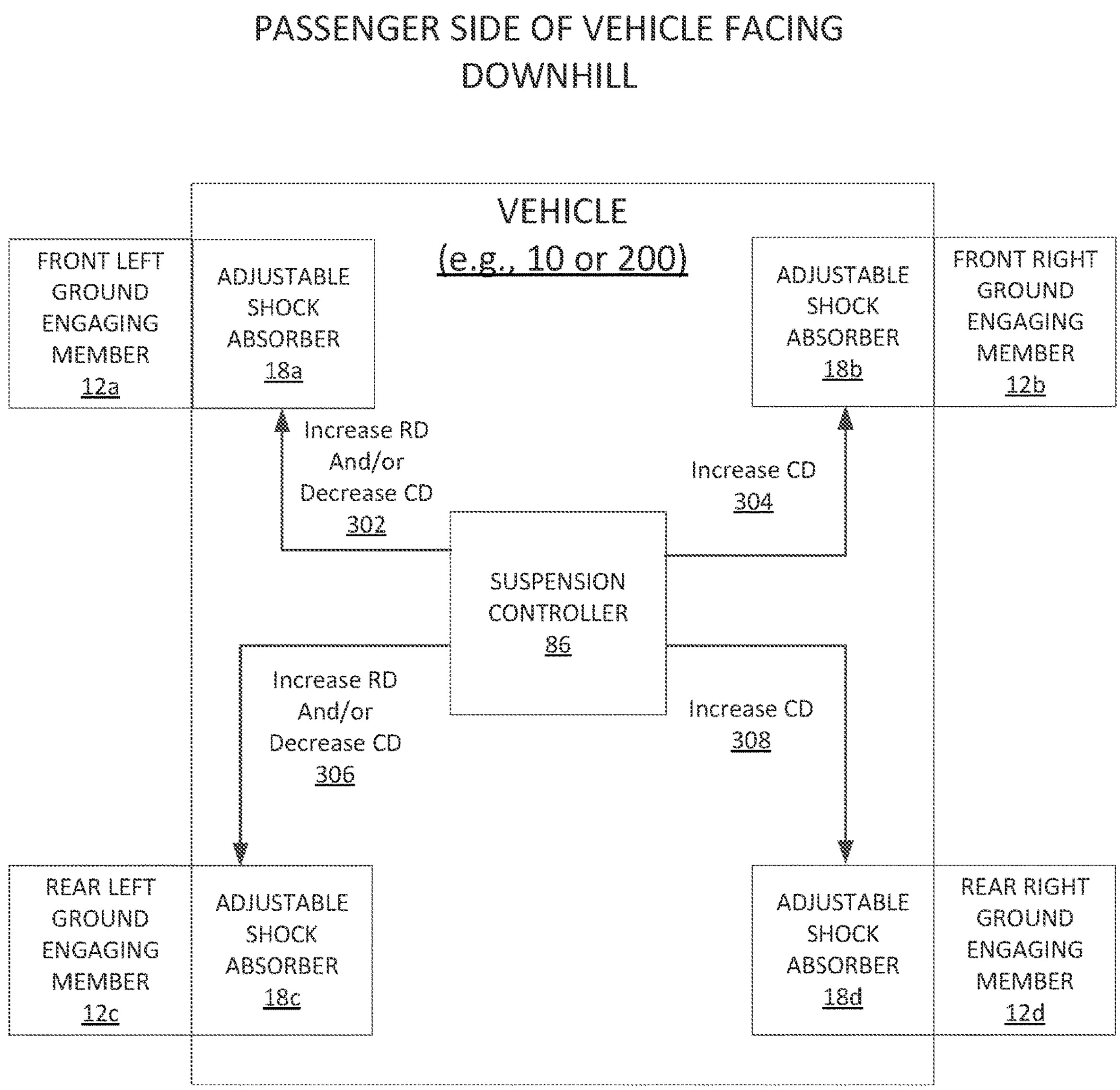

PASSENGER SIDE OF VEHICLE FACING DOWNHILL
VEHICLE
(e.g., 10 or 200)
FRONT LEFT GROUND ENGAGING MEMBER 12a
ADJUSTABLE SHOCK ABSORBER 18a
ADJUSTABLE SHOCK ABSORBER 18b
FRONT RIGHT GROUND ENGAGING MEMBER 12b
Increase RD And/or Decrease CD 302
Increase CD 304
SUSPENSION CONTROLLER 86
Increase RD And/or Decrease CD 306
Increase CD 308
REAR LEFT GROUND ENGAGING MEMBER 12c
ADJUSTABLE SHOCK ABSORBER 18c
ADJUSTABLE SHOCK ABSORBER 18d
REAR RIGHT GROUND ENGAGING MEMBER 12d

Fig. 20

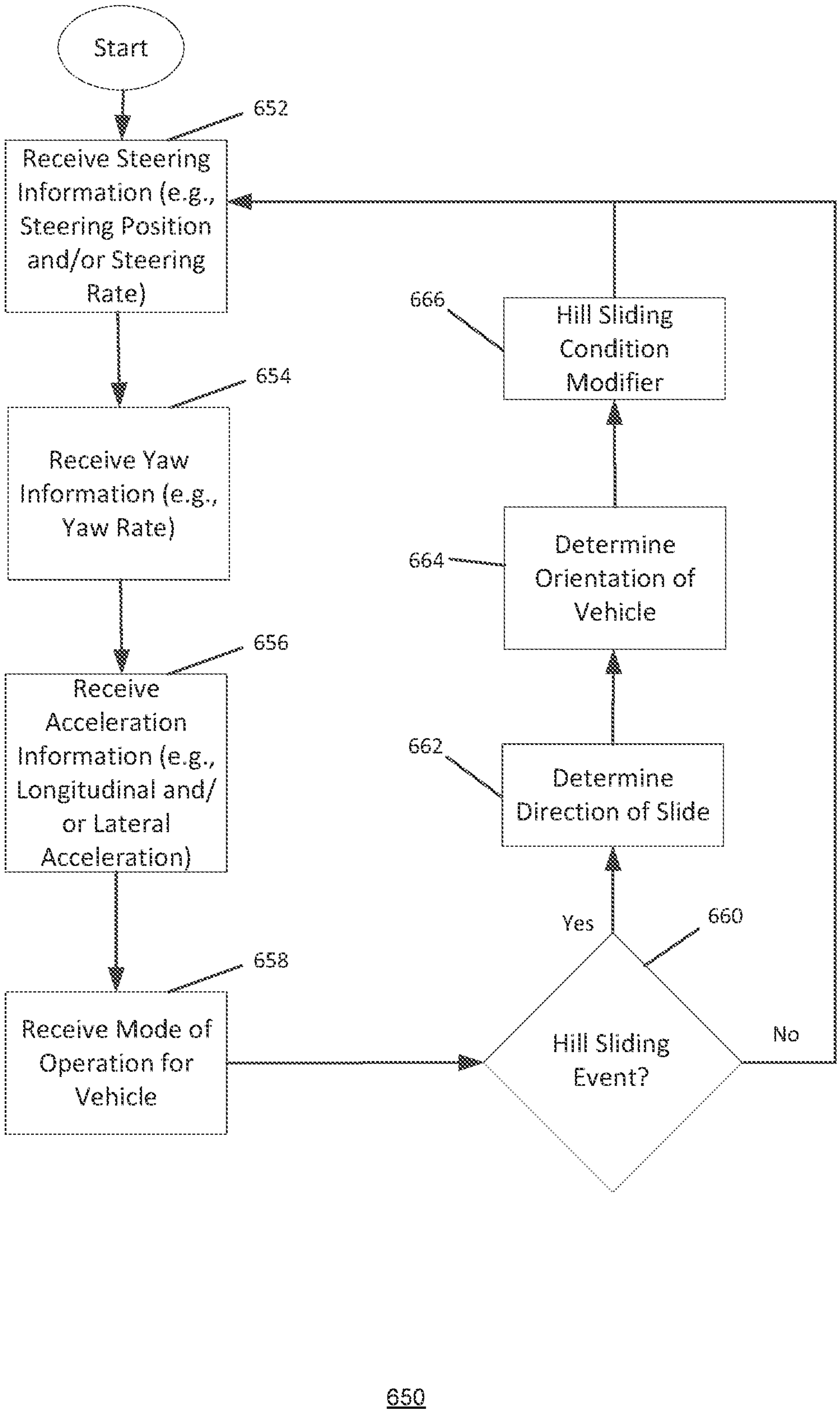

Start
652
Receive Steering Information (e.g., Steering Position and/or Steering Rate)
654
Receive Yaw Information (e.g., Yaw Rate)
656
Receive Acceleration Information (e.g., Longitudinal and/or Lateral Acceleration)
658
Receive Mode of Operation for Vehicle
660
Hill Sliding Event?
Yes
No
662
Determine Direction of Slide
664
Determine Orientation of Vehicle
666
Hill Sliding Condition Modifier
650

Fig. 21

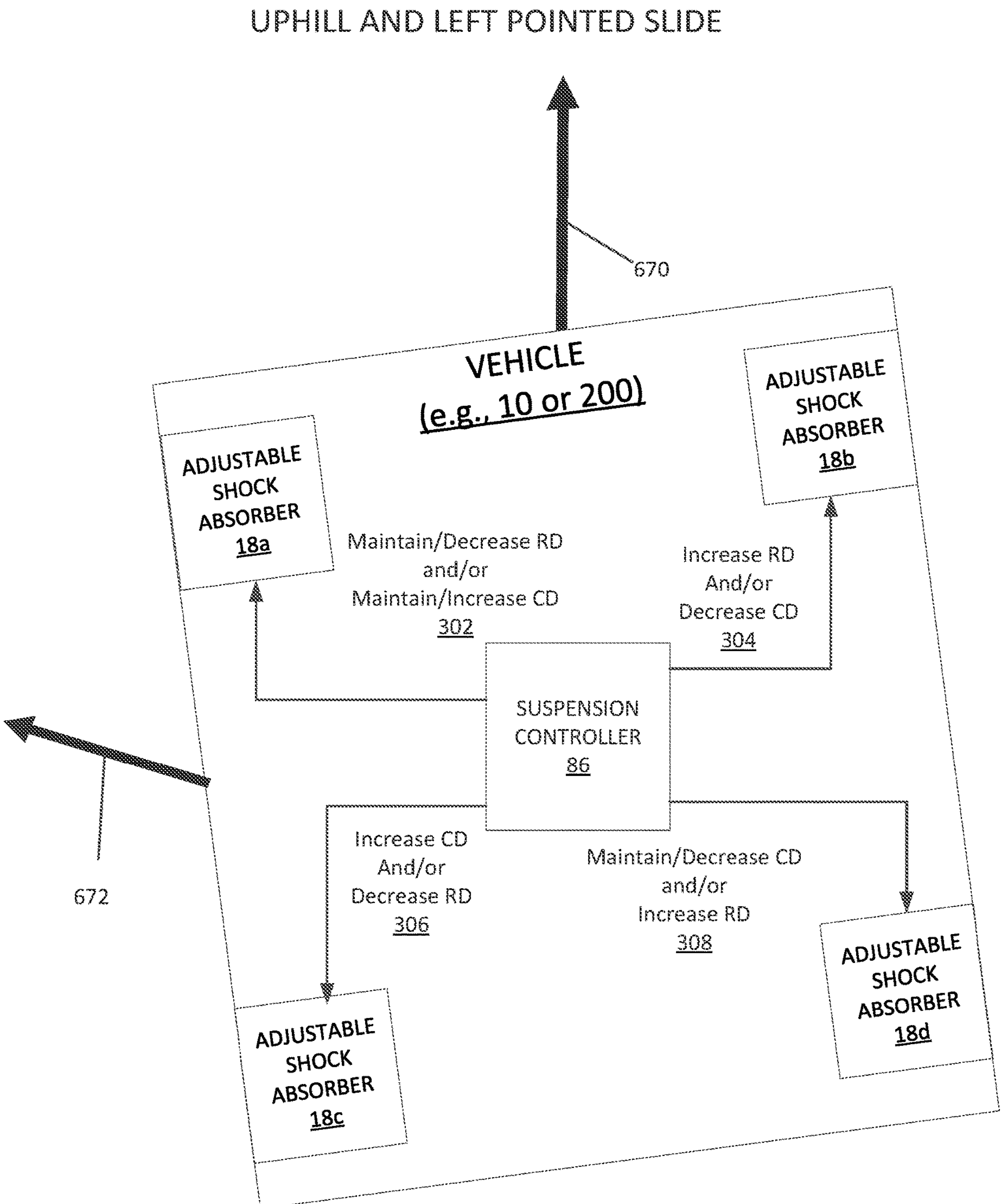

UPHILL AND LEFT POINTED SLIDE
670
VEHICLE
(e.g., 10 or 200)
ADJUSTABLE SHOCK ABSORBER 18a
ADJUSTABLE SHOCK ABSORBER 18b
Maintain/Decrease RD and/or Maintain/Increase CD 302
Increase RD And/or Decrease CD 304
SUSPENSION CONTROLLER 86
672
Increase CD And/or Decrease RD 306
Maintain/Decrease CD and/or Increase RD 308
ADJUSTABLE SHOCK ABSORBER 18c
ADJUSTABLE SHOCK ABSORBER 18d

Fig. 22

DOWNHILL AND RIGHT POINTED SLIDE

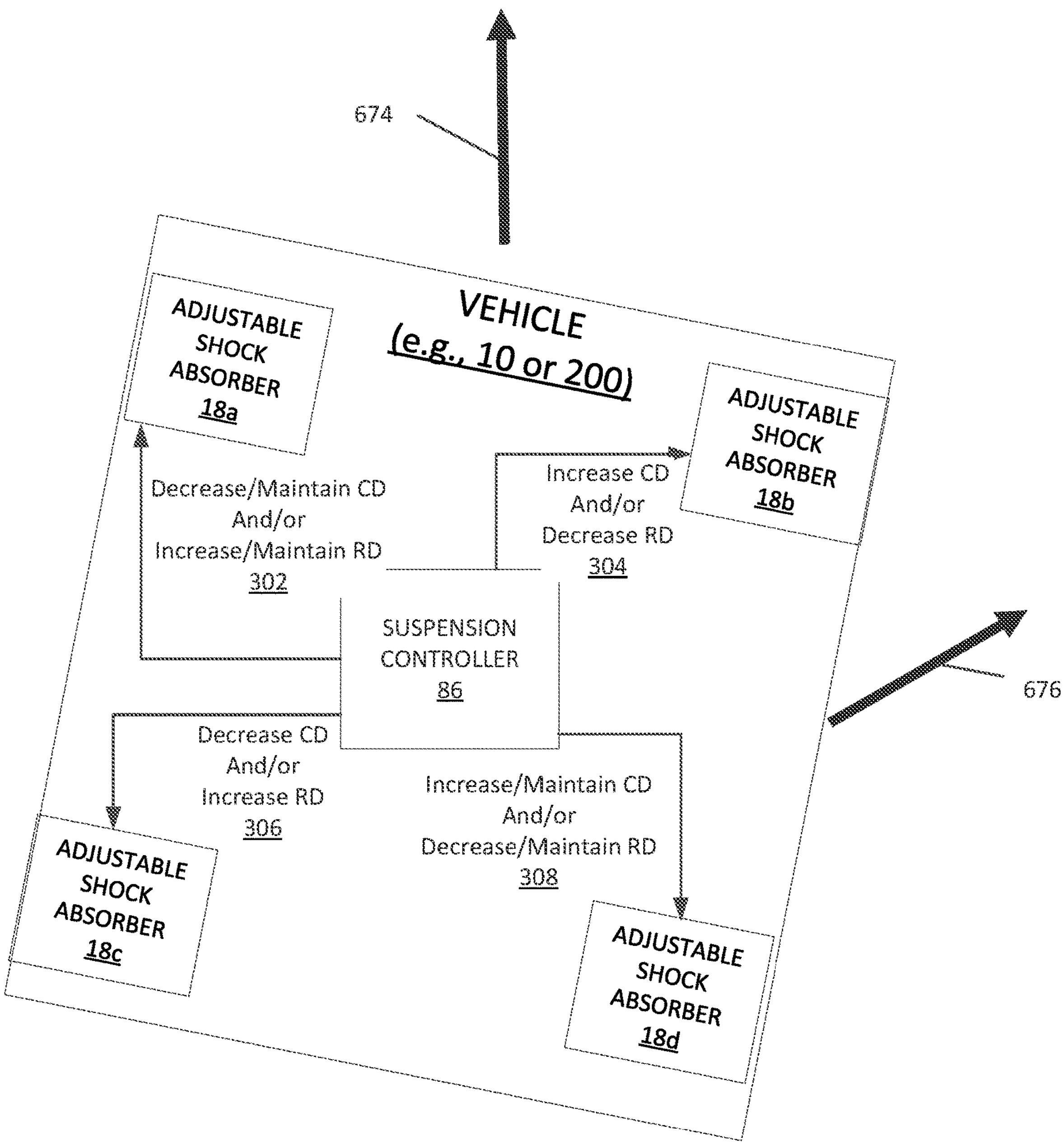


Fig. 23

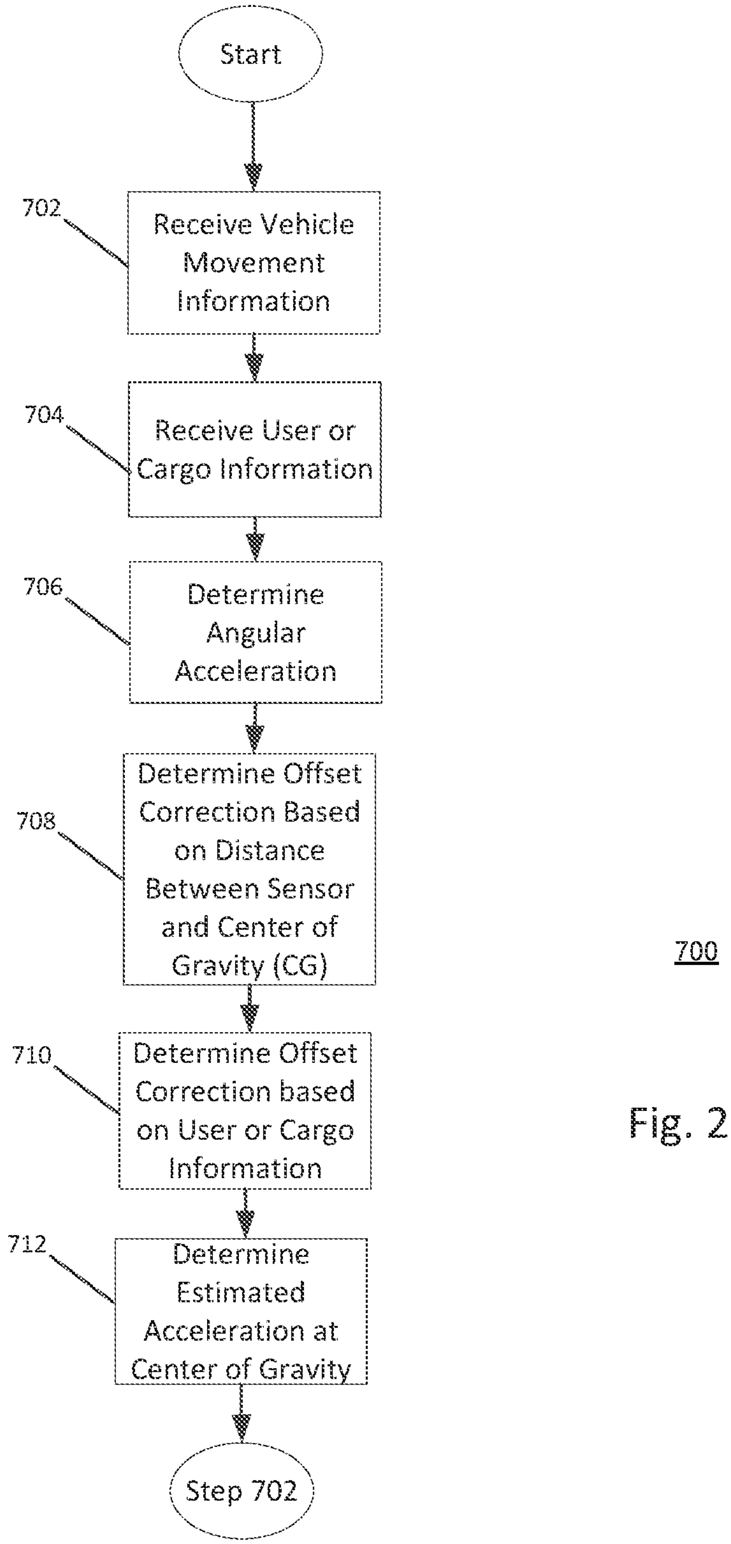

Start
702
Receive Vehicle Movement Information
704
Receive User or Cargo Information
706
Determine Angular Acceleration
708
Determine Offset Correction Based on Distance Between Sensor and Center of Gravity (CG)
710
Determine Offset Correction based on User or Cargo Information
712
Determine Estimated Acceleration at Center of Gravity
Step 702
700

Fig. 24

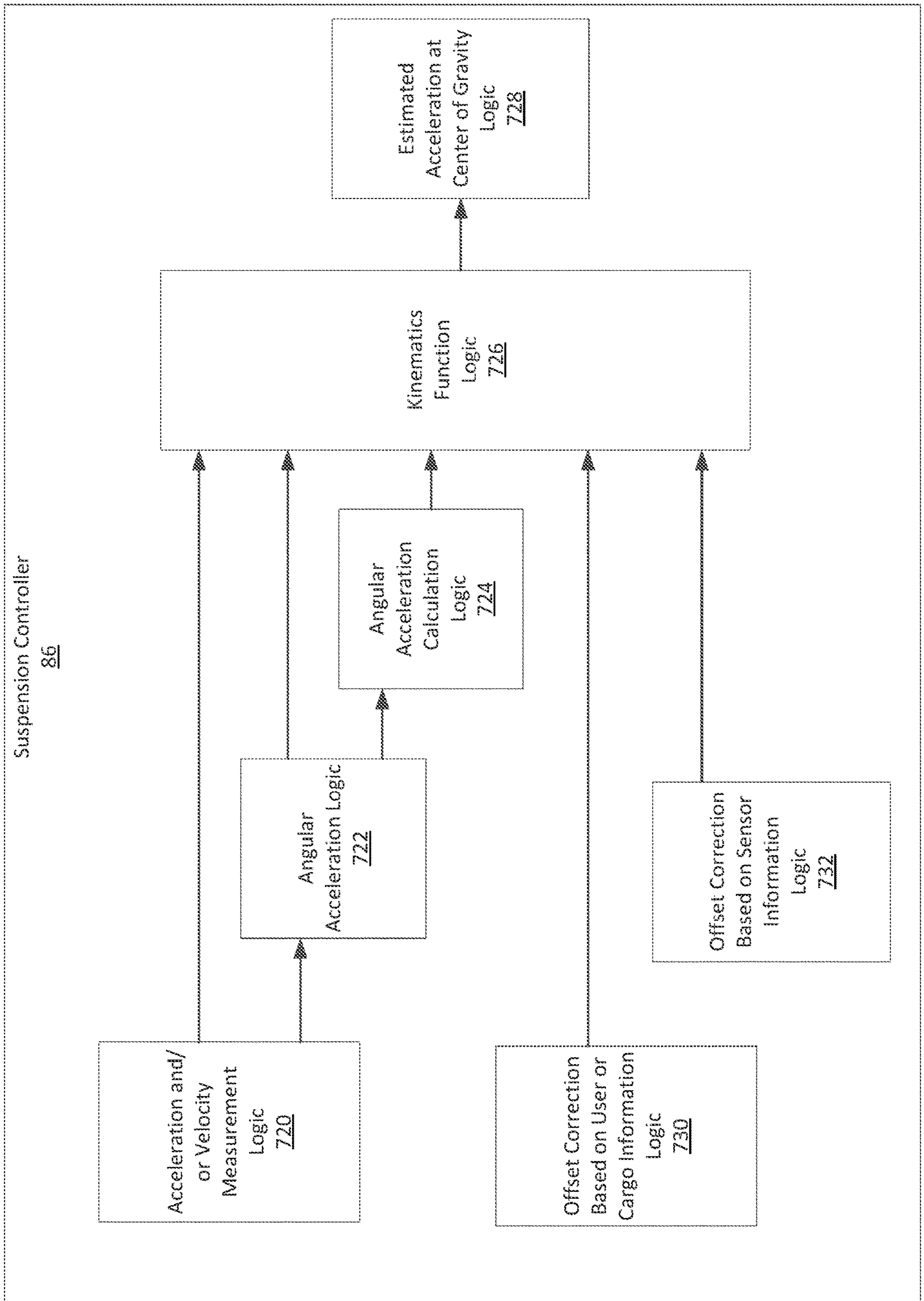

Suspension Controller
86
Acceleration and/ or Velocity Measurement Logic
720
Angular Acceleration Logic
722
Angular Acceleration Calculation Logic
724
Kinematics Function Logic
726
Estimated Acceleration at Center of Gravity Logic
728
Offset Correction Based on User or Cargo Information Logic
730
Offset Correction Based on Sensor Information Logic
732

Fig. 25

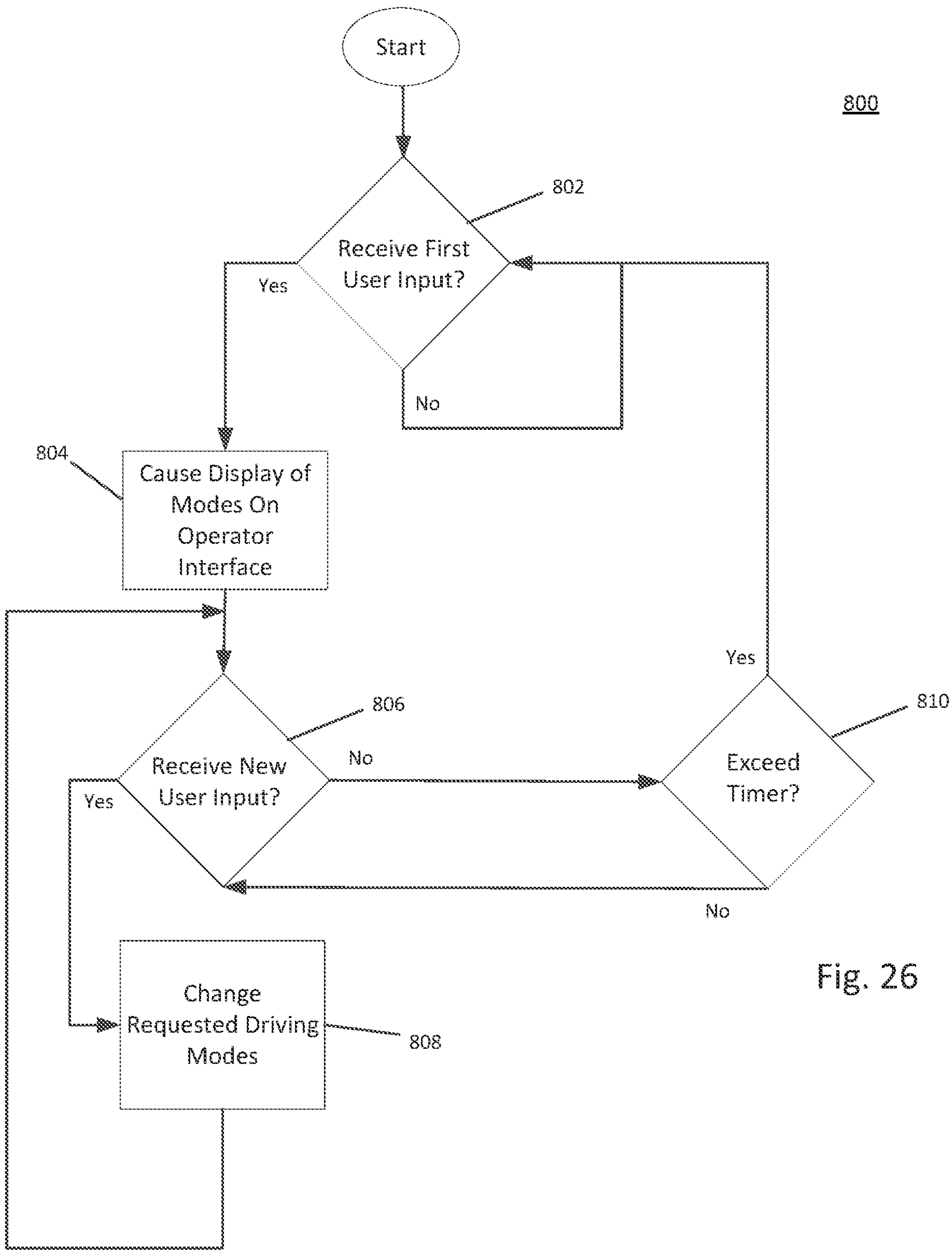

Start
800
802
Receive First User Input?
Yes
No
804
Cause Display of Modes On Operator Interface
806
Receive New User Input?
Yes
No
Yes
810
Exceed Timer?
No
Change Requested Driving Modes
808

Fig. 26

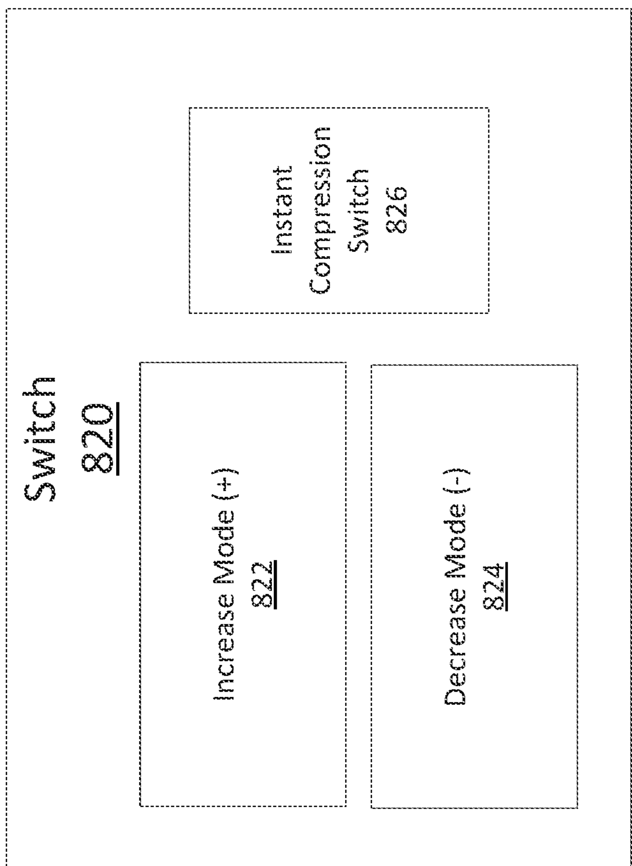

Switch
820
Increase Mode (+)
822
Decrease Mode (-)
824
Instant
Compression
Switch
826

Fig. 27

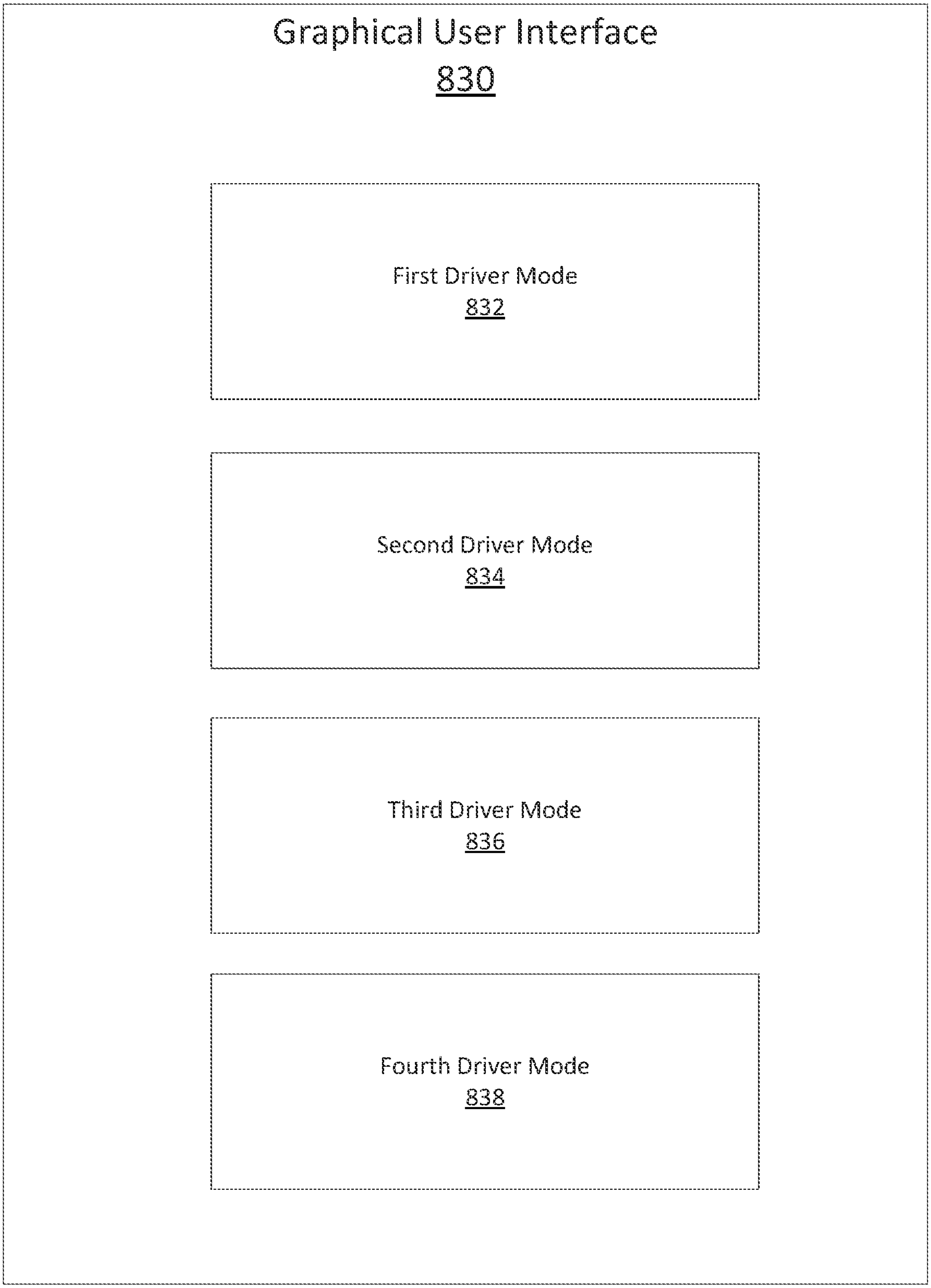

Graphical User Interface
830
First Driver Mode
832
Second Driver Mode
834
Third Driver Mode
836
Fourth Driver Mode
838

Fig. 28

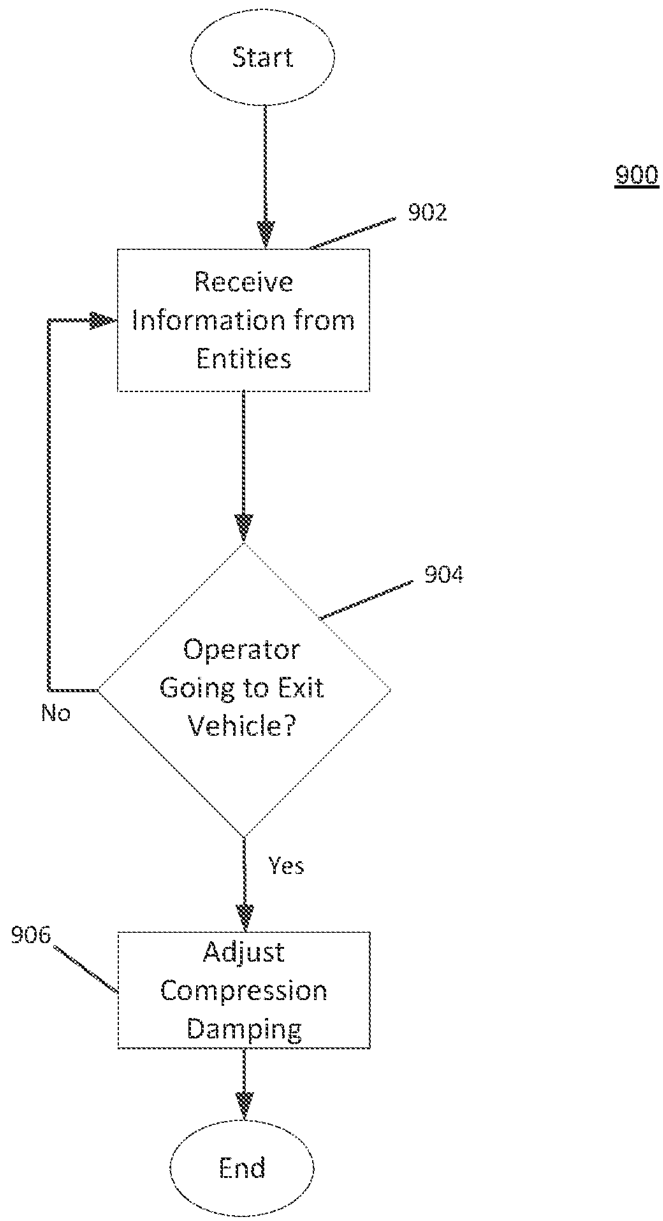

900
Start
902
Receive Information from Entities
904
Operator Going to Exit Vehicle?
No
Yes
906
Adjust Compression Damping
End

Fig. 29

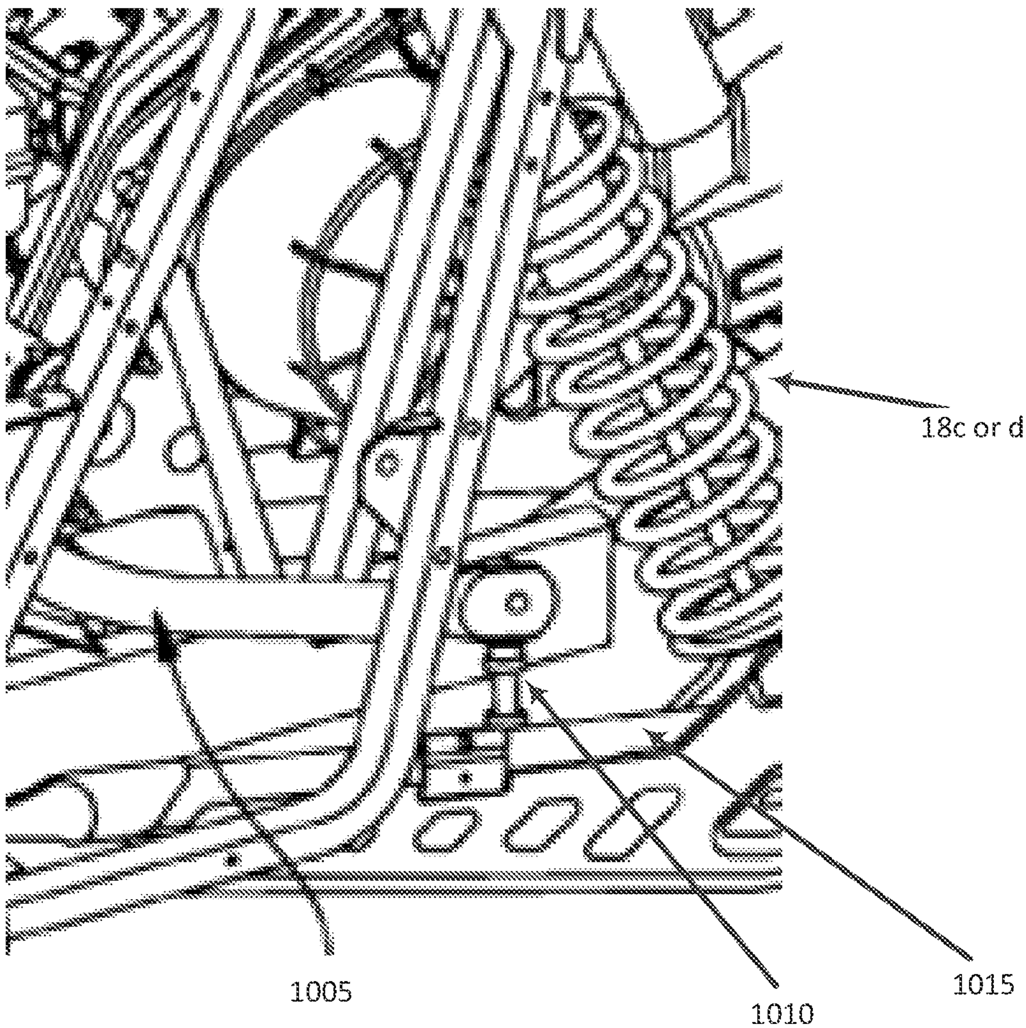


Fig. 30

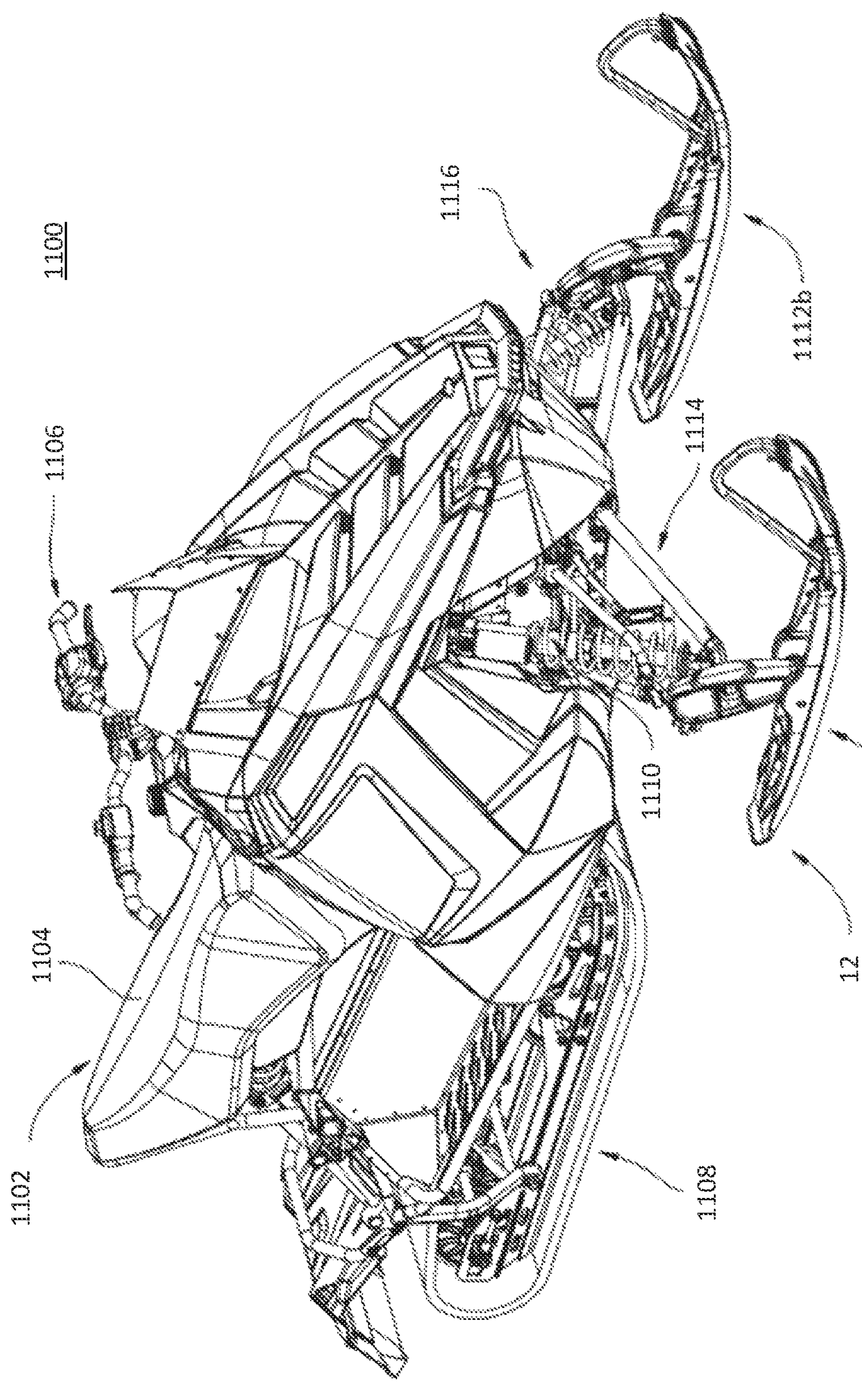


Fig. 31

VEHICLE HAVING ADJUSTABLE COMPRESSION AND REBOUND DAMPING

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a divisional of U.S. patent application Ser. No. 17/888,938, filed Aug. 16, 2022, which is a continuation of U.S. patent application Ser. No. 17/175,888, filed Feb. 15, 2021, titled "VEHICLE HAVING ADJUSTABLE COMPRESSION AND REBOUND DAMPING", which is a divisional of U.S. patent application Ser. No. 16/198,280, filed Nov. 21, 2018, titled "VEHICLE HAVING ADJUSTABLE COMPRESSION AND REBOUND DAMPING", the entire disclosures of which are expressly incorporated herein by reference.

FIELD OF DISCLOSURE

The present disclosure relates to improved suspension characteristics for a vehicle and in particular to systems and methods of damping control, such as compression and/or rebound damping, for shock absorbers.

BACKGROUND OF THE DISCLOSURE

Adjustable shock absorbers are known. Systems and methods for controlling one or more adjustable characteristics of adjustable shock absorbers are disclosed in US Published Patent Application No. 2016/0059660 (filed Nov. 6, 2015, titled VEHICLE HAVING SUSPENSION WITH CONTINUOUS DAMPING CONTROL) and US Published Application 2018/0141543 (filed Nov. 17, 2017, titled VEHICLE HAVING ADJUSTABLE SUSPENSION).

SUMMARY OF THE DISCLOSURE

In an exemplary embodiment of the present disclosure, a recreational vehicle is provided. The recreational vehicle includes a plurality of ground engaging members, a frame supported by the plurality of ground engaging members, a plurality of suspensions, a plurality of adjustable shock absorbers, at least one sensor positioned on the recreational vehicle and configured to provide sensor information to a controller, and a controller operatively coupled to the sensor and the plurality of adjustable shock absorbers. Each of the suspensions couples a ground engaging member to the frame. The controller is configured to receive sensor information from the sensor, determine a cornering event related to the recreational vehicle executing a turn based on the sensor information, and provide, to at least one of the plurality of adjustable shock absorbers and based on the cornering event, one or more commands to result in a decrease of a damping characteristic of the at least one of the plurality of adjustable shock absorbers.

In some instances, the controller is configured determine, based on the sensor information, a direction of the turn corresponding to the cornering event, determine, based on the direction of the turn, at least one inner adjustable shock absorber of the plurality of adjustable shock absorbers, and provide, to the at least one inner adjustable shock absorber, one or more commands to result in a decrease of a compression damping characteristic and an increase of a rebound damping characteristic. In some examples, the at least one sensor comprises an inertial measurement unit (IMU), the sensor information comprises acceleration information indicating a lateral acceleration value, and the controller is configured to determine the cornering event by determining that the recreational vehicle is turning based on comparing the lateral acceleration value to a first threshold.

In some variations, the sensor information further comprises yaw rate information indicating a yaw rate. The controller is further configured to determine the cornering event by determining that the recreational vehicle is turning based on comparing the yaw rate to a second threshold. In some instances, the at least one sensor further comprises a steering sensor and the sensor information further comprises steering information indicating a steering position or a steering rate corresponding to a steering wheel. Further, the controller is configured to determine the cornering event by determining that the recreational vehicle is turning based on comparing the steering position to a third threshold.

In some variations, the sensor information comprises yaw rate information indicating a yaw rate and steering information indicating a steering position or a steering rate. The controller is further configured to prioritize the yaw rate information over the steering information such that the controller is configured to determine the vehicle is executing the turn in a first direction based on the yaw rate indicating the turn in the first direction and even if the steering position or the steering rate indicates the turn in a second direction or does not indicate the turn. In some instances, the sensor information comprises acceleration information indicating a lateral acceleration value and yaw rate information indicating a yaw rate. The controller is further configured to prioritize the acceleration information over the yaw rate information such that the controller is configured to determine the vehicle is executing the turn in a first direction based on the lateral acceleration value indicating the turn in the first direction and even if the yaw rate indicates the turn in a second direction or does not indicate the turn. In some examples, the sensor information comprises steering information indicating a steering position or a steering rate and acceleration information indicating a lateral acceleration value. The controller is further configured to prioritizing the acceleration information over the steering information such that the controller is configured to determine the vehicle is executing the turn in a first direction based on the lateral acceleration value indicating the turn in the first direction and even if the steering position or the steering rate indicates the turn in a second direction or does not indicate the turn.

In another exemplary embodiment of the present disclosure, a recreational vehicle is provided. The recreational vehicle includes a plurality of ground engaging members, a frame supported by the plurality of ground engaging members, a plurality of suspensions, a plurality of adjustable shock absorbers, a first sensor positioned on the recreational vehicle and configured to provide cornering information to a controller, a second sensor positioned on the recreational vehicle and configured to provide acceleration information to the controller and a controller operatively coupled to the sensor and the plurality of adjustable shock absorbers. Each of the suspensions couples a ground engaging member to the frame. The controller is configured to receive, from the first sensor, the cornering information, receive, from the second sensor, the acceleration information, determine, based on the cornering information, a cornering event corresponding to the recreational vehicle executing a turn, determine, based on the acceleration information, a position of the recreational vehicle during the turn, and provide, to the at least one of the plurality of adjustable shock absorbers and based on the cornering event and the position of the recreational vehicle during the turn, one or more commands to result in an adjustment of a damping characteristic of the at least one of the plurality of adjustable shock absorbers.

In some instances, the second sensor is an accelerometer or an IMU. In some examples, the controller is configured to determine the position of the recreational vehicle during the turn by determining, based on the acceleration information indicating a longitudinal deceleration, that the vehicle is entering the turn. Also, the controller is further configured to determine, based on the cornering event, a plurality of damping characteristics for the plurality of adjustable shock absorbers, bias, based on the determining that the vehicle is entering the turn, the plurality of damping characteristics, and generate the one or more commands based on the plurality of biased damping characteristics. In some instances, the controller is configured to bias the plurality of damping characteristics by additionally increasing the compression damping of a front adjustable shock absorber of the plurality of adjustable shock absorbers, and additionally decreasing the compression damping of a rear adjustable shock absorber of the plurality of adjustable shock absorbers. In some examples, the controller is configured to bias the plurality of damping characteristics by additionally increasing the rebound damping of a rear adjustable shock absorber of the plurality of adjustable shock absorbers and additionally decreasing the rebound damping of a front adjustable shock absorber of the plurality of adjustable shock absorbers.

In some instances, the controller is configured to determine the position of the recreational vehicle during the turn by determining, based on the acceleration information indicating a longitudinal acceleration, that the vehicle is exiting the turn. Also, the controller is further configured to determine, based on the cornering event, a plurality of damping characteristics for the plurality of adjustable shock absorbers, bias, based on the determining that the vehicle is exiting the turn, the plurality of damping characteristics, and generate the one or more commands based on the plurality of biased damping characteristics. In some examples, the controller is configured to bias the plurality of damping characteristics by additionally increasing the compression damping of a rear adjustable shock absorber of the plurality of adjustable shock absorbers and additionally decreasing the compression damping of a front adjustable shock absorber of the plurality of adjustable shock absorbers. In some variations, the controller is configured to bias the plurality of damping characteristics by additionally increasing the rebound damping of a front adjustable shock absorber of the plurality of adjustable shock absorbers and additionally decreasing the rebound damping of a rear adjustable shock absorber of the plurality of adjustable shock absorbers.

In another exemplary embodiment of the present disclosure, a recreational vehicle is provided. The recreational vehicle includes a plurality of ground engaging members, a frame supported by the plurality of ground engaging members, a plurality of suspensions, a plurality of adjustable shock absorbers, a sensor positioned on the recreational vehicle and configured to provide sensor information to a controller, and a controller operatively coupled to the sensor and the plurality of adjustable shock absorbers. Each of the suspensions couples a ground engaging member to the frame. The controller is configured to receive sensor information from the sensor, determine a braking event corresponding to the recreational vehicle based on the sensor information, and provide, to at least one of the plurality of adjustable shock absorbers and based on the braking event, one or more commands to decrease a damping characteristic of the at least one of the plurality of adjustable shock absorbers.

In some instances, the controller is configured to provide one or more commands to decrease a rebound damping characteristic of the at least one of the plurality of adjustable shock absorbers. In some examples, the controller is configured to provide, one or more commands to decrease a compression damping characteristic of the at least one of the plurality of adjustable shock absorbers. In some variations, the sensor is a brake sensor, and the sensor information is information indicating actuation of a brake pedal. In some instances, the controller is configured to provide, to a front adjustable shock absorber of the plurality of adjustable shock absorbers, a command to increase a compression damping characteristic and to decrease a rebound damping characteristic. In some examples, the controller is configured to provide, to a rear adjustable shock absorber of the plurality of adjustable shock absorbers, a command to increase a rebound damping characteristic and to decrease a compression damping characteristic.

In another exemplary embodiment of the present disclosure, a recreational vehicle is provided. The recreational vehicle includes a plurality of ground engaging members, a frame supported by the plurality of ground engaging members, a plurality of suspensions, a plurality of adjustable shock absorbers, a first sensor positioned on the recreational vehicle and configured to provide braking information to a controller, a second sensor positioned on the recreational vehicle and configured to provide acceleration information, and a controller operatively coupled to the sensor and the plurality of adjustable shock absorbers. Each of the suspensions couples a ground engaging member to the frame. The controller is configured to receive, from the first sensor, the braking information, receive, from the second sensor, the acceleration information, determine a braking event corresponding to the recreational vehicle based on the braking information, determine, based on the acceleration information, an amount to reduce a damping characteristic of an adjustable shock absorber from the plurality of adjustable shock absorbers, and provide, to the adjustable shock absorber and based on the cornering event, one or more first commands to adjust the damping characteristics of the front adjustable shock absorber to the determined amount.

In some instances, the second sensor is an accelerometer, and the acceleration information indicates a longitudinal deceleration of the recreational vehicle. In some examples, the second sensor is an inertial measurement unit (IMU), and the acceleration information indicates a longitudinal deceleration of the recreational vehicle. In some variations, the second sensor is a brake sensor, and the acceleration information indicates a longitudinal deceleration of the recreational vehicle. In some instances, the adjustable shock absorber is a shock absorber positioned at a rear portion of the recreational vehicle.

In some variations, the controller is configured to determine, based on the acceleration information, a deceleration value, and in response to determining the deceleration value is below a first threshold, maintaining a compression damping of the adjustable shock absorber. In some examples, the controller is configured to determine, based on the acceleration information, a deceleration value, in response to determining the deceleration value is greater than a first threshold and below a second threshold, reducing a compression damping of the adjustable shock absorber to a first value, and in response to determining the deceleration value is greater than the first threshold and the second threshold, reducing the compression damping of the adjustable shock absorber to a second value, wherein the second value is below the first value.

In another exemplary embodiment of the present disclosure, a recreational vehicle is provided. The recreational vehicle includes a plurality of ground engaging members, a frame supported by the plurality of ground engaging members, a plurality of suspensions, a plurality of adjustable shock absorbers, a sensor positioned on the recreational vehicle and configured to provide airborne information and landing information to a controller, and a controller operatively coupled to the sensor and the plurality of adjustable shock absorbers. Each of the suspensions couples a ground engaging member to the frame. The controller is configured to receive, from the sensor, the airborne information, determine, based on the airborne information, an airborne event indicating the recreational vehicle is airborne, provide, based on the airborne event, one or more first commands to result in decreasing a rebound damping characteristic for the plurality of adjustable shock absorbers from a pre-takeoff rebound value to a free-fall rebound value, receive, from the at least one sensor, landing information, determine, based on the landing information, a landing event indicating the recreational vehicle has landed subsequent to the airborne event, determine, based on the airborne event and the landing event, a time duration that the recreational vehicle is airborne, provide, based on the landing event and the time duration the recreational vehicle is airborne, one or more second commands to result in increasing the rebound damping characteristic for the plurality of adjustable shock absorbers from the free-fall rebound value to a post-landing rebound value to prevent a landing hop, and provide one or more third commands to result in decreasing the rebound damping characteristic for the plurality of adjustable shock absorbers from the post-landing rebound value to the pre-takeoff rebound value.

In some instances, the sensor is an accelerometer or an IMU. In some examples, the sensor is an inertial measurement unit (IMU). In some variations, the one or more third commands cause the rebound damping characteristic for the plurality of adjustable shock absorbers to gradually decrease from the post-landing rebound value to the pre-takeoff rebound value. In some examples, the controller is further configured to increase the post-landing rebound value as the time duration that the recreational vehicle is airborne increases. In some variations, the controller is further configured to in response to determining the time duration is below a first threshold, set the post-landing rebound value to a same value as the pre-takeoff rebound value. In some instances, the controller is further configured to in response to determining the time duration is below a first threshold, bias the post-landing rebound value for a front shock absorber of the plurality of adjustable shock absorbers different from a rear shock absorber of the plurality of adjustable shock absorbers, and generate the one or more second commands based on the biasing the post-landing rebound value. In some instances, the controller is configured to bias the post-landing rebound value by additionally increasing the post-landing rebound value for the front shock absorber.

In some examples, the controller is configured to provide, based on the airborne event, one or more fourth commands to result in gradually increasing a compression damping characteristic for the plurality of adjustable shock absorbers from a pre-takeoff compression value to a post-landing compression value, provide, based on the landing event and the time duration, one or more fifth commands to result in maintaining the compression damping characteristic for the plurality of adjustable shock absorbers at the post-landing compression value, and provide one or more sixth commands to result in decreasing the compression damping characteristic for the plurality of adjustable shock absorbers from the post-landing compression value to the pre-takeoff compression value. In some instances, the controller is further configured to in response to determining the time duration is below a first threshold, set the post-landing compression value to a same value as the pre-takeoff compression value. In some examples, the controller is further configured to in response to determining the time duration is below a first threshold, bias the post-landing compression value for a front shock absorber of the plurality of adjustable shock absorbers different from a rear shock absorber of the plurality of adjustable shock absorbers, and generate the one or more fifth commands based on the biasing the post-landing compression value. In some variations, the controller is further configured to in response to determining the time duration is below a first threshold, bias the post-landing compression value for a front shock absorber of the plurality of adjustable shock absorbers different from a rear shock absorber of the plurality of adjustable shock absorbers, and generate the one or more fifth commands based on the biasing the post-landing compression value. In some examples, the free-fall rebound value is substantially zero.

In another exemplary embodiment of the present disclosure, a recreational vehicle is provided. The recreational vehicle includes a plurality of ground engaging members, a frame supported by the plurality of ground engaging members, a plurality of suspensions, a plurality of adjustable shock absorbers, a sensor positioned on the recreational vehicle and configured to provide acceleration information to a controller, and a controller operatively coupled to the sensor and the plurality of adjustable shock absorbers. Each of the suspensions couples a ground engaging member to the frame. The controller is configured to receive, from the at least one sensor, the acceleration information, determine, based on the acceleration information, an orientation of the vehicle, and provide, to at least one of the plurality of adjustable shock absorbers and based on the orientation of the vehicle, one or more commands to result in an adjustment of a damping characteristic of the at least one of the plurality of adjustable shock absorbers.

In some instances, the sensor is an accelerometer or an IMU. In some examples, the sensor is an IMU. In some variations, the recreational vehicle further includes an operator interface configured to provide one or more user inputs indicating mode selections to the controller. The controller is configured to provide the one or more commands to result in the adjustment of the damping characteristic based on receiving, from the operator interface, user input indicating a selection of a rock crawler mode. In some examples, the at least one sensor comprises a second sensor configured to provide vehicle speed information to the controller. The controller is further configured to receive, from the second sensor, vehicle speed information indicating a vehicle speed of the recreational vehicle, and in response to determining that the vehicle speed is greater than a threshold, transitioning the vehicle from the rock crawler mode to a different operating mode.

In some variations, the controller is further configured to determine, based on the acceleration information, a longitudinal acceleration and a lateral acceleration of the recreational vehicle, determine, based on the longitudinal acceleration and the lateral acceleration, a pitch angle and a roll angle of the recreational vehicle, and the controller is configured to determine the orientation of the recreational vehicle based on the pitch angle and the roll angle. In some examples, the controller is configured to determine, based on the longitudinal acceleration and the lateral acceleration, that the orientation of the recreational vehicle is on flat ground, and provide, based on the determination that the recreational vehicle is on flat ground, one or more commands to result in an increase of a compression damping characteristic and a decrease of a rebound damping characteristic for the at least one of the plurality of adjustable shock absorbers.

In some instances, the controller is configured to determine, based on the longitudinal acceleration and the lateral acceleration, at least one uphill adjustable shock absorber and at least one downhill adjustable shock absorber from the plurality of adjustable shock absorbers, and provide, to the at least one uphill adjustable shock absorber, one or more commands to result in an increase of a rebound damping characteristic and a decrease of a compression damping characteristic. In some examples, the controller is further configured to determine, based on the longitudinal acceleration and the lateral acceleration, at least one uphill adjustable shock absorber and at least one downhill adjustable shock absorber from the plurality of adjustable shock absorbers, and provide, to the at least one downhill adjustable shock absorber, one or more commands to result in an increase of a compression damping characteristic.

In another exemplary embodiment of the present disclosure, a recreational vehicle is provided. The recreational vehicle includes a plurality of ground engaging members, a frame supported by the plurality of ground engaging members, a plurality of suspensions, a plurality of adjustable shock absorbers, a sensor positioned on the recreational vehicle and configured to provide sensor information to a controller, and a controller operatively coupled to the sensor and the plurality of adjustable shock absorbers. Each of the suspensions couples a ground engaging member to the frame. The controller is configured to receive, from the at least one sensor, the sensor information, determine, based on the sensor information, a sliding event corresponding to the recreational vehicle sliding while traversing a slope, and provide, to at least one of the plurality of adjustable shock absorbers and based on the determining of the sliding event, one or more commands to result in adjusting a damping characteristic for the at least one of the plurality of adjustable shock absorbers.

In some instances, the at least one sensor comprises an inertial measurement unit (IMU). Also, the sensor information comprises acceleration information indicating a lateral acceleration value. The controller is configured to determine the sliding event by determining that the recreational vehicle is sliding based on comparing the lateral acceleration value to a first threshold. In some examples, the sensor information further comprises yaw rate information indicating a yaw rate. The controller is further configured to determine the cornering based on the yaw rate. In some variations, the sensor information comprises acceleration information indicating a lateral acceleration value and yaw rate information indicating a yaw rate. The controller is further configured to prioritize the acceleration information over the yaw rate information such that the controller is configured to determine the vehicle is sliding while traversing the slope based on the lateral acceleration value exceeding a first threshold and even if the yaw rate does not exceed a second threshold.

In some instances, the sensor information comprises steering information indicating a steering position or a steering rate and acceleration information indicating a lateral acceleration value. The controller is further configured to prioritizing the acceleration information over the steering information such that the controller is configured to determine the vehicle is sliding while traversing the slope based on the lateral acceleration value exceeding a first threshold and even if the steering position or the steering rate does not exceed a second threshold. In some variations, the sensor information indicates a lateral acceleration. The controller is further configured to determine, based on the lateral acceleration, a direction of a slide corresponding to the sliding event, determining, based on the direction of the slide, at least one leading adjustable shock absorber of the plurality of adjustable shock absorbers, and the controller is configured to provide the one or more commands by providing, to the at least one leading adjustable shock absorber, one or more commands to result in an increase of a compression damping characteristic and a decrease of a rebound damping characteristic.

In some examples, the sensor information indicates a lateral acceleration. The controller is further configured to determine, based on the lateral acceleration, a direction of a slide corresponding to the sliding event, determining, based on the direction of the slide, at least one trailing adjustable shock absorber of the plurality of adjustable shock absorbers, and the controller is configured to provide the one or more commands by providing, to the at least one trailing adjustable shock absorber, one or more commands to result in a decrease of a compression damping characteristic and an increase of a rebound damping characteristic. In some variations, the sensor information indicates a longitudinal acceleration. The controller is further configured to determine, based on the longitudinal acceleration, an orientation of the vehicle, determine, based on the orientation of the vehicle, a plurality of damping characteristics for the plurality of adjustable shock absorbers, bias, based on the orientation of the vehicle, the plurality of damping characteristics, and generate the one or more commands based on the plurality of biased damping characteristics.

In some instances, the controller is configured to bias the plurality of damping characteristics by additionally increasing a compression damping of a downhill adjustable shock absorber of the plurality of adjustable shock absorbers and additionally decreasing a compression damping of an uphill adjustable shock absorber of the plurality of adjustable shock absorbers. In some examples, the controller is configured to bias the plurality of damping characteristics by additionally increasing the rebound damping of an uphill adjustable shock absorber of the plurality of adjustable shock absorbers.

In another exemplary embodiment of the present disclosure, a method and vehicle of adjusting a plurality of adjustable shock absorbers of a recreational vehicle traveling over terrain including at least one dune is provided. For example, the method and vehicle determines an orientation of the recreational vehicle based on a longitudinal acceleration of the vehicle, determines a slide of the recreational vehicle based on a lateral acceleration of the vehicle, detects the recreational vehicle is sliding across the dune based on the orientation and the slide of the recreational vehicle, and adjusts at least one damping characteristic of the plurality of adjustable shock absorbers to turn the recreational vehicle into the dune as the recreational vehicle continues across the dune.

In some instances, the method and vehicle determines, based on the slide, at least one leading adjustable shock absorber and at least one trailing adjustable shock absorber of the plurality of adjustable shock absorbers and provides, to the at least one leading adjustable shock absorber, one or more commands to result in an increase of a compression damping characteristic and a decrease in a rebound damping characteristic. In some examples, the method and vehicle determines, based on the slide, at least one leading adjustable shock absorber and at least one trailing adjustable shock absorber of the plurality of adjustable shock absorbers and provides, to the at least one trailing adjustable shock absorber, one or more commands to result in an increase of a rebound damping characteristic and a decrease of a compression damping characteristic.

In some variations, the method and vehicle determines, based on the orientation, at least one uphill adjustable shock absorber and at least one downhill adjustable shock absorber of the plurality of adjustable shock absorbers and provides, to the at least one uphill adjustable shock absorber, one or more commands to result in an increase of a compression damping characteristic. In some instances, the method and vehicle determines, based on the orientation, at least one uphill adjustable shock absorber and at least one downhill adjustable shock absorber of the plurality of adjustable shock absorbers and provides, to the at least one downhill adjustable shock absorber, one or more commands to result in an increase of a rebound damping characteristic and a decrease of the compression damping characteristic.

Additional features of the present disclosure will become apparent to those skilled in the art upon consideration of the following detailed description of illustrative embodiments exemplifying the best mode of carrying out the invention as presently perceived.

BRIEF DESCRIPTION OF THE DRAWINGS

The foregoing aspects and many additional features of the present system and method will become more readily appreciated and become better understood by reference to the following detailed description when taken in conjunction with the accompanying drawings, where:

FIG. **1** shows a representative view of components of a vehicle of the present disclosure having a suspension with a plurality of continuous damping control shock absorbers and a plurality of sensors integrated with a controller of the vehicle;

FIG. **2** shows an adjustable damping shock absorber coupled to a vehicle suspension;

FIG. **3** shows an x-axis, a y-axis, and a z-axis for a vehicle, such as an ATV;

FIG. **4** shows a representative view of an exemplary power system for the vehicle of FIG. **1**;

FIG. **5** shows a representative view of an exemplary controller of the vehicle of FIG. **1**;

FIG. **6** shows a first, perspective view of an exemplary vehicle;

FIG. **7** shows a second, perspective view of the exemplary vehicle of FIG. **6**;

FIG. **8** shows a side view of the exemplary vehicle of FIG. **6**;

FIG. **9** shows a bottom view of the exemplary vehicle of FIG. **6**;

FIG. **10** shows an exemplary control system for controlling the damping of one or more shock absorbers;

FIG. **11** shows an exemplary flowchart describing the operation of the suspension controller during a cornering event and/or a braking event;

FIG. **12** shows an example of the suspension controller adjusting the adjustable shock absorbers for the vehicle during a braking event;

FIG. **13** shows an example of the suspension controller adjusting the adjustable shock absorbers for the vehicle during a cornering event;

FIG. **14** shows an example flowchart describing the operation of the suspension controller **86** during an airborne event and a landing event;

FIG. **15** shows an example representation of the compression and rebound damping characteristics of a vehicle during a pre-takeoff period, a free-fall period, and a post-landing period;

FIG. **16** shows an exemplary flowchart describing the suspension controller performing in a rock crawl operation;

FIG. **17** shows an example of the suspension controller adjusting the adjustable shock absorbers for the vehicle performing in a rock crawl operation;

FIG. **18** shows another example of the suspension controller adjusting the adjustable shock absorbers for the vehicle performing in a rock crawl operation;

FIG. **19** shows another example of the suspension controller adjusting the adjustable shock absorbers for the vehicle performing in a rock crawl operation;

FIG. **20** shows another example of the suspension controller adjusting the adjustable shock absorbers for the vehicle performing in a rock crawl operation;

FIG. **21** shows an exemplary flowchart describing the operation of the suspension controller during a sliding event;

FIG. **22** shows another example of the suspension controller adjusting the adjustable shock absorbers during the sliding event;

FIG. **23** shows another example of the suspension controller adjusting the adjustable shock absorbers during the sliding event;

FIG. **24** shows an exemplary flowchart illustrating a method for performing real-time correction of the inertial measurement of a vehicle;

FIG. **25** shows an exemplary schematic block diagram illustrating an example of logic components in the suspension controller;

FIG. **26** shows an exemplary flowchart describing the operation of the suspension controller when switching between driver modes;

FIG. **27** shows an exemplary physical switch for adjusting driving modes;

FIG. **28** shows an exemplary graphical user interface for adjusting driver modes;

FIG. **29** shows an exemplary flowchart describing a method for implementing egress aid for a vehicle;

FIG. **30** illustrates a representative view of a sway bar of the vehicle of FIG. **1**; and

FIG. **31** illustrates a view of another exemplary vehicle.

DETAILED DESCRIPTION OF EMBODIMENTS

For the purposes of promoting an understanding of the principles of the present disclosure, reference will now be made to the embodiments illustrated in the drawings, which are described below. The embodiments disclosed below are not intended to be exhaustive or limited to the precise form disclosed in the following detailed description. Rather, the embodiments are chosen and described so that others skilled in the art may utilize their teachings.

Referring now to FIG. **1**, the present disclosure relates to a vehicle **10** having a suspension system **11** located between a plurality of ground engaging members **12** and a vehicle frame **14**. Exemplary ground engaging members **12** include wheels, skis, guide tracks, treads or other suitable devices for supporting the vehicle relative to the ground. The suspension typically includes springs **16** and shock absorbers **18** coupled between the ground engaging members **12** and the frame **14**. The springs **16** may include, for example, coil springs, leaf springs, air springs or other gas springs. The air or gas springs **16** may be adjustable. See, for example, U.S. Pat. No. 7,950,486, assigned to the current assignee, the entire disclosure of which is incorporated herein by reference.

The adjustable shock absorbers **18** are often coupled between the vehicle frame **14** and the ground engaging members **12** through an A-arm linkage **70** (See FIG. **2**) or other type linkage. Springs **16** are also coupled between the ground engaging members **12** and the vehicle frame **14**. FIG. **2** illustrates an adjustable shock absorber **18** mounted on an A-arm linkage **70** having a first end pivotably coupled to the vehicle frame **14** and a second end pivotably coupled to A-arm linkage **70** which moves with wheel **12**. A damping control activator **74** is coupled to controller **20** by one or more wires **76**. An exemplary damping control activator is an electronically controlled valve which is activated to increase or decrease the damping characteristics of adjustable shock absorber **18**.

In one embodiment, the adjustable shock absorbers **18** include solenoid valves mounted at the base of the shock body or internal to a damper piston of the shock absorber **18**. The stiffness of the shock is increased or decreased by introducing additional fluid to the interior of the shock absorber, removing fluid from the interior of the shock absorber, and/or increasing or decreasing the ease with which fluid can pass from a first side of a damping piston of the shock absorber to a second side of the damping piston of the shock absorber. In another embodiment, the adjustable shock absorbers **18** include a magnetorheological fluid internal to the shock absorber **18**. The stiffness of the shock is increased or decreased by altering a magnetic field experienced by the magnetorheological fluid. Additional details on exemplary adjustable shocks are provided in US Published Patent Application No. 2016/0059660, filed Nov. 6, 2015, titled VEHICLE HAVING SUSPENSION WITH CONTINUOUS DAMPING CONTROL, assigned to the present assignee, the entire disclosure of which is expressly incorporated by reference herein.

In one embodiment, a spring **16** and shock **18** are located adjacent each of the ground engaging members **12**. In an all-terrain vehicle (ATV), for example, a spring **16** and an adjustable shock **18** are provided adjacent each of the four wheels **12**. In a snowmobile, for example, one or more springs **16** and one or more adjustable shocks **18** are provided for each of the two front skis and the rear tread. Some manufacturers offer adjustable springs **16** in the form of either air springs or hydraulic preload rings. These adjustable springs **16** allow the operator to adjust the ride height on the go. However, a majority of ride comfort comes from the damping provided by shock absorbers **18**.

In an illustrated embodiment, controller **20** provides signals and/or commands to adjust damping characteristics of the adjustable shocks **18**. For example, the controller **20** provides signals to adjust damping of the shocks **18** in a continuous or dynamic manner. In other words, adjustable shocks **18** may be adjusted to provide differing compression damping, rebound damping, or both. In one embodiment, adjustable shocks **18** include a first controllable valve to adjust compression damping and a second controllable valve to adjust rebound damping. In another embodiment, adjustable shocks include a combination valve which controls both compression damping and rebound damping.

In an illustrated embodiment of the present disclosure, an operator interface **22** is provided in a location easily accessible to the driver operating the vehicle. For example, the operator interface **22** is either a separate user interface mounted adjacent the driver's seat on the dashboard or integrated onto a display within the vehicle. Operator interface **22** includes user input devices to allow the driver or a passenger to manually adjust shock absorber **18** damping during operation of the vehicle based on road conditions that are encountered or to select a preprogrammed active damping profile for shock absorbers **18** by selecting a ride mode. In one embodiment, a selected ride mode (e.g., a selected driver mode) alters characteristics of suspension system **11** alone, such as the damping profile for shock absorbers **18**. In one embodiment, a selected ride mode alters characteristics of suspension system **11** and other vehicle systems, such as a driveline torque management system **50** or a steering system **104**.

Exemplary input devices for operator interface **22** include levers, buttons, switches, soft keys, and other suitable input devices. Operator interface **22** may also include output devices to communicate information to the operator. Exemplary output devices include lights, displays, audio devices, tactile devices, and other suitable output devices. In another illustrated embodiment, the user input devices are on a steering wheel, handle bar, or other steering control of the vehicle **10** to facilitate actuation of the damping adjustment. For example, referring to FIG. **27**, a physical switch **820** may be located on the steering wheel, handle bar, or other steering control of the vehicle **10**. A display **24** is also provided on or next to the operator interface **22** or integrated into a dashboard display of vehicle **10** to display information related to the compression and/or rebound damping characteristics.

As explained in further detail below, controller **20** receives user inputs from operator interface **22** and adjusts the damping characteristics of the adjustable shocks **18** accordingly. The operator may independently adjust front and rear shock absorbers **18** to adjust the ride characteristics of the vehicle **10**. In certain embodiments, each of the shocks **18** is independently adjustable so that the damping characteristics of the shocks **18** are changed from one side of the vehicle **10** to another. Side-to-side adjustment is desirable during sharp turns or other maneuvers in which different damping profiles for shock absorbers **18** on opposite sides of the vehicle improves the handling characteristics of the vehicle. The damping response of the shock absorbers **18** can be changed in a matter of milliseconds to provide nearly instantaneous changes in damping for potholes, dips in the road, or other driving conditions. Additionally, and/or alternatively, controller **20** may independently adjust the damping characteristics of front and/or rear shocks **18**. An advantage, among others, of adjusting the damping characteristics of the front and/or rear shocks **18** is that the vehicle **10** may be able to operate more efficiently in rough terrain.

The controller **20** communicates with (e.g., provides, transmits, receives, and/or obtains) multiple vehicle condition sensors **40**. For example, a wheel accelerometer **25** is coupled adjacent each ground engaging member **12**. The controller **20** communicates with each of the accelerometers **25**. For instance, the accelerometers **25** may provide information indicating movement of the ground engaging members and the suspension components **16** and **18** as the vehicle traverses different terrain. Further, the controller **20** may communicate with additional vehicle condition sensors **40**, such as a vehicle speed sensor **26**, a steering sensor **28**, a chassis supported accelerometer **30**, a chassis supported gyroscope **31**, an inertial measurement unit (IMU) **37** (shown on FIG. **10**), a physical switch **820** (shown on FIG. **10**), and other sensors which monitor one or more characteristics of vehicle **10**.

Accelerometer **30** is illustratively a three-axis accelerometer supported on the chassis of the vehicle **10** to provide information indicating acceleration forces of the vehicle **10** during operation. In some instances, accelerometer **30** is located at or close to a center position (e.g., a center of gravity position) of vehicle **10**. In other instances, the accelerometer **30** is located at a position that is not near the center of gravity of the vehicle **10**. In the exemplary vehicle **200** illustrated in FIGS. **6**-**9**, the chassis accelerometer **30** is located along a longitudinal centerline plane **122** of vehicle **200**. The x-axis, y-axis, and z-axis for a vehicle **10**, illustratively an ATV, are shown in FIG. **3**.

Gyroscope **31** is illustratively a three-axis gyroscope supported on the chassis to provide indications of inertial measurements, such as roll rates, pitch rates, and/or yaw rates, of the vehicle during operation. In one embodiment, accelerometer **30** is not located at a center of gravity of vehicle **10** and the readings of gyroscope **31** are used by controller **20** to determine the acceleration values of vehicle **10** at the center of gravity of vehicle **10**. In one embodiment, accelerometer **30** and gyroscope **31** are integrated into the controller **20**, such as a suspension controller **86**.

In some examples and referring to FIG. **10**, an IMU, such as the IMU **37** is supported on the chassis to provide indications of the inertial measurements, including the angular rate and/or the acceleration forces, of the vehicle **10** during operation. The IMU **37** may include the functionalities of the accelerometer **30** and/or the gyroscope **31**. As such, in some instances, the accelerometer **30** and/or the gyroscope **31** are optional and might not be included in the vehicle **10**. In other instances, the vehicle **10** may include the gyroscope **31** and the accelerometer **30** instead of the IMU **37**.

The controller **20** may also communicate with additional vehicle condition sensors **40**, such as a brake sensor **32**, a throttle position sensor **34**, a wheel speed sensor **36**, and/or a gear selection sensor **38**.

Referring to FIG. **4**, one embodiment of a driveline torque management system **50** of vehicle **10** is illustrated. Driveline torque management system **50** controls the amount of torque exerted by each of ground engaging members **12**. Driveline torque management system **50** provides a positive torque to one or more of ground engaging members **12** to power the movement of vehicle **10** through a power system **60**. Driveline torque management system **50** further provides a negative torque to one or more of ground engaging members **12** to slow or stop a movement of vehicle **10** through a braking system **75**. In one example, each of ground engaging members **12** has an associated brake of braking system **75**.

Power system **60** includes a prime mover **62**. Exemplary prime movers **62** include internal combustion engines, two stroke internal combustion engines, four stroke internal combustion engines, diesel engines, electric motors, hybrid engines, and other suitable sources of motive force. To start the prime mover **62**, a power supply system **64** is provided. The type of power supply system **64** depends on the type of prime mover **62** used. In one embodiment, prime mover **62** is an internal combustion engine and power supply system **64** is one of a pull start system and an electric start system. In one embodiment, prime mover **62** is an electric motor and power supply system **64** is a switch system which electrically couples one or more batteries to the electric motor.

A transmission **66** is coupled to prime mover **62**. Transmission **66** converts a rotational speed of an output shaft **61** of prime mover **62** to one of a faster rotational speed or a slower rotational speed of an output shaft **63** of transmission **66**. It is contemplated that transmission **66** may additionally rotate output shaft **63** at the same speed as output shaft **61**.

In the illustrated embodiment, transmission **66** includes a shiftable transmission **68** and a continuously variable transmission ("CVT") **71**. In one example, an input member of CVT **71** is coupled to prime mover **62**. An input member of shiftable transmission **68** is in turn coupled to an output member of CVT **71**. In one embodiment, shiftable transmission **68** includes a forward high setting, a forward low setting, a neutral setting, a park setting, and a reverse setting. The power communicated from prime mover **62** to CVT **71** is provided to a drive member of CVT **71**. The drive member in turn provides power to a driven member through a belt. Exemplary CVTs are disclosed in U.S. Pat. Nos. 3,861,229; 6,176,796; 6,120,399; 6,860,826; and 6,938,508, the disclosures of which are expressly incorporated by reference herein. The driven member provides power to an input shaft of shiftable transmission **68**. Although transmission **66** is illustrated as including both shiftable transmission **68** and CVT **71**, transmission **66** may include only one of shiftable transmission **68** and CVT **71**. Further, transmission **66** may include one or more additional components.

Transmission **66** is further coupled to at least one differential **73** which is in turn coupled to at least one ground engaging members **12**. Differential **73** may communicate the power from transmission **66** to one of ground engaging members **12** or multiple ground engaging members **12**. In an ATV embodiment, one or both of a front differential and a rear differential are provided. The front differential powering at least one of two front wheels of the ATV and the rear differential powering at least one of two rear wheels of the ATV. In a side-by-side vehicle embodiment having seating for at least an operator and a passenger in a side-by-side configuration, one or both of a front differential and a rear differential are provided. The front differential powering at least one of two front wheels of the side-by-side vehicle and the rear differential powering at least one of multiple rear wheels of the side-by-side vehicle. In one example, the side-by-side vehicle has three axles and a differential is provided for each axle. An exemplary side-by-side vehicle **200** is illustrated in FIGS. **6**-**9**.

In one embodiment, braking system **75** includes anti-lock brakes. In one embodiment, braking system **75** includes active descent control and/or engine braking. In one embodiment, braking system **75** includes a brake and in some embodiments a separate parking brake. Braking system **75** may be coupled to any of prime mover **62**, transmission **66**, differential **73**, and ground engaging members **12** or the connecting drive members therebetween. Brake sensor **32**, in one example, monitors when braking system **75** is applied. In one example, brake sensor **32** monitors when a user actuatable brake input, such as brake pedal **232** (see FIG. **7**) in vehicle **200**, is applied.

Referring to FIG. **5**, controller **20** has at least one associated memory **76**. Controller **20** provides the electronic control of the various components of vehicle **10**. Further, controller **20** is operatively coupled to a plurality of vehicle condition sensors **40** as described above, which monitor various parameters of the vehicle **10** or the environment surrounding the vehicle **10**. Controller **20** performs certain operations (e.g., provides commands) to control one or more subsystems of other vehicle components. In certain embodiments, the controller **20** forms a portion of a processing subsystem including one or more computing devices having memory, processing, and communication hardware. Controller **20** may be a single device or a distributed device, and the functions of the controller **20** may be performed by hardware and/or as computer instructions on a non-transitory computer readable storage medium, such as memory **76**.

As illustrated in the embodiment of FIG. **5**, controller **20** is represented as including several controllers. These controllers may each be single devices or distributed devices or one or more of these controllers may together be part of a single device or distributed device. The functions of these controllers may be performed by hardware and/or as computer instructions on a non-transitory computer readable storage medium, such as memory **76**.

In one embodiment, controller **20** includes at least two separate controllers which communicate over a network **78**. In one embodiment, network **78** is a CAN network. Details regarding an exemplary CAN network are disclosed in U.S. patent application Ser. No. 11/218,163, filed Sep. 1, 2005, the disclosure of which is expressly incorporated by reference herein. Of course any suitable type of network or data bus may be used in place of the CAN network. In one embodiment, two wire serial communication is used for some connections.

Referring to FIG. **5**, controller **20** includes an operator interface controller **80** which controls communication with an operator through operator interface **22**. A prime mover controller **82** controls the operation of prime mover **62**. A transmission controller **84** controls the operation of transmission system **66**.

A suspension controller **86** controls adjustable portions of suspension system **11**. Exemplary adjustable components include adjustable shocks **18**, adjustable springs **16**, and/or configurable stabilizer bars. Additional details regarding adjustable shocks, adjustable springs, and configurable stabilizer bars is provided in US Published Patent Application No. 2016/0059660, filed Nov. 6, 2015, titled VEHICLE HAVING SUSPENSION WITH CONTINUOUS DAMPING CONTROL, assigned to the present assignee, the entire disclosure of which is expressly incorporated by reference herein.

Communication controller **88** controls communications between a communication system **90** of vehicle **10** and remote devices, such as other vehicles, personal computing devices, such as cellphones or tablets, a centralized computer system maintaining one or more databases, and other types of devices remote from vehicle **10** or carried by riders of vehicle **10**. In one embodiment, communication controller **88** of vehicle **10** communicates with paired devices over a wireless network. An exemplary wireless network is a radio frequency network utilizing a BLUETOOTH protocol. In this example, communication system **90** includes a radio frequency antenna. Communication controller **88** controls the pairing of devices to vehicle **10** and the communications between vehicle **10** and the remote device. In one embodiment, communication controller **88** of vehicle **10** communicates with remote devices over a cellular network. In this example, communication system **90** includes a cellular antenna and communication controller **88** receives and sends cellular messages from and to the cellular network. In one embodiment, communication controller **88** of vehicle **10** communicates with remote devices over a satellite network. In this example, communication system **90** includes a satellite antenna and communication controller **88** receives and sends messages from and to the satellite network. In one embodiment, vehicle **10** is able to communicate with other vehicles **10** over a Radio Frequency mesh network and communication controller **88** and communication system **90** are configured to enable communication over the mesh network. An exemplary vehicle communication system is disclosed in U.S. patent application Ser. No. 15/262,113, filed Sep. 12, 2016, titled VEHICLE TO VEHICLE COMMUNICATIONS DEVICE AND METHODS FOR RECREATIONAL VEHICLES, the entire disclosure of which is expressly incorporated by reference herein.

A steering controller **102** controls portions of a steering system **104**. In one embodiment, steering system **104** is a power steering system and includes one or more steering sensors **28** (shown in FIG. **1**). Exemplary sensors and electronic power steering units are provided in U.S. patent application Ser. No. 12/135,107, assigned to the assignee of the present application, titled VEHICLE, docket PLR-06-22542.02P, the disclosure of which is expressly incorporated by reference herein. A vehicle controller **92** controls lights, loads, accessories, chassis level functions, and other vehicle functions. A ride height controller **96** controls the preload and operational height of the vehicle. In one embodiment, ride height controller controls springs **16** to adjust a ride height of vehicle **10**, either directly or through suspension controller **86**. In one example, ride height controller **96** provides more ground clearance in a comfort ride mode compared to a sport ride mode.

An agility controller **100** controls a braking system of vehicle **10** and the stability of vehicle **10**. Control methods of agility controller **100** may include integration into braking circuits (ABS) such that a stability control system can improve dynamic response (vehicle handling and stability) by modifying the shock damping in conjunction with electronic braking control.

In one embodiment, controller **20** either includes a location determiner **110** and/or communicates via network **78** to a location determiner **110**. The location determiner **110** determines a current geographical location of vehicle **10**. An exemplary location determiner **110** is a GPS unit which determines the position of vehicle **10** based on interaction with a global satellite system.

Referring to FIGS. **6-9**, an exemplary side-by-side vehicle **200** is illustrated. Vehicle **200**, as illustrated, includes a plurality of ground engaging members **12**. Illustratively, ground engaging members **12** are wheels **204** and associated tires **206**. As mentioned herein, one or more of ground engaging members **12** are operatively coupled to power system **60** (see FIG. **4**) to power the movement of vehicle **200** and braking system **75** to slow movement of vehicle **200**.

Referring to the illustrated embodiment in FIG. **6**, a first set of wheels, one on each side of vehicle **200**, generally correspond to a front axle **208**. A second set of wheels, one on each side of vehicle **200**, generally correspond to a rear axle **210**. Although each of front axle **208** and rear axle **210** are shown having a single ground engaging member **12** on each side, multiple ground engaging members **12** may be included on each side of the respective front axle **208** and rear axle **210**. As configured in FIG. **6**, vehicle **200** is a four wheel, two axle vehicle.

Referring to FIG. **9**, wheels **204** of front axle **208** are coupled to a frame **212** of vehicle **200** through front independent suspensions **214**. Front independent suspensions **214** in the illustrated embodiment are double A-arm suspensions. Other types of suspensions systems may be used for front independent suspensions **214**. The wheels **204** of rear axle **210** are coupled to frame **212** of vehicle **200** through rear independent suspensions **216**. Other types of suspensions systems may be used for rear independent suspensions **216**.

Returning to FIG. **6**, vehicle **200** includes a cargo carrying portion **250**. Cargo carrying portion **250** is positioned rearward of an operator area **222**. Operator area **222** includes seating **224** and a plurality of operator controls. In the illustrated embodiment, seating **224** includes a pair of bucket seats. In one embodiment, seating **224** is a bench seat. In one embodiment, seating **224** includes multiple rows of seats, either bucket seats or bench seats or a combination thereof. Exemplary operator controls include a steering wheel **226**, a gear selector **228**, an accelerator pedal **230** (see FIG. **7**), and a brake pedal **232** (see FIG. **7**). Steering wheel **226** is operatively coupled to the wheels of front axle **208** to control the orientation of the wheels relative to frame **212**. Gear selector **228** is operatively coupled to the shiftable transmission **68** to select a gear of the shiftable transmission **68**. Exemplary gears include one or more forward gears, one or more reverse gears, and a park setting. Accelerator pedal **230** is operatively coupled to prime mover **62** to control the speed of vehicle **200**. Brake pedal **232** is operatively coupled to brake units associated with one or more of wheels **204** to slow the speed of vehicle **200**.

Operator area **222** is protected with a roll cage **240**. Referring to FIG. **6**, side protection members **242** are provided on both the operator side of vehicle **200** and the passenger side of vehicle **200**. In the illustrated embodiment, side protection members **262** are each a unitary tubular member.

In the illustrated embodiment, cargo carrying portion **250** includes a cargo bed **234** having a floor **256** and a plurality of upstanding walls. Floor **256** may be flat, contoured, and/or comprised of several sections. Portions of cargo carrying portion **250** also include mounts **258** which receive an expansion retainer (not shown). The expansion retainers which may couple various accessories to cargo carrying portion **250**. Additional details of such mounts and expansion retainers are provided in U.S. Pat. No. 7,055,454, to Whiting et al., filed Jul. 13, 2004, titled "Vehicle Expansion Retainers," the entire disclosure of which is expressly incorporated by reference herein.

Front suspensions **214**A and **214**B each include a shock absorber **260**, respectively. Similarly, rear suspensions **216**A and **216**B each include a shock absorber **262**. In one embodiment each of shock absorbers **260** and shock absorbers **262** are electronically adjustable shocks **18** which are controlled by a controller **20** of vehicle **200**.

Additional details regarding vehicle **200** are provided in U.S. Pat. Nos. 8,827,019 and 9,211,924, assigned to the present assignee, the entire disclosures of which are expressly incorporated by reference herein. Other exemplary recreational vehicles include ATVs, utility vehicles, snowmobiles, other recreational vehicles designed for off-road use, on-road motorcycles, and other suitable vehicles (e.g., FIG. **31**).

FIG. **10** shows an exemplary control system **300** for controlling the damping of shock absorbers **18**. In some instances, the control system **300** may be included in the vehicle **10** and/or the vehicle **200** described above. For example, the suspension controller **86** may communicate with (e.g., receive and/or provide) one or more entities (e.g., sensors, devices, controllers, and/or subsystems) from the vehicle **10** and/or **200** described above. Additionally, and/or alternatively, the vehicle **10** and **200** may be the same vehicle (e.g., the vehicle **200** may include entities from vehicle **10**, such as the suspension controller **86**).

Additionally, the controller **20** may include a suspension controller **86** as described above in FIG. **5**. The suspension controller **86** may communicate with the plurality of vehicle condition sensors **40** as described above. Further, the suspension controller **86** may provide information (e.g., one or more commands) to each of the adjustable shock absorbers **18***a*, **18***b*, **18***c*, and **18***d*. For example, the suspension controller **86** may provide commands to adjust the compression damping characteristic and/or the rebound damping characteristic for the adjustable shock absorbers **18***a*, **18***b*, **18***c*, and **18***d*.

While exemplary sensors, devices, controllers, and/or subsystems are provided in FIG. **10**, additional exemplary sensors, devices, controllers, and/or subsystems used by the suspension controller **86** to adjust shock absorbers **18** are provided in US Published Patent Application No. 2016/0059660 (filed Nov. 6, 2015, titled VEHICLE HAVING SUSPENSION WITH CONTINUOUS DAMPING CONTROL) and US Published Application 2018/0141543 (filed Nov. 17, 2017, titled VEHICLE HAVING ADJUSTABLE SUSPENSION), both assigned to the present assignee and the entire disclosures of each expressly incorporated by reference herein.

The illustrative control system **300** is not intended to suggest any limitation as to the scope of use or functionality of embodiments of the present disclosure. Neither should the illustrative control system **300** be interpreted as having any dependency or requirement related to any single entity or combination of entities illustrated therein. Additionally, various entities depicted in FIG. **10**, in embodiments, may be integrated with various ones of the other entities depicted therein (and/or entities not illustrated). For example, the suspension controller **86** may be included within the controller **20**, and may communicate with one or more vehicle condition sensors **40** as described above. The functionalities of the suspension controller **86** and/or other entities in control system **300** will be described below.

Cornering Event and Braking Event

FIG. **11** shows an example flowchart describing the operation of the suspension controller **86** during a cornering event and/or a braking event. FIGS. **12** and **13** show examples of the suspension controller **86** adjusting the shock absorbers **18** for the vehicle **10** during a cornering event and/or braking event. As will be described in more detail below, for better performance during a braking and/or cornering event, a vehicle, such as vehicle **10**, may use a controller **20** (e.g., the suspension controller **86**) to adjust the rebound and/or compression damping. For instance, during a cornering event, the suspension controller **86** may reduce (e.g., decrease) the compression damping of the inner adjustable shock absorbers **18**, which may cause the vehicle **10** to ride lower during cornering and/or to be less upset when the inner ground engaging members **12** encounter bumps. Further, the suspension controller **86** may reduce the rebound damping of the outside adjustable shock absorbers **18**, which may promote shock extensions to flatten the vehicle **10** and/or cause the wheels of the vehicle **10** to follow the ground better to provide better traction. Additionally, and/or alternatively, the suspension controller **86** may increase the compression damping of the outside adjustable shocks **18** and/or increase the rebound damping of the inside adjustable shock absorbers **18**.

During a braking event, the suspension controller **86** may reduce the compression damping for the rear adjustable shock absorbers **18** based on a deceleration rate (e.g., deceleration value), which may cause increased stability of the vehicle **10** during bumps and/or rough trails. Further, by reducing the compression damping for the rear adjustable shock absorbers **18**, the suspension controller **86** may cause the rear ground engaging members **12** to absorb events, such as bumps, better. Also, the suspension controller **86** may also reduce the rebound damping for the front adjustable shock absorbers **18**, which may promote shock extensions to flatten the vehicle **10** and/or cause the wheels of the vehicle **10** to follow the ground better to provide better traction. Additionally, and/or alternatively, the suspension controller **86** may increase the compression damping of the front adjustable shocks **18** and/or increase the rebound damping of the rear adjustable shock absorbers **18**. By increasing the rebound damping of the rear adjustable shock absorbers **18**, the vehicle **10** may be better able to control the pitch body movement and/or weight transfer during a braking event. Further, the suspension controller **86** may hold the vehicle **10** flatter and/or more stable when the vehicle **10** encounters bumps while braking.

In operation, at step **402**, the suspension controller **86** may receive steering information from one or more sensors, such as the steering sensor **28**. The steering information may indicate a steering position, steering angle, and/or steering rate of a steering wheel, such as steering wheel **226**. The steering position and/or angle may indicate a position and/or an angle of the steering wheel for the vehicle **10**. The steering rate may indicate a change of the position and/or angle of the steering wheel over a period of time.

At step **404**, the suspension controller **86** may receive yaw rate information from one or more sensors, such as the gyroscope **31** and/or the IMU **37**. The yaw information indicates the yaw rate of the vehicle **10**.

At step **406**, the suspension controller **86** may receive acceleration information indicating an acceleration rate or deceleration rate of the vehicle **10** from one or more sensors, such as the IMU **37** and/or the chassis accelerometer **30**. The acceleration information may indicate multi-axis acceleration values of the vehicle, such as a longitudinal acceleration and/or a lateral acceleration. In some examples, the suspension controller **86** may receive information from another sensor, such as the throttle position sensor **34** and/or the accelerator pedal sensor **33**. The suspension controller **86** use the information to determine the acceleration rate. For example, the suspension controller **86** may use the throttle position from the throttle position sensor and/or the position of the accelerator pedal **230** from the acceleration pedal sensor **33** to determine whether the vehicle **10** is accelerating and/or decelerating.

At step **408**, the suspension controller **86** may receive brake information from a sensor, such as the brake sensor **32**. The brake information may indicate a position (e.g., braking and/or not braking) of the brake pedal **232**. Additionally, and/or alternatively, the brake information may indicate an amount of brake pressure on the brake pedal **232**.

At step **410**, the suspension controller **86** may receive an operating mode of the vehicle **10** from the operator interface **22** and/or one or more other controllers (e.g., controller **20**). Each mode may include a different setting for the rebound and/or compression damping. An operator (e.g., user) may input a mode of operation on the operator interface **22**. The operator interface **22** may provide the user input indicating the mode of operation to the controller **20** and/or the suspension controller **86**. The suspension controller **86** may use the user input to determine the mode of operation for the vehicle **10**.

At step **412**, the suspension controller **86** may determine whether a cornering event (e.g., a turn) is occurring. Further, the suspension controller **86** may determine a direction of the turn (e.g., a left turn or a right turn). For example, the suspension controller **86** may determine the cornering event and/or direction of the turn based on the steering information indicating a steering rate, angle, and/or position, yaw rate information indicating a yaw rate, and/or the acceleration information indicating a lateral acceleration. The suspension controller **86** may compare the steering rate, steering angle, steering position, yaw rate, and/or lateral acceleration with one or more corresponding thresholds (e.g., pre-determined, pre-programmed, and/or user-defined) to determine the cornering event. The suspension controller **86** may use the positive and/or negative values of the steering rate, angle, position, yaw rate, and/or lateral acceleration to determine the direction of the turn.

In some examples, the suspension controller **86** may determine the cornering event based on the steering rate, angle, and/or position being greater than a threshold. In such examples, the method **400** may move to step **414**. Otherwise, if the suspension controller **86** determines that the steering rate, angle and/or position is below the threshold, the method **400** may move to step **418**. In some variations, the suspension controller **86** may determine the cornering event based on the yaw rate. For example, based on the yaw rate being greater than a threshold, the suspension controller **86** may determine a cornering event is occurring. In some instances, the suspension controller **86** may determine the cornering event based on a lateral acceleration. For example, based on the lateral acceleration being greater than a threshold, the suspension controller **86** may determine that a cornering event is occurring.

In some variations, the suspension controller **86** may prioritize the steering information, the yaw rate information, and/or the acceleration information, and determine the cornering event based on the priorities. For instance, in some examples, the steering position and/or angle, steering rate, yaw rate, and lateral acceleration may all indicate a turn. In other examples, the steering position and/or angle, steering rate, yaw rate, and/or lateral acceleration may conflict (e.g., the steering position might not indicate a turn and the yaw rate may indicate a turn; the yaw rate might not indicate a turn and the lateral acceleration may indicate a turn). For example, in a counter steer or a slide, the vehicle **10** may be turning in one direction, such as a left turn (e.g., indicated by a yaw rate and/or lateral acceleration); however, the steering position may indicate a turn in the opposite direction, such as a right turn, or might not indicate a turn. In such examples, the suspension controller **86** may prioritize the lateral acceleration and/or the yaw rate over the steering rate, angle, and/or position. For example, the suspension controller **86** may determine the vehicle **10** is turning and the turn is a left turn.

In some variations, the suspension controller **86** may prioritize the lateral acceleration over the yaw rate and the yaw rate over the steering rate, angle, and/or position. In other words, the suspension controller **86** may determine the cornering event is occurring and/or a direction of turn based on the lateral acceleration indicating a turn even if the yaw rate, steering rate, angle, and/or position do not indicate a turn. Further, the suspension controller **86** may determine the cornering event is occurring and/or a direction of turn based on the yaw rate indicating a turn even if the steering rate, angle, and/or position do not indicate a turn.

At step **414**, the suspension controller **86** may determine a cornering state of the vehicle **10**. The cornering state may indicate whether the vehicle **10** is entering, in the middle of, and/or exiting a corner event. Additionally, and/or alternatively, the cornering state may indicate whether the vehicle

10 is braking, accelerating, and/or decelerating through the cornering event. For example, the suspension controller 86 may determine the cornering state based on the acceleration information (e.g., the longitudinal acceleration) and/or the brake information (e.g., the position of the brake pedal 232 and/or amount of pressure on the brake pedal 232). As will be explained below, the suspension controller 86 may adjust and/or bias the adjustable shock absorbers 18 based on the cornering event, the braking of the vehicle 10, and/or the acceleration/deceleration of the vehicle 10.

At step 416, the suspension controller 86 may execute a corner condition modifier for one or more of the adjustable shock absorbers 18. For example, the suspension controller 86 may adjust (e.g., increase and/or decrease) the rebound and/or compression damping for one or more of the adjustable shock absorbers 18 based on detecting a cornering event (e.g., from step 412 indicating that the vehicle 10 is turning). FIG. 13 shows the suspension of the vehicle 10 during a cornering event. For example, in response to detecting the cornering event, the suspension controller 86 may provide information (e.g., one or more commands) to the adjustable shock absorbers 18. For instance, the suspension controller 86 may determine that the vehicle 10 is turning left (e.g., based on the steering position or rate, yaw rate, and/or lateral acceleration). In response, the suspension controller 86 may provide information 302 and/or 306 to the inner adjustable shock absorbers 18*a* and/or 18*c* to decrease the compression damping (CD) and/or increase the rebound damping (RD). By decreasing the compression damping of the inner adjustable shock absorbers 18*a* and/or 18*c*, the vehicle 10 may absorb bumps on that side better and/or be more stable when cornering in rough terrain. Further, the vehicle 10 may sit lower when cornering because the inside will move through the compression stroke easier when the vehicle is "on the sway bar" torsion. Additionally, by increasing the rebound on the inside adjustable shock absorbers 18*a* and/or 18*c*, the vehicle 10 may control the roll gradient and/or rate during the cornering event. Also, the suspension controller 86 may provide information 304 and/or 308 to the outer adjustable shock absorbers 18*b* and/or 18*d* to increase the compression damping (CD) and/or decrease the rebound damping (RD). After the suspension controller 86 determines and/or executes a corner condition modifier, the method 400 may move back to step 402 and repeat continuously.

In some variations, during the cornering event, the suspension controller 86 may adjust (e.g., increase and/or decrease) the rebound and/or compression damping for one or more of the adjustable shock absorbers 18 based on the cornering state (e.g., the braking of the vehicle 10 and/or the acceleration/deceleration of the vehicle 10). For example, during a cornering event, the suspension controller 86 may adjust the compression and/or rebound damping for the inner and/or outer adjustable shock absorbers 18 as described above. Additionally, and/or alternatively, based on the braking, acceleration, and/or deceleration rate, the suspension controller 86 may further bias (e.g., further increase by a value or percentage and/or further decrease by a value or percentage) the compression damping and/or rebound damping for the front and rear adjustable shock absorbers 18. By biasing the compression and/or rebound damping, the vehicle 10 may improve weight transfer on a corner entry and exit and may also improve the vehicle yaw response.

In other words, in response to detecting the cornering event, the suspension controller 86 may adjust the inner/outer adjustable shock absorbers 18 as described above. Then, based on the vehicle 10 slowing down, the suspension controller 86 may detect and/or determine a longitudinal deceleration (e.g., negative longitudinal acceleration) of the vehicle. Based on detecting the longitudinal deceleration, the suspension controller 86 may further bias (e.g., change and/or determine) the compression and/or rebound damping for the front/rear shock absorbers 18. For example, based on the longitudinal deceleration, the suspension controller 86 may additionally increase (e.g., increase by a percentage or a value) the compression damping of the front adjustable shock absorbers 18*a* and 18*b* and/or additionally decrease (e.g., decrease by a percentage or a value) the compression damping of rear adjustable shock absorbers 18*c* and 18*d*. Further, the suspension controller 86 may additionally increase the rebound damping of the rear shock absorbers 18*c* and 18*d* and/or additionally decrease the rebound damping of the front shock absorbers 18*a* and 18*b*.

For instance, the suspension controller 86 may adjust the compression and/or rebound damping for a shock absorber 18 based on whether the shock absorber is an inner or outer shock absorber as described above (e.g., set the compression damping at a value of 73). Then, based on the cornering state (e.g., a positive/negative longitudinal acceleration), the suspension controller 86 may further bias the compression and/or rebound damping. For example, the suspension controller 86 may decrease the front compression damping (e.g., set the compression damping from 73 to 62) if the suspension controller 86 detects a positive acceleration and increase the front compression damping (e.g., set the compression damping from 73 to 80) if the suspension controller 86 detects a negative acceleration. The suspension controller 86 may operate similarly for rebound damping.

Additionally, and/or alternatively, the suspension controller 86 may bias the compression and/or rebound damping differently based on the positive/negative longitudinal acceleration being greater than or less than one or more thresholds. For example, if acceleration is greater than a first threshold, then the suspension controller 86 may set the compression damping from a first value (e.g., 73) to a second value (e.g., 62). If acceleration is greater than a second threshold, then the suspension controller 86 may set the compression damping from the first value (e.g., 73) to a third value (e.g., 59). Additionally, and/or alternatively, if acceleration is greater than a third threshold, then the suspension controller 86 may set the compression damping from the first value (e.g., 73) to a fourth value (44). The suspension controller 86 may operate similarly for negative accelerations and/or braking as well.

At the end of the cornering event (e.g., a turn exit), the vehicle 10 may speed up (e.g., an operator may actuate the accelerator pedal 230). Based on detecting a positive acceleration (e.g., a longitudinal acceleration), the suspension controller 86 may further bias the front/rear shock absorbers 18. For example, based on the positive longitudinal acceleration, the suspension controller 86 may additionally increase the compression damping of rear adjustable shock absorbers 18*c* and 18*d* and/or additionally decrease the compression damping of the front adjustable shock absorbers 18*a* and 18*b*. Further, the suspension controller 86 may additionally increase the rebound damping of the front shock absorbers 18*a* and 18*b* and/or additionally decrease rebound damping of the rear shock absorbers 18*c* and 18*d*. After the executing the corner condition modifier, the method 400 may move back to step 402.

If the suspension controller 86 does not detect a cornering event, the method 400 may move to step 418. At step 418, the suspension controller 86 may determine whether a braking event is occurring. For example, the suspension controller **86** may determine whether the brake pedal **232** is actuated. In other words, the suspension controller **86** may determine or detect a braking event (e.g., whether the vehicle **10** is braking). If the suspension controller **86** determines that the brake pedal **232** is not actuated, the method **400** moves back to **402** and then repeats. If the suspension controller **86** determines that the brake pedal **232** is actuated, the method **400** moves to step **410**.

At step **420**, the suspension controller **86** may determine the deceleration rate of the vehicle **10** (e.g., two tenths of a gravitational constant (G) or half of a G). For example, the suspension controller **86** may determine the deceleration rate of the vehicle **10** from the chassis accelerometer **30** and/or the IMU **37**. Additionally, and/or alternatively, the suspension controller **86** may determine and/or predict the deceleration rate of the vehicle **10** based on the amount of brake pressure on the brake pedal **232**. As mentioned above, the brake sensor **32** may provide the amount of brake pressure on the brake pedal **232** to the suspension controller **86**. Additionally, and/or alternatively, the suspension controller **86** may determine and/or predict the deceleration rate of the vehicle **10** based on an engine torque reduction from an engine torque sensor.

At step **422**, the suspension controller **86** may execute a brake condition modifier for one or more of the adjustable shock absorbers **18**. For example, the suspension controller **86** may adjust (e.g., increase and/or decrease) the rebound and/or compression damping for one or more of the adjustable shock absorbers **18** based on detecting a braking event (e.g., from step **408** indicating that the brake pedal has been actuated). FIG. **12** shows the suspension of the vehicle **10** during a braking event. For example, in response to detecting the braking event, the suspension controller **86** may provide information (e.g., one or more commands) to the adjustable shock absorbers **18** during the braking event. For instance, the suspension controller **86** may provide information **302** and/or **304** to the front adjustable shock absorbers **18***a* and/or **18***b* to increase the compression damping (CD) and decrease the rebound damping (RD). Further, the suspension controller **86** may provide information **306** and/or **308** to the rear adjustable shock absorbers **18***c* and/or **18***d* to increase the rebound damping (RD) and decrease the compression damping (CD). After the suspension controller **86** determines and/or executes a brake condition modifier, the method **400** may move back to step **402** and repeat continuously.

In some examples, in response to detecting the braking event, the suspension controller **86** may adjust the rebound and/or compression damping for the one or more adjustable shock absorbers **18** using the deceleration rate from step **410**. For example, based on comparing the deceleration rate with a threshold, the suspension controller **86** may adjust the rebound and/or compression damping. For instance, if the deceleration rate is above a first threshold (e.g., above two tenths of a G), then the suspension controller **86** may reduce the compression damping of the rear adjustable shock absorbers **18***c* and/or **18***d*. If the deceleration is below the first threshold, then the suspension controller **86** might not reduce the compression damping (e.g., maintain the current compression damping) of the rear adjustable shock absorbers **18***c* and/or **18***d*. Additionally, and/or alternatively, if the deceleration rate is above the first threshold (e.g., above two tenths of a G), but below a second threshold (e.g., half of a G), then the suspension controller **86** may reduce the compression damping to a first value. If the deceleration is above the second threshold, then the suspension controller **86** may reduce to a second value that is below the first value (e.g., a softer compression damping value).

In some instances, the suspension controller **86** may separate operation of the braking event and/or cornering event. For example, step **412**, **414**, and **416** may be optional, and the method **400** may move directly to step **418**.

Landing Hop Prevention

FIG. **14** shows an example flowchart describing the operation of the suspension controller **86** during an airborne event and a landing event. FIG. **14** will be described with reference to FIG. **15**. FIG. **15** shows an example graph **525** illustrating the compression and rebound damping characteristics of a vehicle, such as vehicle **10** and/or **200**, during a pre-takeoff period **530**, a free-fall period **535**, and a post-landing period **540**. As will be described in more detail below, by adjusting the compression damping and/or rebound damping of one or more adjustable shock absorbers **18** during a free-fall and/or post-landing event, the suspension controller **86** may increase the stability of vehicle **10** post-landing by preventing the vehicle **10** from hoping, unloading, and/or decreasing the weight on the ground engaging members **12**. Additionally, and/or alternatively, this may also provide a faster vehicle **10** in a race environment since after a landing event, the throttle may be applied quicker without tire (e.g., ground engaging members **12**) spin. Additionally, and/or alternatively, this may also provide a more stable vehicle because the vehicle **10** will have better traction, and thus better user control.

In operation, at step **502**, the suspension controller **86** may receive information (e.g., inputs) from one or more entities (e.g., sensors, devices, and/or subsystems) of vehicle **10**. For example, the suspension controller **86** may receive (e.g., retrieve and/or obtain) information (e.g., data packets and/or signals indicating sensor readings) from the one or more sensors, devices, and/or subsystems. In some instances, the suspension controller **86** may receive information indicating the x, y, and/or z-axis acceleration from the chassis accelerometer **30** and/or IMU **37**. For example, referring back to FIG. **3**, the chassis accelerometer **30** and/or IMU **37** may measure the x, y, and/or z-axis acceleration values for the vehicle **10**, and may provide the acceleration values to the suspension controller **86**.

At step **506**, the suspension controller **86** may determine whether an airborne event is occurring. For example, the suspension controller **86** may determine whether an airborne event is occurring based on the magnitude of the x-axis acceleration value, the y-axis acceleration value, and/or the z-axis acceleration value. For instance, the suspension controller **86** may compare the x, y, and/or z-axis acceleration values with one or more thresholds (e.g., one or more pre-defined, pre-programmed, and/or user-defined thresholds) to determine whether the vehicle **10** is in free-all. If the x, y, and/or z-axis acceleration values are less than the one or more thresholds, the method **500** may move to step **508**. If the x, y, and/or z-axis acceleration values are greater than the thresholds, the method **500** may move back to step **502** and repeat.

In some examples, the suspension controller **86** may use two or three different thresholds. For example, the suspension controller **86** may compare the magnitudes for each of the x, y, and/or z-axis acceleration with a different threshold. Additionally, and/or alternatively, the suspension controller **86** may compare the magnitudes for the x and y-axis acceleration with a same threshold. For example, in free-fall, the chassis accelerometer **30** and/or IMU **37** may measure the vehicle **10** to have zero or substantially zero acceleration (e.g., wind, air resistance, and/or other factors may account for substantially zero acceleration) for the x and y-axis, and 1 G or substantially 1 G on the z-axis. As such, the suspension controller **86** may use a first threshold for the x-axis and y-axis acceleration values and a second threshold for the z-axis acceleration value. In some variations, the first threshold for the x-axis and the y-axis may be 0.3 Gs. In some instances, the suspension controller **86** may combine (e.g., calculate) a magnitude of acceleration for the x, y, and z-axis acceleration, and compare the combined magnitude with a threshold to determine whether the vehicle **10** is in free-fall.

Exemplary detection of an airborne event is described in US Published Patent Application No. 2016/0059660 (filed Nov. 6, 2015, titled VEHICLE HAVING SUSPENSION WITH CONTINUOUS DAMPING CONTROL) and US Published Application 2018/0141543 (filed Nov. 17, 2017, titled VEHICLE HAVING ADJUSTABLE SUSPENSION), both assigned to the present assignee and the entire disclosures of each expressly incorporated by reference herein.

At step **508**, the suspension controller **86** may determine an airborne condition modifier for one or more of the adjustable shock absorbers **18**. For example, the suspension controller **86** may provide information (e.g., one or more commands) to the adjustable shock absorbers **18** to increase and/or gradually increase the compression damping. By increasing the adjustable shock absorbers **18**, the vehicle **10** may increase the energy absorption upon landing. Further, the suspension controller **86** may provide information (e.g., one or more commands) to decrease the rebound damping for the adjustable shock absorbers **18**. By decreasing the rebound damping, the adjustable shock absorbers **18** may achieve full shock extension faster, increase energy absorption upon landing, and/or increase the travel of the shocks **18** available for landing. For example, the suspension controller **86** may reduce the rebound damping for the adjustable shock absorbers **18** to zero or substantially zero.

FIG. **15** shows an example of a graph **525** indicating compression and rebound damping characteristics for a vehicle, such as vehicle **10**. The graph **525** is merely one example of a possible compression and rebound damping characteristics set by the suspension controller **86** over different periods of time. Graph **525** shows three periods, a pre-takeoff (e.g., pre-airborne) period **530**, a free-fall period **535**, and a post-landing period **540**. Initially, during the pre-takeoff period, the rebound damping **550** and the compression damping **555** for the adjustable shock absorbers **18** are steady. When an airborne event occurs, the suspension controller **86** gradually increases the compression damping **55** over the free-fall period **535** and decreases the rebound damping **550** to substantially zero. The post-landing period **540** will be discussed below.

Referring back to FIG. **14**, at step **510**, the suspension controller **86** may determine whether a landing event has occurred. For example, similar to step **506**, the suspension controller **86** may determine whether a landing event is occurring based on the magnitude of the x-axis acceleration value, the y-axis acceleration value, and/or the z-axis acceleration value. For instance, the suspension controller **86** may compare the x, y, and/or z-axis acceleration values with one or more thresholds (e.g., one or more pre-defined, pre-programmed, and/or user-defined thresholds) to determine whether the vehicle **10** has landed. If the x, y, and/or z-axis acceleration values are greater than the one or more thresholds, the method **500** may move to step **508**. If the x, y, and/or z-axis acceleration values are less than the thresholds, then step **510** may repeat.

In some examples, the suspension controller **86** may use the same thresholds as in step **506**. For example, as mentioned above, in free-fall, the chassis accelerometer **30** and/or IMU **37** may measure the vehicle **10** to have zero or substantially zero acceleration for the x and y-axis, and 1 G or substantially 1 G on the z-axis. As such, the suspension controller **86** may determine whether a landing event is occurring based on determining whether the acceleration values for the x and/or y-axis is greater than the first threshold (e.g., 0.3 Gs) and/or whether the acceleration values for the z-axis is greater than a second threshold. In some instances, the suspension controller **86** may combine (e.g., calculate) a magnitude of acceleration for the x, y, and z-axis acceleration, and compare the combined magnitude with a threshold to determine whether the vehicle **10** has landed.

At step **512**, the suspension controller **86** may determine a duration for the airborne event (e.g., a duration that the vehicle **10** is in free-fall). For example, the suspension controller **86** may include a timer, and may use the timer to determine the duration between detecting the airborne event at step **506** and detecting the landing event at step **510**.

At step **514**, the suspension controller **86** may execute a landing condition modifier for one or more of the adjustable shock absorbers **18**. For example, the suspension controller **86** may provide information (e.g., one or more commands) to the adjustable shock absorbers **18** to adjust the compression damping and/or the rebound damping to a post-landing set-point for a period of time. For instance, the suspension controller **86** may increase the rebound damping for a period of time. Also, the suspension controller **86** may maintain the compression damping at the post-landing setting for a period of time. By maintaining the compression damping, the vehicle **10** may be easier able to get through the landing compression stroke of the shock.

At step **516**, the suspension controller **86** may execute a normal condition modifier. For example, the suspension controller **86** may adjust the compression damping and/or rebound damping back to normal (e.g., back to the pre-takeoff position). For example, the suspension controller may decrease and/or gradually decrease the rebound damping back to the normal (e.g., the pre-takeoff rebound damping setting). Further, the suspension controller **86** may decrease the compression damping back to normal (e.g., the pre-takeoff compression damping setting).

In some variations, the suspension controller **86** may adjust the post-landing compression damping settings and/or rebound damping settings based on the duration of the airborne event. Referring back to FIG. **15**, the compression damping **555** is gradually increasing, and increase as the duration of the airborne event increases. As such, during the post-landing period **540**, the suspension controller **86** may maintain an increased compression damping **555** based on the airborne event. Additionally, and/or alternatively, the suspension controller **86** may change the duration to maintain the increased compression damping **555** based on the duration of the airborne event. For example, the vehicle **10** may encounter a small bump (e.g., the airborne event is less than three hundred milliseconds). In such instances, the suspension controller **86** may compare the duration of the airborne event with a threshold (e.g., pre-programmed, pre-defined, and/or user-defined). If the duration of the airborne event is less than the threshold, the suspension controller **86** might not maintain the increased compression damping **555**. Instead, the suspension controller **86** may reduce and/or gradually reduce the compression damping back to the pre-takeoff **530** compression damping **555** in response to detecting the landing event.

In some examples, if the duration of the airborne event is greater than the threshold, the suspension controller **86** may maintain the increased compression damping **555** for a period of time. In some instances, the suspension controller **86** may determine the period of time to maintain the increased compression damping **555** based on the duration of the airborne event (e.g., the longer the duration for the airborne event, the longer the period of time to maintain the increased compression damping **555**).

In response to detecting the landing event (e.g., the start of the post-landing period **540**), the suspension controller **86** may increase the rebound damping **550** based on the duration of the airborne event. For example, traditionally, the vehicle **10** tends to unload the tires (e.g., wheels are light) and/or perform a hop upon landing. To prevent the hop or unloading of the tires, the suspension controller **86** increases the rebound damping **550** for the adjustable shock absorbers **18** based on the duration of the airborne event (e.g., as the duration of the airborne event increases, the rebound damping **550** after landing increases). After a set period of time (e.g., after the vehicle **10** has completed its first shock compression and rebound cycle), the suspension controller **86** decreases and/or gradually decreases the rebound damping **550** back to the pre-takeoff period's **530** rebound damping setting. By increasing the rebound damping **550**, the suspension controller **86** reduces the hop/unloading of the vehicle **10** causing increases in the stability and/or traction on the landing events.

In some examples, the vehicle **10** may encounter a small bump or hill, the vehicle **10** might not perform a hop upon landing or performs a small hop upon landing. In such examples, the suspension controller **86** may adjust the compression and/or rebound damping characteristics based on comparing the duration of the airborne event with a threshold. For example, the suspension controller **86** might not increase the rebound damping **550**, and instead set the rebound damping **550** to the pre-takeoff period's **530** rebound damping setting in response to detecting the landing event.

Additionally, and/or alternatively, based on the duration of the airborne event, the suspension controller **86** may bias the compression damping and/or rebound damping for the front and/or rear adjustable shock absorbers **18**. For example, based on detecting the duration of the airborne event is below a threshold (e.g., the vehicle **10** is encountering a small bump or hill), the suspension controller **86** may additionally increase the rebound damping for the post-landing rebound value for the rear adjustable shock absorbers **18***c* and **18***d*. Biasing the front/rear shock absorbers **18** in response to detecting the airborne event may increase stability of the vehicle **10** and/or assist the vehicle **10** when traveling through rough terrain.

Rock Crawler Operation

FIG. **16** shows an example flowchart describing the operation of the suspension controller **86** during a rock crawler operation. FIGS. **17-20** show examples of the suspension controller **86** adjusting the adjustable shock absorbers **18** for the vehicle **10** during the rock crawler operation.

By adjusting the compression damping and/or rebound damping of one or more adjustable shock absorbers **18** during the rock crawl operation, the suspension controller **86** may increase vehicle **10** stability when encountering rocks. For example, based on received information (e.g., longitudinal acceleration, lateral acceleration vehicle speed, engine speed, and/or roll angle), the suspension controller **86** may determine the orientation of the vehicle **10** (e.g., whether the vehicle **10** is on flat ground, facing uphill, facing downhill, passenger side facing downhill, and/or driver side facing downhill). Based on the orientation of the vehicle **10**, the suspension controller **86** may adjust the compression and/or rebound damping to lean the vehicle **10** into the hill, and/or rock, causing the vehicle **10** to reduce the overall pitch/roll angle when traversing an obstacle. Further, by adjusting the compression and/or rebound damping when the vehicle **10** is on flat ground, the vehicle **10** has increased ground clearance.

In operation, at step **602**, the suspension controller **86** may receive information (e.g., inputs) from one or more entities (e.g., sensors, devices, and/or subsystems) of vehicle **10**. For example, the suspension controller **86** may receive (e.g., retrieve and/or obtain) information (e.g., data packets and/or signals indicating sensor readings) from the one or more sensors, devices, and/or subsystems. In some instances, the suspension controller **86** may receive longitudinal and/or lateral acceleration rates from a sensor, such as the IMU **37** and/or the chassis accelerometer **30**. In some examples, the suspension controller **86** may receive pitch rate, roll rate, pitch angles, roll angles, and/or yaw rates from a sensor, such as the gyroscope **31**. In some instances, the suspension controller **86** may receive a vehicle speed from a sensor, such as the vehicle speed sensor **26**.

In some examples, the suspension controller **86** may receive information indicating a mode of operation for the vehicle **10**. For example, the suspension controller **86** may receive information indicating the mode of operation from an operator interface **22** and/or another controller (e.g., controller **20**). For instance, the operator interface **22** may receive user input indicating a selection of the rock crawler mode. The operator interface **22** may provide the user input to the suspension controller **86**. The suspension controller **86** may be configured to operate in a rock crawler mode based on the user input.

At step **604**, the suspension controller **86** may determine whether the vehicle **10** is in a rock crawler mode. For example, based on information indicating the mode of operation, the suspension controller **86** may determine whether the vehicle **10** is in the rock crawler mode. If the vehicle **10** is in the rock crawling mode, the method **600** may move to step **606**. If not, the method **600** may move back to step **602**.

At step **606**, the suspension controller **86** may determine whether the vehicle speed is below a threshold. For example, the suspension controller **86** may compare the vehicle speed with a threshold. If the vehicle speed is less than the threshold, then the method **600** may move to step **608**. If not, then the method **600** may move back to step **602**. In other words, the suspension controller **86** may operate in the rock crawler mode and provide rock crawler condition modifiers at low vehicle speeds. At higher vehicle speeds, the suspension controller **86** might not operate in the rocker crawler mode. Instead, at higher vehicle speeds, the suspension controller **86** may operate in a different mode of operation, such as a normal or comfort mode. In some instances, the suspension controller **86** may use the engine speed rather than the vehicle speed to determine whether to operate in the rock crawler mode and/or provide rock crawler condition modifiers.

At step **608**, the suspension controller **86** may determine an orientation of the vehicle **10**. For example, based on the magnitude of the x-axis acceleration value, the y-axis acceleration value, and/or the z-axis acceleration value, the suspension controller **86** may determine a longitudinal acceleration and/or a lateral acceleration of the vehicle **10**. Using the longitudinal and/or lateral acceleration, the suspension controller **86** may determine a roll and/or pitch angle of the vehicle **10**. Then, based on the roll and/or pitch angle of the vehicle, the suspension controller **86** may determine whether the vehicle **10** is on a flat surface, the front of the vehicle **10** facing uphill, the front of the vehicle **10** is facing downhill, the passenger side of the vehicle **10** is facing downhill, and/or the driver side of the vehicle **10** is facing downhill. Additionally, and/or alternatively, the suspension controller **86** may determine the orientation, such as whether the passenger side of the vehicle **10** is facing downhill and/or the driver side of the vehicle **10** is facing downhill, based on the roll rates from the gyroscope **31** and/or the IMU **37**.

At step **610**, the suspension controller **86** may execute a rock crawler condition modifier. For example, based on the orientation of the vehicle **10**, the suspension controller **86** may provide information (e.g., one or more commands) to adjust the compression damping and/or the rebound damping for one or more of the adjustable shock absorbers **18**. In some examples, the suspension controller **86** may increase the compression damping for the downhill adjustable shock absorbers **18**. Additionally, and/or alternatively, the suspension controller **86** may increase the rebound damping and/or decrease the compression damping for the uphill adjustable shock absorbers **18**.

FIGS. **17**-**20** show examples of the rebound and compression damping characteristics of the adjustable shock absorbers **18** using method **600**. FIG. **17** shows the compression damping and/or rebound damping characteristics of the vehicle **10** on flat ground. For example, at step **608**, the suspension controller **86** may determine the vehicle **10** is on flat ground (e.g., based on the longitudinal acceleration value and/or the lateral acceleration value. Based on the orientation of the vehicle **10**, the suspension controller **86** may provide information (e.g., one or more commands) to adjust the damping characteristics of the vehicle **10**. For instance, the suspension controller **86** may provide information **302**, **304**, **306**, **308** to increase the compression damping and to decrease the rebound damping for the adjustable shock absorber **18***a*, **18***b*, **18***c*, and **18***d*. By adjusting the compression damping and rebound damping, the suspension controller **86** may maximize the ground clearance for obstacle avoidance, may avoid full stiff compression when on flat ground to remove unnecessary harshness, and may allow the ground engaging members **12** to fall into holes in the ground more quickly, thus not upsetting the vehicle **10**.

FIG. **18** shows the compression damping and/or rebound damping characteristics of the vehicle **10** when the front of the vehicle **10** is facing downhill. For example, at step **608**, the suspension controller **86** may determine the vehicle **10** is facing downhill (e.g., based on the longitudinal acceleration value and/or lateral acceleration value). Based on the orientation of the vehicle **10**, the suspension controller **86** may provide information (e.g., one or more commands) to adjust the damping characteristics of the vehicle **10**. For instance, the suspension controller **86** may provide information **302** and **304** to increase the compression damping for the adjustable shock absorbers **18***a* and **18***b* (e.g., the downhill facing shock absorbers). Further, the suspension controller **86** may provide information **306** and **308** to increase the rebound damping and/or decrease the compression damping for the adjustable shock absorber **18***c* and **18***d* (e.g., the uphill facing shock absorbers).

FIG. **19** shows the compression damping and/or rebound damping characteristics of the vehicle **10** when the front of the vehicle **10** is facing uphill. For example, at step **608**, the suspension controller **86** may determine the vehicle **10** is facing uphill (e.g., based on the longitudinal acceleration value and/or lateral acceleration value). Based on the orientation of the vehicle **10**, the suspension controller **86** may provide information (e.g., one or more commands) to adjust the damping characteristics of the vehicle **10**. For instance, the suspension controller **86** may provide information **302** and **304** to increase the rebound damping and/or decrease the compression damping for the adjustable shock absorbers **18***a* and **18***b* (e.g., the uphill facing shock absorbers). Further, the suspension controller **86** may provide information **306** and **308** to increase the compression damping for the adjustable shock absorber **18***c* and **18***d* (e.g., the downhill facing shock absorbers).

FIG. **20** shows the compression damping and/or rebound damping characteristics of the vehicle **10** when the passenger side of the vehicle **10** is facing downhill. For example, at step **608**, the suspension controller **86** may determine the passenger side of the vehicle **10** is facing downhill (e.g., based on the lateral acceleration value and/or longitudinal acceleration value). Based on the orientation of the vehicle **10**, the suspension controller **86** may provide information (e.g., one or more commands) to adjust the damping characteristics of the vehicle **10**. For instance, the suspension controller **86** may provide information **304** and **308** to increase the compression damping for the adjustable shock absorbers **18***b* and **18***d* (e.g., the downhill facing shock absorbers). Further, the suspension controller **86** may provide information **302** and **306** to increase the rebound damping and/or decrease the compression damping for the adjustable shock absorber **18***a* and **18***c* (e.g., the uphill facing shock absorbers).

In some variations, a portion of the vehicle **10**, such as the passenger side of the front of the vehicle **10**, is facing downhill. In other words, the vehicle **10** may be angled so one wheel (e.g., the front right ground engaging member **12***b*) is facing downhill and one wheel (e.g., the rear left ground engaging member **12***c*) is facing uphill. In such instances, at step **608**, the suspension controller **86** may determine a pitch and roll angle based on the longitudinal and/or lateral acceleration values. Based on the pitch and roll angles, the suspension controller **86** may determine the orientation of the vehicle (e.g., that the front right ground engaging member **12***b* is facing). Based on the orientation of the vehicle **10**, the suspension controller **86** may provide information **304** to increase the compression damping for the adjustable shock absorbers **18***b* (e.g., the downhill facing shock absorber). Further, the suspension controller **86** may provide information **306** to increase the rebound damping and/or decrease the compression damping for the adjustable shock absorber **18***c* (e.g., the uphill facing shock absorber). Additionally, and/or alternatively, based on the angle of the orientation, the suspension controller **86** may provide information **302** to maintain and/or decrease the rebound damping and/or maintain and/or increase the compression damping for the adjustable shock absorber **18***a* (e.g., a shock absorber that is neither uphill or downhill). Additionally, and/or alternatively, based on the angle of the orientation, the suspension controller **86** may provide information **308** to decrease the rebound damping and/or maintain the compression damping for the adjustable shock absorber **18***d* (e.g., a shock absorber that is neither uphill or downhill).

In some instances, when the vehicle **10** is in the rock crawler mode, the suspension controller **86** may increase, decrease, and/or maintain the compression and/or rebound damping as shown in FIG. **17**. Further, in response to determining the orientation of the vehicle **10** (e.g., step **608**), the suspension controller **86** may additionally increase and/or decrease the compression and/or rebound damping as shown in FIGS. **18-20**. In other words, in such instances, the increasing, decreasing, and/or maintain of the compression/rebound damping as shown in FIGS. **18-20** are based off of the compression/rebound damping from FIG. **17**.

Hill (Dune) Sliding Operation

FIG. **21** shows an example flowchart describing the operation of the suspension controller **86** during a hill sliding event. FIGS. **22** and **23** show examples of the suspension controller **86** adjusting the adjustable shock absorbers **18** for the vehicle **10** during the hill sliding event. By adjusting the damping during an front uphill sliding event, the vehicle **10** may lean better into the hill or dune, hold onto a steeper slope better, and/or provide a smoother ride (e.g., less chop) to the user when traversing the hill or dune. By adjusting the damping during a front downhill sliding event, the vehicle **10** may lean better into the hill or dune, increase vehicle stability, and/or make it easier to traverse back uphill. A hill sliding event is any event where the vehicle **10** is traversing a slope (e.g., hill, dune, ramp) and begins to slide in at least one direction.

In operation, at step **652**, the suspension controller **86** may receive steering information from one or more sensors, such as the steering sensor **28**. The steering information may indicate a steering position (e.g., steering angle) of a steering wheel, such as steering wheel **226**. The steering position and/or angle may indicate a position and/or an angle of the steering wheel for the vehicle **10**. The steering rate may indicate a change of the position and/or angle of the steering wheel over a period of time.

At step **654**, the suspension controller **86** may receive yaw rate information from one or more sensors, such as the gyroscope **31** and/or the IMU **37**. The yaw information indicates the yaw rate of the vehicle **10**.

At step **656**, the suspension controller **86** may receive acceleration information indicating an acceleration rate or deceleration rate of the vehicle **10** from one or more sensors, such as the IMU **37** and/or the chassis accelerometer **30**. The acceleration information may indicate multi-axis acceleration values of the vehicle, such as a longitudinal acceleration and/or a lateral acceleration. In some examples, the suspension controller **86** may receive information from another sensor, such as the throttle position sensor **34** and/or the accelerator pedal sensor **33**. The suspension controller **86** use the information to determine the acceleration rate. For example, the suspension controller **86** may use the throttle position from the throttle position sensor and/or the position of the accelerator pedal **230** from the acceleration pedal sensor to determine whether the vehicle **10** is accelerating and/or decelerating.

At step **658**, the suspension controller **86** may receive an operating mode of the vehicle **10** from the operator interface **22** and/or one or more other controllers (e.g., controller **20**). Each mode may include a different setting for the rebound and/or compression damping. An operator (e.g., user) may input a mode of operation on the operator interface **22**. The operator interface **22** may provide the user input indicating the mode of operation to the controller **20** and/or the suspension controller **86**. The suspension controller **86** may use the user input to determine the mode of operation for the vehicle **10**.

At step **660**, the suspension controller **86** may determine whether a hill sliding event (e.g., whether the vehicle **10** is sliding while traversing a slope) is occurring. For example, the suspension controller **86** may determine the hill sliding event based on the steering information indicating a steering rate, angle, and/or position, yaw rate information indicating a yaw rate, the acceleration information indicating a lateral and/or longitudinal acceleration, and/or other information (e.g., a pitch angle or rate). The suspension controller **86** may compare the steering rate, steering angle, steering position, yaw rate, lateral acceleration, and/or longitudinal acceleration with one or more thresholds (e.g., pre-determined, pre-programmed, and/or user-defined) to determine the hill sliding event. If the suspension controller **86** determines the vehicle **10** is in a hill sliding event, the method **650** may move to step **662**. If not, the method may move to step **652**.

In some instances, the detection of the hill sliding event may be similar to the detection of the cornering event. Further, the commands to adjust the damping of the shock absorbers **18** based on the detection of the hill sliding event may be similar to the cornering event. For example, the suspension controller **86** may prioritize the steering information, the yaw rate information, and/or the acceleration information to determine the hill sliding event. In other words, the steering position and/or angle, steering rate, yaw rate, and/or lateral acceleration may conflict, and the suspension controller **86** may prioritize the lateral acceleration over the steering position, angle, rate and/or the yaw rate. For example, the suspension controller **86** may determine the hill sliding event is occurring based on the lateral acceleration value exceeding (e.g., above or below) a lateral acceleration threshold even if the yaw rate, steering rate, angle, and/or position do not exceed their corresponding thresholds.

In some examples, the suspension controller **86** may use a vehicle speed and/or an engine speed to determine whether the vehicle **10** is in a hill sliding event. For example, the suspension controller **86** may compare the vehicle speed and/or the engine speed with a threshold. If the suspension controller **86** determines the vehicle speed and/or engine speed is greater than a threshold (e.g., the vehicle **10** is traveling at a high vehicle speed), then the method **650** may move to **662**. Otherwise, the method **650** may move back to **652**. In other words, the suspension controller **86** may execute the hill sliding condition modifier when the vehicle is moving at a high vehicle speed.

At step **662**, the suspension controller **86** may determine a direction of the slide. For example, based on the lateral acceleration, the suspension controller **86** may determine whether the vehicle **10** is hill sliding to the left side or the right side.

At step **664**, the suspension controller **86** may determine an orientation of the vehicle **10**. For example, similar to step **608**, based on the longitudinal and/or lateral acceleration, the suspension controller **86** may determine whether the front of the vehicle **10** is facing uphill, the front of the vehicle **10** is facing downhill, the passenger side of the vehicle **10** is facing downhill, and/or the driver side of the vehicle **10** is facing downhill.

At step **666**, the suspension controller **86** may execute a hill sliding condition modifier. For example, based on the direction of the slide and/or the orientation of the vehicle **10**, the suspension controller **86** may provide information (e.g., one or more commands) to adjust the compression damping and/or the rebound damping for the one or more shock absorbers **18**. For example, based on the direction of the slide, the suspension controller **86** may increase the compression damping and/or decrease the rebound damping on the leading adjustable shock absorbers **18**. Additionally, and/or alternatively, the suspension controller **86** may decrease the compression damping and/or increase the rebound damping on the trailing adjustable shock absorbers **18**. Also, based on the orientation (e.g., the front of the vehicle **10** facing uphill and/or downhill), the suspension controller **86** may further bias the compression damping and/or rebound damping for the front and rear shock absorbers **18**. In other words, the suspension controller **86** may additionally increase the compression damping on the downhill adjustable shock absorbers **18**. Additionally, and/or alternatively, the suspension controller **86** may additionally decrease the compression damping and/or additionally increase the rebound damping of the uphill shock absorbers **18**. Afterwards, the method **650** may move back to step **652**. FIGS. **22** and **23** will describe the hill condition modifier in further detail.

FIG. **22** shows the vehicle **10** traveling uphill and in a left pointed slide. For example, the suspension controller **86** may determine that the vehicle **10** is traveling uphill **670** and sliding left **672** based on the lateral and/or longitudinal acceleration. The suspension controller **86** may adjust the compression and/or rebound damping characteristics for a left pointed slide **672** similar to a right turn during a cornering event. In other words, the suspension controller **86** may increase the compression damping and/or decrease the rebound damping for the leading shock absorbers **18***a* and **18***c* (e.g., the outside shock absorbers). Further, the suspension controller **86** may decrease the compression damping and/or increase the rebound damping for the leading shock absorbers **18***b* and **18***d* (e.g., the inside shock absorbers). For a right pointed slide, the suspension controller **86** may reverse the compression damping and/or rebound damping for the shock absorbers (e.g., decrease the compression damping and/or increase the rebound damping for the shock absorbers **18***a* and **18***c*; increase the compression damping and/or decrease the rebound damping for shock absorbers **18***b* and **18***d*).

Additionally, and/or alternatively, the suspension controller **86** may bias the front and/or rear adjustable shock absorbers **18** for the uphill orientation **670** similar to detecting an acceleration during a cornering event. For example, based on the uphill orientation of the vehicle **10**, the suspension controller **86** may additionally increase (e.g., by a value or percentage) the compression damping and/or additionally decrease (e.g., by a value or a percentage) the rebound damping of rear adjustable shock absorbers **18***c* and **18***d* (e.g., the downhill shock absorbers). Further, the suspension controller **86** may additionally decrease the compression damping and/or additionally increase the rebound damping of the front adjustable shock absorbers **18***a* and **18***b* (e.g., the uphill shock absorbers).

After determining the slide **672** and the orientation **670** of the vehicle, the suspension controller **86** may provide information (e.g., one or more commands) to adjust the shock absorbers **18**. For example, in an uphill **670** and left pointed slide **672**, the suspension controller **86** may provide information **302** to maintain or decrease the rebound damping and/or maintain or increase the compression damping for the adjustable shock absorber **18***a*. Further, the suspension controller **86** may provide information **304** to increase the rebound damping and/or decrease the compression damping for the adjustable shock absorber **18***b*. Also, the suspension controller **86** may provide information **306** to increase the compression damping and/or to decrease the rebound damping for the adjustable shock absorber **18***c*. Additionally, the suspension controller **86** may provide information **308** to maintain or decrease the compression damping and/or increase the rebound damping for the adjustable shock absorber **18***d*.

FIG. **23** shows the vehicle **10** traveling downhill and in a right pointed slide. For example, the suspension controller **86** may determine that the vehicle **10** is traveling downhill **674** and sliding right **676** based on the lateral and/or longitudinal acceleration. The suspension controller **86** may adjust the compression and/or rebound damping characteristics for a right pointed slide **676** similar to a left turn during a cornering event. In other words, the suspension controller **86** may increase the compression damping and/or decrease the rebound damping for the leading shock absorbers **18***b* and **18***d* (e.g., the outside shock absorbers). Further, the suspension controller **86** may decrease the compression damping and/or increase the rebound damping for the trailing shock absorbers **18***a* and **18***c* (e.g., the inside shock absorbers). The left pointed slide **672** is described above.

Additionally, and/or alternatively, the suspension controller **86** may bias the front and/or rear adjustable shock absorbers **18** for the downhill orientation **674** similar to detecting a deceleration (e.g., braking) during a cornering event. For example, based on the downhill orientation of the vehicle **10**, the suspension controller **86** may additionally increase (e.g., by a value or percentage) the compression damping and/or additionally decrease (e.g., by a value or a percentage) the rebound damping of front adjustable shock absorbers **18***a* and **18***b* (e.g., the downhill shock absorbers). Further, the suspension controller **86** may additionally decrease the compression damping and/or additionally increase the rebound damping of the rear adjustable shock absorbers **18***c* and **18***d* (e.g., the uphill shock absorbers).

After determining the slide and the orientation of the vehicle **10**, the suspension controller **86** may provide information (e.g., one or more commands) to adjust the shock absorbers **18**. For example, in a downhill orientation **674** and right pointed slide **676**, the suspension controller **86** may provide information **302** to decrease or maintain the compression damping and/or increase or maintain the rebound damping for the adjustable shock absorber **18***a*. Further, the suspension controller **86** may provide information **304** to increase the compression damping and/or decrease the rebound damping for the adjustable shock absorber **18***b*. Also, the suspension controller **86** may provide information **306** to decrease the compression damping and/or to increase the rebound damping for the adjustable shock absorber **18***c*. Additionally, the suspension controller **86** may provide information **308** to maintain or increase the compression damping and/or decrease or maintain the rebound damping for the adjustable shock absorber **18***d*.

Realtime Correction of Inertial measurement at a Center of Gravity of a Vehicle

FIG. **24** shows an exemplary flowchart illustrating a method for performing real-time correction of the inertial measurement of a vehicle. For example, the IMU **37**, the gyroscope **31**, and/or the chassis accelerometer **30** may measure the inertial measurements of the vehicle **10** (e.g., the acceleration and/or angular velocity or rate of the vehicle **10**). In some examples, these sensors might not be at the center of gravity (CG) of the vehicle **10**, and a kinematic transformation may be used to account for the sensors not being at the CG. For example, a controller, such as the suspension controller **86**, may use a distance between a sensor (e.g., the IMU **37**, the gyroscope **31**, and/or the chassis accelerometer **30**) and the CG of the vehicle **10** to determine offset correction information. Using the offset correction information, the suspension controller **86** may transform (e.g., determine and/or calculate) the measured acceleration values to CG acceleration values (e.g., an estimated CG acceleration). Additionally, and/or alternatively, the CG of the vehicle **10** may also be affected by the number of users and/or the cargo. For example, a driver and a passenger may change the CG of the vehicle **10**. As such, the suspension controller **86** may receive user and/or cargo information, and use the user and/or cargo information to determine second offset correction information. The suspension controller **86** may use the first and/or second offset corrections to determine the CG acceleration values.

By providing real-time correction of the inertial measurement of the vehicle **10**, the suspension controller **86** may allow for the IMU **37** and/or other sensors to reside at a location other than the CG of the vehicle **10** while still providing an accurate vehicle inertial measurement estimation for controls. Additionally, and/or alternatively, the suspension controller **86** may allow for a more accurate estimation given the additional information regarding the weight (e.g., riders) and/or location of the weight (e.g., location of the riders) of the vehicle **10**. FIG. **24** will be described below with reference to FIG. **25**.

FIG. **25** shows a schematic block diagram illustrating an example of logic components in the suspension controller **86**. For example, the suspension controller **86** includes one or more logic components, such as an acceleration and/or velocity measurement logic **720**, angular acceleration logic **722**, angular acceleration calculation logic **724**, kinematics function logic **726**, estimated acceleration at center of gravity (CG) logic **728**, offset correction based on user or cargo information logic **730**, and/or offset correction based on sensor information logic **732**. The logic **720**, **722**, **724**, **726**, **728**, **730**, and/or **732** is any suitable logic configuration including, but not limited to, one or more state machines, processors that execute kernels, and/or other suitable structure as desired.

Further, in some examples, the suspension controller **86** and/or the controller **20** may include memory and one or more processors. The memory may store computer-executable instructions that when executed by the one or more processors cause the processors to implement the method and procedures discussed herein, including the method **700**. Additionally, various components (e.g., logic) depicted in FIG. **25** are, in embodiments, integrated with various ones of the other components depicted therein (and/or components not illustrated), all of which are considered to be within the ambit of the present disclosure.

In operation, at step **702**, the suspension controller **86** (e.g., logic **720**) may receive vehicle movement information from one or more entities (e.g., sensors, devices, and/or subsystems) of vehicle **10**. For example, the suspension controller **86** may receive vehicle movement information indicating the acceleration and/or velocity of the vehicle **10** from one or more sensors. For instance, the logic **720** may receive x, y, and/or z acceleration values (e.g., linear acceleration values) from a sensor, such as the IMU **37** and/or the chassis accelerometer **30**. The logic **720** may also receive x, y, and/or z angular velocity values from a sensor, such as the IMU **37** and/or the gyroscope **31**.

At step **704**, the suspension controller **86** (e.g., logic **730**) may receive user or cargo information from one or more entities (e.g., sensors, devices, and/or subsystems) of vehicle **10**. In some instances, the suspension controller **86** may receive user or cargo information from the operator interface **22**. For example, the operator interface **22** may provide, to the suspension controller **86**, user information indicating a number of riders in the vehicle **10** and/or a weight or mass corresponding to the riders in the vehicle **10**. Additionally, and/or alternatively, the operator interface **22** may provide, to the suspension controller **86**, cargo information indicating a weight or mass of cargo that the vehicle **10** is carrying.

In some examples, the suspension controller **86** may receive the user or cargo information from one or more sensors. For example, one or more sensors, such as seat belt sensors, may provide, to the suspension controller **86**, information indicating the number of riders in the vehicle **10** (e.g., when a user puts on their seat belt, the seat belt sensor may provide information indicating the seat belt is engaged to the suspension controller **86**).

At step **706**, the suspension controller **86** (e.g., the logic **722** and/or **724**) may determine the angular acceleration for the vehicle **10**. For example, the logic **722** may filter the angular velocity and/or rates received from a sensor, such as the IMU **37** and/or gyroscope **31**. The logic **724** may differentiate the angular rates to determine the angular acceleration.

At step **708**, the suspension controller **86** (e.g., the logic **732**) may determine a first offset correction based on a distance between a sensor (e.g., the IMU **37**, the chassis accelerometer **30**, and/or the gyroscope **31**) and the CG of the vehicle **10**. The CG of the vehicle **10** may be pre-programmed and/or pre-defined. Additionally, and/or alternatively, the suspension controller **86** (e.g., the logic **732**) may determine the first offset correction based on a distance between a sensor (e.g., the IMU **37**, the chassis accelerometer **30**, and/or the gyroscope **31**) and the suspension controller **86**.

At step **710**, the suspension controller **86** (e.g., the logic **730**) may determine a second offset correction based on the user and/or cargo information received at step **704**. For example, based on the number of users, weight/mass of users, and/or weight/mass of the cargo, the logic **730** may determine a second offset correction.

At step **712**, the suspension controller **86** (e.g., the logic **726** and/or **728**) may determine an estimated acceleration at the CG of the vehicle **10**. For example, the logic **726** may receive information from the logic **720** (e.g., the linear acceleration values), the logic **724** (e.g., the determined angular acceleration values), the logic **730** (e.g., the second offset based on the user or cargo information), and/or the logic **732** (e.g., the first offset based on the sensor information). The logic **726** may use a kinematics function to determine the 3-axis estimated acceleration at the CG of the vehicle **10**. The logic **726** may provide the 3-axis estimated acceleration at the CG of the vehicle **10** to the logic **728**. Then, the method **700** may return to step **702**, and may repeat continuously.

In some examples, the logic **726** may receive and/or store the 3-axis (e.g., x, y, and/or z-axis) estimated acceleration values at the CG of the vehicle **10**. Additionally, and/or alternatively, the methods **400**, **500**, and/or **700** may use the determined estimated acceleration values at the CG of the vehicle **10** to execute corner condition modifiers (e.g., step **418**), brake condition modifiers (e.g., step **412**), landing condition modifiers (e.g., step **415**), and/or sand dune condition modifiers (e.g., step **608**). For example, the suspension controller **86** may use the estimated acceleration values at the CG of the vehicle **10** to determine and/or detect events (e.g., cornering events, braking events, landing events, airborne events). Additionally, and/or alternatively, the suspension controller **86** may use the estimated acceleration values at the CG of the vehicle **10** to adjust the compression damping and/or rebound damping of the adjustable shock absorbers **18**.

Changing Drive Modes, Active Vehicle Events, and Suspension Controller Architecture

FIG. **26** shows an exemplary flowchart describing the operation of the suspension controller **86** when switching between driver modes. For example, an operator may use the operator interface **22** (e.g., a touch screen) and/or a physical switch to adjust the driving modes. For instance, each driving mode may indicate a different compression and/or rebound damping setting. As will be explained below, the suspension controller **86** may receive one or more user inputs, and based on the user inputs may change the driving modes. In response to changing the driving modes, the suspension controller **86** may provide one or more commands to adjust the compression and/or rebound damping for the adjustable shock absorbers **18**. FIG. **26** will be described with reference to FIGS. **27** and **28** below.

FIG. **27** shows an exemplary physical switch **820** for adjusting driving modes. In some examples, the physical switch **820** may be located on a steering wheel, such as the steering wheel **226**. The physical switch **820** may be operatively coupled to and/or communicate with the suspension controller **86**. For example, the suspension controller **86** may receive user input from the physical switch **820**. The switch **820** may include an increase mode (+) button **822** (e.g., a first mode changing input device), a decrease mode (−) button **824** (e.g., a second mode changing input device), and/or an instant compression switch **826**. When actuated and/or pressed by a user, the instant compression switch **826** may provide information to the suspension controller **86**. Based on the information from the instant compression switch **826**, the suspension controller **86** may provide momentary stiffness to the adjustable shock absorbers **18**. In other words, the suspension controller **86** may increase and/or significantly increase the compression damping of the adjustable shock absorbers **18** for a brief time duration.

FIG. **28** shows an exemplary graphical user interface **830**. The graphical user interface **830** may be displayed on an interface, such as the operator interface **22**. For example, the suspension controller **86** may cause display of the graphical user interface **830** on the operator interface **22**. The graphical user interface **830** may include one or more modes, such as a first driver mode **832**, a second driver mode **834**, a third driver mode **836**, and/or a fourth driver mode **838**. The graphical user interface **830** is merely exemplary, and the suspension controller **86** may include multiple driver modes, including more or less than four driver modes. As will be explained below, the suspension controller **86** may receive multiple user inputs, and may adjust the compression and/or rebound damping characteristics based on the number of received user inputs. Exemplarily driving modes may include, but are not limited to, a comfort mode, a rough trial mode, a handling mode, and/or a rock crawl mode.

In operation, at step **802**, the suspension controller **86** may determine whether it has received a first user input from the physical switch **820**. For example, a user may press (e.g., actuate) the increase mode (+) button **822** and/or the decrease mode (−) button **824**. The physical switch **820** may provide information indicating the actuation of the increase mode (+) button **822** and/or the decrease mode (−) button **824**. If the suspension controller **86** has received a first user input, then the method **800** may move to step **804**. If not, then the method **800** may remain at step **802**.

At step **804**, the suspension controller **86** may cause display of the driving modes on an operator interface, such as operator interface **22**. For example, based on receiving the first user input from the physical switch **820**, the suspension controller **86** may cause display of the graphical user interface **830** on the operator interface **22**. Additionally, and/or alternatively, the graphical user interface **830** may indicate a current selected driver mode.

At step **806**, the suspension controller **86** may determine whether it has received a new user input. For example, initially, when the suspension controller **86** receives the first user input, the suspension controller **86** might not change the current driving mode, such as the second driver mode **834**. Instead, the suspension controller **86** may change the current driving mode if it receives two or more user inputs (e.g., the first user input and one or more new user inputs). For example, at step **802**, the suspension controller **86** may receive a first user input indicating an actuation of the increase mode (+) button **822** and/or the decrease mode (−) button **824**. At step **806**, the suspension controller **86** may receive a new user input indicating a second actuation of the increase mode (+) button **822** and/or the decrease mode (−) button **824**.

At step **808**, based on the new user input, the suspension controller **86** may change the requested driving mode. For example, after receiving the new user input (e.g., a second actuation of button **822** and/or button **824**), the suspension controller **86** may change the requested driving mode to a higher or lower driving mode. For instance, if the current driving mode is the second driver mode **834** and the suspension controller **86** receives an activation of button **822** at step **806**, the suspension controller **86** may change the requested driving mode to the third driver mode **836**. If the suspension controller **86** receives an activation of button **824**, the suspension controller **86** may change the requested driving mode to the first driver mode **832**. The method **800** may move back to step **806**, and repeat. In the next iteration, the suspension controller **86** may receive another new user input (e.g., a third actuation of the button **822** and/or button **824**). At step **808**, the suspension controller **86** may change the requested driving mode again based on whether button **822** or button **824** was actuated.

In some instances, prior to changing the driving modes and/or adjusting the adjustable shock absorbers **18**, the suspension controller **86** may determine whether an event described above, such as a cornering event, a braking event, an airborne event, and/or a landing event, is occurring. If the suspension controller **86** determines an event is occurring, the suspension controller **86** may delay and/or not change the active driving mode until the event has ended. For example, if the requested driving mode is the second driver mode **834** and an event occurs, the suspension controller **86** may delay and/or not change to the active driver mode (e.g., the second driver mode **834**) until the event ends.

Referring back to step **806**, if the suspension controller **86** does not receive a new input, the method **800** may move to step **810**. At step **810**, the suspension controller **86** may determine whether the amount of time that it has not received a new user input exceeds a timer, such as three seconds. If it has not exceeded the timer, then the method **800** may move back to **806**. If it has exceeded the timer, then the method may move to step **802**, and repeat. In other words, after a period of time (e.g., 3 seconds), the change driving mode feature may time out, and the user may need to provide another first input to begin the process again of switching driver modes.

In some examples, the suspension controller **86** may include a wall for the highest and lowest driving modes (e.g., the first driver mode **832** and fourth driver mode **838**). For example, if the current driving mode is the second driver mode **834** and the suspension controller **86** receives more than two user inputs, the suspension controller **86** may change the requested driving mode to the highest driving mode, such as the fourth driver mode **838**, and remain at the highest driving mode regardless of additional user inputs. In other words, even if a user actuates the increase mode (+) button **822** ten times, the suspension controller **86** may continue to select the highest driving mode (e.g., the further driver mode **838**).

In some examples, the controller **20** may include a steering controller **102** (e.g., a power steering controller). The steering controller **102** may include a plurality of power steering modes for the vehicle **10**. Based on the method **800**, the suspension controller **86** may determine a driving mode (e.g., a suspension mode and/or a power steering mode) as described above. The suspension controller **86** may adjust the adjustable shock absorbers **18** based on the driving mode. Additionally, and/or alternatively, the suspension controller **86** may provide the driving mode (e.g., the power steering mode) to the steering controller **102**. The steering controller **102** may change the power steering characteristics based on the received driving mode from the suspension controller **86**. In some instances, each suspension mode corresponds to one or more power steering modes. As such, by the operator selecting a suspension mode (e.g., using method **800**), the operator may also select a corresponding power steering mode. Thus, the suspension controller **86** may provide the selected driving mode to the steering controller **102**, and the steering controller **102** may implement the power steering mode characteristics.

Push and Pull Instant Compression Button Activation

The vehicle **10** may include a sensor on the steering shaft (e.g., a shaft connecting the steering wheel, such as steering wheel **226**, to the vehicle frame). The sensor, such as a strain gauge or a spring-loaded contact sensor, may detect a force (e.g., a push or pull) on the steering wheel **226**. The sensor may be operatively coupled to and/or communicate with the suspension controller **86**, and may provide information indicating the push or pull on the steering wheel **226** to the suspension controller **86**.

The suspension controller **86** may receive the information indicating the push or pull (e.g., the force exerted on the steering wheel **226**), and compare the force exerted on the steering with a threshold. Based on the comparison, the suspension controller **86** may provide one or more commands to momentarily increase the compression damping (e.g., set the compression damping to a stiff damping level) for the adjustable shock absorbers **18**. In other words, instead of using a button, such as the instant compression switch **826** shown on FIG. **27**, an operator may push or pull the steering wheel **226** to cause a momentary stiff compression damping for the adjustable shock absorbers **18**.

Egress Aid

FIG. **29** shows an exemplary flowchart describing a method for implementing egress aid for a vehicle. For example, an operator may seek to exit the vehicle, such as vehicle **10**. In such examples, the suspension controller **86** may reduce the damping (e.g., the compression damping) for the adjustable shock absorbers **18**, causing the vehicle **10** to lower in height. By lowering the vehicle height, the operator may more easily exit the vehicle **10**.

In operation, at step **902**, the suspension controller **86** may receive information (e.g., inputs) from one or more entities (e.g., sensors, devices, and/or subsystems) of vehicle **10**. For example, the suspension controller **86** may receive (e.g., retrieve and/or obtain) information (e.g., data packets and/or signals indicating sensor readings) from the one or more sensors, devices, and/or subsystems.

At step **904**, the suspension controller **86** may determine whether the operator is going to (e.g., intending to) exit the vehicle **10**. For example, the suspension controller **86** may receive, from the gear selection sensor **38**, information indicating the vehicle has shifted to park. Based on the information, the suspension controller **86** may determine the operator is intending to exit the vehicle **10** and the method **900** may move to step **906**. If not, the method **900** may move to step **902**.

In some instances, the suspension controller **86** may determine the operator is intending to exit the vehicle based on the engine speed. For example, the suspension controller **86** may receive information indicating the engine speed is zero or substantially zero. As such, the suspension controller **86** may determine the operator is intending to exit the vehicle. In some examples, the suspension controller **86** may determine the operator is intending to exit the vehicle based on information indicating a vehicle key in the off position and/or a vehicle speed (e.g., from the vehicle speed sensor **26**).

In some variations, the suspension controller **86** may receive information indicating the vehicle **10** is operating in the rock crawling mode described above. In the rock crawling mode, the vehicle speed may be low and/or substantially zero and/or may be over a rock. As such, based on the vehicle **10** operating in the rock crawling mode, the suspension controller **86** may determine to deactivate the egress aid (e.g., to not cause the vehicle **10** to become stuck), and the method **900** may move back to step **902**.

At step **906**, the suspension controller **86** may adjust the compression damping of the adjustable shock absorbers **18**. For example, the suspension controller **86** may reduce the compression damping of the adjustable shock absorbers **18**. By reducing the compression damping, the vehicle **10** may provide egress aid to the operator (e.g., based on lowering the height of the vehicle **10**).

Sway Bar

FIG. **30** illustrates a representative view of a sway bar of a vehicle, such as vehicle **10**. For example, the vehicle **10** may include a sway bar **1005**. Further, the sway bar **1005** may include one or more sway bar adjustable shock absorbers **1010**. The sway bar adjustable shock absorber **1010** may be coupled to the sway bar **1005** and a steering bar **1015**. The sway bar **1005** may be in the rear of the vehicle, and located close to a rear adjustable shock absorber, such as shock absorber **18***c*. Further, the vehicle **10** may include more than one sway bar **1005** and/or sway bar adjustable shock absorber **1010**. For example, on the other side of the vehicle **10** (e.g., the side with the adjustable shock absorber **18***d*), the vehicle **10** may include a second sway bar **1005** and/or a second sway bar adjustable shock absorber **1010**.

Also, similar to the adjustable shock absorbers **18**, the suspension controller **86** may provide one or more commands to adjust the damping characteristics of the sway bar adjustable shock absorbers **1010**. For example, referring to FIG. **11**, in response to detecting a cornering event, the suspension controller **86** may provide one or more commands to increase and/or engage the compression damping for the sway bar adjustable shock absorbers **1010**. Additionally, and/or alternatively, the suspension controller **86** may provide different damping characteristics based on the driving modes. For instance, in some driving modes, the suspension controller **86** may increase and/or engage the compression damping for the sway bar adjustable shock absorbers **1010** in response to detecting events. In other driving modes, the suspension controller **86** might not increase nor engage the compression damping for the sway bar adjustable shock absorbers **1010** in response to detecting events. For example, referring to FIG. **26**, based on changing the driving modes, the suspension controller **86** may provide one or more commands to adjust the compression and/or rebound damping for the sway bar adjustable shock absorbers **1010**.

Exemplary sway bars are described in U.S. Pat. No. 9,365,251 (filed Jun. 14, 2016, titled Side-by-side vehicle), which is assigned to the present assignee and the entire disclosure is expressly incorporated by reference herein.

FIG. **31** illustrates a view of an exemplary vehicle **1100**, such as a snowmobile. Vehicle **1100**, as illustrated, includes a plurality of ground engaging members **12**. Illustratively, the ground engaging members **12** are the endless track assembly **1108** and a pair of front skis **1112***a* and **1112***b*. The endless track assembly **1108** is operatively coupled to power system **60** (see FIG. **4**) to power the movement of vehicle **1100**. The vehicle may also include a seat **1102** and a seat surface **1104**. Also, the vehicle may include handlebars **1106**.

Further, the suspensions **1114** and **1116** are coupled to the frame of the vehicle **1100** and the pair of front skis **1112***a* and **1112***b*. The suspensions **1114** and **1116** may include adjustable shock absorbers, such as the adjustable shock absorber **1110**. Also, the endless track assembly **1108** may also be coupled to one or more suspensions and/or adjustable shock absorbers.

The vehicle **1100** may be the same and/or include components of vehicle **10**. For example, the vehicle **1100** may include the plurality of vehicle condition sensors **40** described above, and may also include one or more controllers, such as the suspension controller **86**. The suspension controller **86** may receive information from the plurality of vehicle condition sensors **40**. Using the received information, the suspension controller **86** may adjust the compression and/or rebound damping of the adjustable shock absorbers, such as adjustable shock absorber **1110** as described above. Additional details regarding vehicle **1100** are provided in U.S. Pat. Nos. 9,809,195 and 8,994,494, assigned to the present assignee, the entire disclosures of which are expressly incorporated by reference herein.

In embodiments, substantially zero is any value which is effectively zero. For example, a substantially zero value does not provide an appreciable difference in the operation compared to when the value is zero.

The above detailed description of the present disclosure and the examples described therein have been presented for the purposes of illustration and description only and not by limitation. It is therefore contemplated that the present disclosure covers any and all modifications, variations or equivalents that fall within the scope of the basic underlying principles disclosed above and claimed herein.

What is claimed is:

1. A recreational vehicle, comprising:
a plurality of ground engaging members;
a frame supported by the plurality of ground engaging members;
a plurality of suspensions, wherein each of the plurality of suspensions couples a ground engaging member, from the plurality of ground engaging members, to the frame, wherein the plurality of suspensions includes a plurality of adjustable shock absorbers;
at least one sensor positioned on the recreational vehicle and configured to provide sensor information to a controller, the at least one sensor comprises an acceleration sensor and the sensor information comprises acceleration information indicating a lateral acceleration value; and
the controller operatively coupled to the at least one sensor and the plurality of adjustable shock absorbers, wherein the controller is configured to:
receive, from the at least one sensor, the sensor information;
determine, based on a comparison of the lateral acceleration value to a first threshold, a cornering event corresponding to the recreational vehicle executing a turn; and
provide, to at least one of the plurality of adjustable shock absorbers and based on the determining of the cornering event, one or more commands to result in a decrease of a damping characteristic of the at least one of the plurality of adjustable shock absorbers.

2. The recreational vehicle of claim **1**, wherein the controller is further configured to:
determine, based on the sensor information, a direction of the turn corresponding to the cornering event;
determining, based on the direction of the turn, at least one inner adjustable shock absorber of the plurality of adjustable shock absorbers, and
wherein the controller is configured to provide the one or more commands by providing, to the at least one inner adjustable shock absorber, one or more commands to result in a decrease of a compression damping characteristic and an increase of a rebound damping characteristic.

3. The recreational vehicle of claim **2**, wherein the controller is further configured to:
determine, based on the direction of the turn, at least one outer adjustable shock absorber of the plurality of adjustable shock absorbers, and
wherein the controller is configured to provide the one or more commands by providing, to the least one outer adjustable shock absorber one or more commands to result in a decrease of a rebound damping characteristic and an increase of a compression damping characteristic.

4. The recreational vehicle of claim **1**, wherein:
the sensor information further comprises yaw rate information indicating a yaw rate; and
the controller is further configured to determine the cornering event by determining that the recreational vehicle is turning based on comparing the yaw rate to a second threshold.

5. The recreational vehicle of claim **4**, wherein:
the at least one sensor further comprises a steering sensor;
the sensor information further comprises steering information indicating a steering position or a steering rate corresponding to a steering wheel; and
the controller is configured to determine the cornering event by determining that the recreational vehicle is turning based on comparing the steering position to a third threshold.

6. The recreational vehicle of claim **1**, wherein the sensor information comprises yaw rate information indicating a yaw rate and steering information indicating a steering position or a steering rate, and wherein the controller is further configured to:
prioritize the yaw rate information over the steering information such that the controller is configured to determine the vehicle is executing the turn in a first direction based on the yaw rate indicating the turn in

the first direction and even if the steering position or the steering rate indicates the turn in a second direction or does not indicate the turn.

7. The recreational vehicle of claim **1**, wherein the sensor information comprises acceleration information indicating a lateral acceleration value and yaw rate information indicating a yaw rate, and wherein the controller is further configured to:

prioritize the acceleration information over the yaw rate information such that the controller is configured to determine the vehicle is executing the turn in a first direction based on the lateral acceleration value indicating the turn in the first direction and even if the yaw rate indicates the turn in a second direction or does not indicate the turn.

8. The recreational vehicle of claim **1**, wherein the sensor information comprises steering information indicating a steering position or a steering rate and acceleration information indicating a lateral acceleration value, and wherein the controller is further configured to:

prioritizing the acceleration information over the steering information such that the controller is configured to determine the vehicle is executing the turn in a first direction based on the lateral acceleration value indicating the turn in the first direction and even if the steering position or the steering rate indicates the turn in a second direction or does not indicate the turn.

9. The recreational vehicle of claim **1**, wherein the acceleration sensor is an inertial measurement unit (IMU).

10. A recreational vehicle comprising:

a plurality of ground engaging members;

a frame supported by the plurality of ground engaging members;

a plurality of suspensions, wherein each of the plurality of suspensions couples a ground engaging member, from the plurality of ground engaging members, to the frame, wherein the plurality of suspensions includes a plurality of adjustable shock absorbers;

a first sensor positioned on the recreational vehicle and configured to provide cornering information to a controller; and

a second sensor positioned on the recreational vehicle and configured to provide acceleration information to the controller;

the controller operatively coupled to the first sensor, the second sensor, and the plurality of adjustable shock absorbers, wherein the controller is configured to:

receive, from the first sensor, the cornering information;

receive, from the second sensor, the acceleration information;

determine, based on the cornering information, a cornering event corresponding to the recreational vehicle executing a turn;

determine, based on the acceleration information indicating a longitudinal deceleration, that the recreational vehicle is entering the turn;

provide, to the at least one of the plurality of adjustable shock absorbers and based on the cornering event and the position of the recreational vehicle during the turn, one or more commands to result in an adjustment of a damping characteristic of the at least one of the plurality of adjustable shock absorbers;

determine, based on the cornering event, a plurality of damping characteristics for the plurality of adjustable shock absorbers;

bias, based on the determining that the vehicle is entering the turn, the plurality of damping characteristics; and

generate the one or more commands based on the plurality of biased damping characteristics.

11. The recreational vehicle of claim **10**, wherein the second sensor is an accelerometer or an inertial measurement unit (IMU).

12. The recreational vehicle of claim **10**, wherein the controller is configured to bias the plurality of damping characteristics by:

additionally increasing the compression damping of a front adjustable shock absorber of the plurality of adjustable shock absorbers; and

additionally decreasing the compression damping of a rear adjustable shock absorber of the plurality of adjustable shock absorbers.

13. The recreational vehicle of claim **10**, wherein the controller is configured to bias the plurality of damping characteristics by:

additionally increasing the rebound damping of a rear adjustable shock absorber of the plurality of adjustable shock absorbers; and

additionally decreasing the rebound damping of a front adjustable shock absorber of the plurality of adjustable shock absorbers.

14. The recreational vehicle of claim **10**, wherein the controller is configured to determine the position of the recreational vehicle during the turn by determining, based on the acceleration information indicating a longitudinal acceleration, that the vehicle is exiting the turn, and

wherein the controller is further configured to:

determine, based on the cornering event, a plurality of damping characteristics for the plurality of adjustable shock absorbers;

bias, based on the determining that the vehicle is exiting the turn, the plurality of damping characteristics; and

generate the one or more commands based on the plurality of biased damping characteristics.

15. The recreational vehicle of claim **14**, wherein the controller is configured to bias the plurality of damping characteristics by:

additionally increasing the compression damping of a rear adjustable shock absorber of the plurality of adjustable shock absorbers; and

additionally decreasing the compression damping of a front adjustable shock absorber of the plurality of adjustable shock absorbers.

16. The recreational vehicle of claim **14**, wherein the controller is configured to bias the plurality of damping characteristics by:

additionally increasing the rebound damping of a front adjustable shock absorber of the plurality of adjustable shock absorbers; and

additionally decreasing the rebound damping of a rear adjustable shock absorber of the plurality of adjustable shock absorbers.

17. A recreational vehicle, comprising:

a plurality of ground engaging members;

a frame supported by the plurality of ground engaging members;

a plurality of suspensions, wherein each of the plurality of suspensions couples a ground engaging member, from the plurality of ground engaging members, to the frame, wherein the plurality of suspensions includes a plurality of adjustable shock absorbers;

at least one sensor positioned on the recreational vehicle and configured to provide sensor information to a controller, and the sensor information comprises yaw rate information indicate a yaw rate and steering information indicating a steering position or a steering rate; and

the controller operatively coupled to the at least one sensor and the plurality of adjustable shock absorbers, wherein the controller is configured to:

receive, from the at least one sensor, the sensor information;

determine, based on the sensor information, a cornering event corresponding to the recreational vehicle executing a turn;

provide, to at least one of the plurality of adjustable shock absorbers and based on the determining of the cornering event, one or more commands to result in a decrease of a damping characteristic of the at least one of the plurality of adjustable shock absorbers; and

prioritize the yaw rate information over the steering information such that the controller is configured to determine the vehicle is executing the turn in a first direction based on the yaw rate indicating the turn in the first direction and even if the steering position or the steering rate indicates the turn in a second direction or does not indicate the turn.

18. The recreational vehicle of claim **17**, wherein the controller is further configured to:

determine, based on the sensor information, a direction of the turn corresponding to the cornering event;

determining, based on the direction of the turn, at least one inner adjustable shock absorber of the plurality of adjustable shock absorbers, and

wherein the controller is configured to provide the one or more commands by providing, to the at least one inner adjustable shock absorber, one or more commands to result in a decrease of a compression damping characteristic and an increase of a rebound damping characteristic.

19. The recreational vehicle of claim **18**, wherein the controller is further configured to:

determine, based on the direction of the turn, at least one outer adjustable shock absorber of the plurality of adjustable shock absorbers, and

wherein the controller is configured to provide the one or more commands by providing, to the least one outer adjustable shock absorber one or more commands to result in a decrease of a rebound damping characteristic and an increase of a compression damping characteristic.

20. The recreational vehicle of claim **17**, wherein the at least one sensor is an acceleration sensor.

21. The recreational vehicle of claim **17**, wherein the one or more commands results in an increase of a first damping characteristic and a decrease of a second damping characteristic of the at least one of the plurality of adjustable shock absorbers.

22. A recreational vehicle, comprising:

a plurality of ground engaging members;

a frame supported by the plurality of ground engaging members;

a plurality of suspensions, wherein each of the plurality of suspensions couples a ground engaging member, from the plurality of ground engaging members, to the frame, wherein the plurality of suspensions includes a plurality of adjustable shock absorbers;

at least one sensor positioned on the recreational vehicle and configured to provide sensor information to a controller, and the sensor information comprises acceleration information indicating a lateral acceleration value and yaw rate information indicating a yaw rate; and

the controller operatively coupled to the at least one sensor and the plurality of adjustable shock absorbers, wherein the controller is configured to:

receive, from the at least one sensor, the sensor information;

determine, based on the sensor information, a cornering event corresponding to the recreational vehicle executing a turn;

provide, to at least one of the plurality of adjustable shock absorbers and based on the determining of the cornering event, one or more commands to result in a decrease of a damping characteristic of the at least one of the plurality of adjustable shock absorbers; and

prioritize the acceleration information over the yaw rate information such that the controller is configured to determine the vehicle is executing the turn in a first direction based on the lateral acceleration value indicating the turn in the first direction and even if the yaw rate indicates the turn in a second direction or does not indicate the turn.

23. The recreational vehicle of claim **22**, wherein the controller is further configured to:

determine, based on the sensor information, a direction of the turn corresponding to the cornering event;

determining, based on the direction of the turn, at least one inner adjustable shock absorber of the plurality of adjustable shock absorbers, and

wherein the controller is configured to provide the one or more commands by providing, to the at least one inner adjustable shock absorber, one or more commands to result in a decrease of a compression damping characteristic and an increase of a rebound damping characteristic.

24. The recreational vehicle of claim **22**, wherein the controller is further configured to:

determine, based on the direction of the turn, at least one outer adjustable shock absorber of the plurality of adjustable shock absorbers, and

wherein the controller is configured to provide the one or more commands by providing, to the least one outer adjustable shock absorber one or more commands to result in a decrease of a rebound damping characteristic and an increase of a compression damping characteristic.

25. The recreational vehicle of claim **22**, wherein the at least one sensor is an acceleration sensor.

26. The recreational vehicle of claim **22**, wherein the one or more commands results in an increase of a first damping characteristic and a decrease of a second damping characteristic of the at least one of the plurality of adjustable shock absorbers.

27. A recreational vehicle, comprising:

a plurality of ground engaging members;

a frame supported by the plurality of ground engaging members;

a plurality of suspensions, wherein each of the plurality of suspensions couples a ground engaging member, from the plurality of ground engaging members, to the frame, wherein the plurality of suspensions includes a plurality of adjustable shock absorbers;

at least one sensor positioned on the recreational vehicle and configured to provide sensor information to a controller, and the sensor information comprises steering information indicating a steering position or a steering rate and acceleration information indicating a lateral acceleration value; and

the controller operatively coupled to the at least one sensor and the plurality of adjustable shock absorbers, wherein the controller is configured to:

receive, from the at least one sensor, the sensor information;

determine, based on the sensor information, a cornering event corresponding to the recreational vehicle executing a turn;

provide, to at least one of the plurality of adjustable shock absorbers and based on the determining of the cornering event, one or more commands to result in a decrease of a damping characteristic of the at least one of the plurality of adjustable shock absorbers, and

prioritize the acceleration information over the steering information such that the controller is configured to determine the vehicle is executing the turn in a first direction based on the lateral acceleration value indicating the turn in the first direction and even if the steering position or the steering rate indicates the turn in a second direction or does not indicate the turn.

28. The recreational vehicle of claim **27**, wherein the controller is further configured to:

determine, based on the sensor information, a direction of the turn corresponding to the cornering event;

determining, based on the direction of the turn, at least one inner adjustable shock absorber of the plurality of adjustable shock absorbers, and

wherein the controller is configured to provide the one or more commands by providing, to the at least one inner adjustable shock absorber, one or more commands to result in a decrease of a compression damping characteristic and an increase of a rebound damping characteristic.

29. The recreational vehicle of claim **27**, wherein the controller is further configured to:

determine, based on the direction of the turn, at least one outer adjustable shock absorber of the plurality of adjustable shock absorbers, and

wherein the controller is configured to provide the one or more commands by providing, to the least one outer adjustable shock absorber one or more commands to result in a decrease of a rebound damping characteristic and an increase of a compression damping characteristic.

30. The recreational vehicle of claim **27**, wherein the at least one sensor is an acceleration sensor.

31. The recreational vehicle of claim **27**, wherein the one or more commands results in an increase of a first damping characteristic and a decrease of a second damping characteristic of the at least one of the plurality of adjustable shock absorbers.

32. A recreational vehicle comprising:

a plurality of ground engaging members;

a frame supported by the plurality of ground engaging members;

a plurality of suspensions, wherein each of the plurality of suspensions couples a ground engaging member, from the plurality of ground engaging members, to the frame, wherein the plurality of suspensions includes a plurality of adjustable shock absorbers;

a first sensor positioned on the recreational vehicle and configured to provide cornering information to a controller; and

a second sensor positioned on the recreational vehicle and configured to provide acceleration information to the controller;

the controller operatively coupled to the first sensor, the second sensor, and the plurality of adjustable shock absorbers, wherein the controller is configured to:

receive, from the first sensor, the cornering information;

receive, from the second sensor, the acceleration information;

determine, based on the cornering information, a cornering event corresponding to the recreational vehicle executing a turn;

determine, based on the acceleration information indicating a longitudinal acceleration, that the recreational vehicle is exiting the turn;

provide, to the at least one of the plurality of adjustable shock absorbers and based on the cornering event and the position of the recreational vehicle during the turn, one or more commands to result in an adjustment of a damping characteristic of the at least one of the plurality of adjustable shock absorbers;

determine, based on the cornering event, a plurality of damping characteristics for the plurality of adjustable shock absorbers;

bias, based on the determining that the vehicle is exiting the turn, the plurality of damping characteristics; and

generate the one or more commands based on the plurality of biased damping characteristics.

33. The recreational vehicle of claim **32**, wherein the controller is further configured to:

determine, based on the sensor information, a direction of the turn corresponding to the cornering event;

determining, based on the direction of the turn, at least one inner adjustable shock absorber of the plurality of adjustable shock absorbers, and

wherein the controller is configured to provide the one or more commands by providing, to the at least one inner adjustable shock absorber, one or more commands to result in a decrease of a compression damping characteristic and an increase of a rebound damping characteristic.

34. The recreational vehicle of claim **32**, wherein the controller is further configured to:

determine, based on the direction of the turn, at least one outer adjustable shock absorber of the plurality of adjustable shock absorbers, and

wherein the controller is configured to provide the one or more commands by providing, to the least one outer adjustable shock absorber one or more commands to result in a decrease of a rebound damping characteristic and an increase of a compression damping characteristic.

35. The recreational vehicle of claim **32**, wherein the at least one sensor is an acceleration sensor.

36. The recreational vehicle of claim **32**, wherein the one or more commands results in an increase of a first damping characteristic and a decrease of a second damping characteristic of the at least one of the plurality of adjustable shock absorbers.

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